



सत्यमेव जयते

Government of India

Ministry of Jal Shakti

Department of Drinking Water & Sanitation

National Jal Jeevan Mission

Operational Guidelines
Jal Jeevan Mission 2.0
Har Ghar Jal

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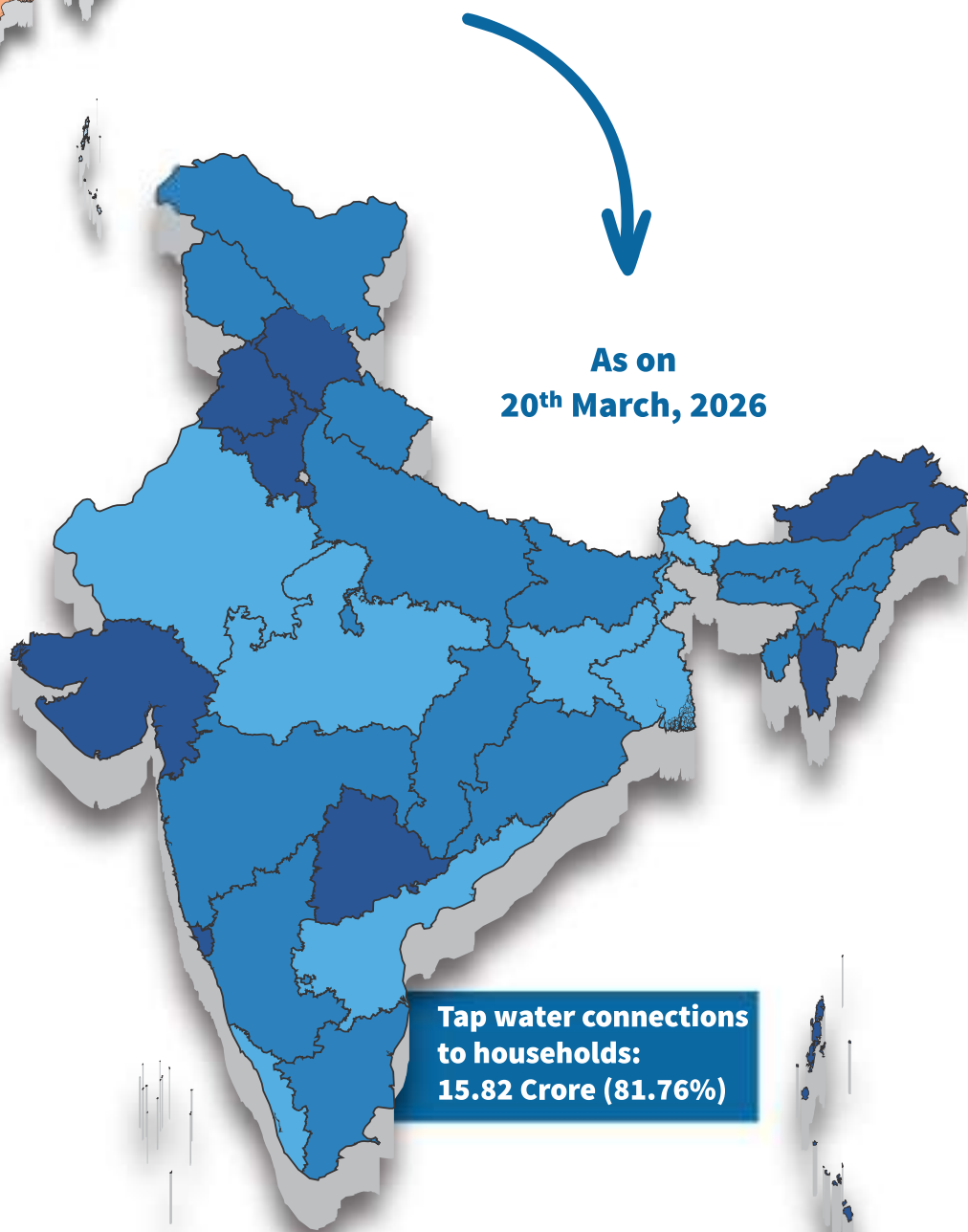
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Abbreviations

AAP	Annual Action Plan
AICTE	All India Council for Technical Education
AI	Artificial Intelligence
ANM	Auxiliary Nurse Midwife
API	Application Programming Interface
ASHA	Accredited Social Health Activist
ATR	Action Taken Report
AWG	Atmospheric Water Generator
AWW	Anganwadi Worker
AWS	Automatic Weather Station
BCC	Behaviour Change Communication
BIS	Bureau of Indian Standards
BISAG-N	Bhaskaracharya National Institute for Space Applications and Geo-informatics
BWS	Bulk Water Supply
C&AG	Comptroller and Auditor-General of India
CA	Chartered Accountant
CBO	Community-Based Organisation
CGWA	Central Ground Water Authority
CGWB	Central Ground Water Board
CII	Confederation of Indian Industry
CIPET	Central Institute of Petrochemicals Engineering and Technology
CIRP	Comprehensive Implementation and Reform Plan
CPHEEO	Central Public Health and Environmental Engineering Organisation
CPO	Central Prabhari Officer
CPWD	Central Public Works Department
CSIR	Council of Scientific and Industrial Research
CSR	Corporate Social Responsibility
CWPP	Community Water Purification Plant
DA-JGUA	Dharti Abha Janjatiya Gram Utkarsh Abhiyan

DBOT	Design-Build-Operate-Transfer
DCSC	District Command and Support Centre
DDWS	Department of Drinking Water & Sanitation
DIP	District Improvement Plan
DLE	Departmental Level Expenditure
DLI	Disbursement Linked Indicator
DMF	District Mineral Development Fund
DPI	Digital Public Infrastructure
DPIIT	Department for Promotion of Industry and Internal Trade
DPR	Detailed Project Report
DSS	Decision Support System
DSSAP	District Source Sustainability Action Plan
DST	Department of Science and Technology
DTU	District Technical Unit
DWSM	District Water and Sanitation Mission
EPC	Engineering, Procurement and Construction
FGD	Focus Group Discussion
FHTC	Functional Household Tap Connection
FICCI	Federation of Indian Chambers of Commerce and Industry
FRC	Free Residual Chlorine
FTK	Field Test Kit
GIS	Geographic Information System
GP	Gram Panchayat
HAM	Hybrid Annuity Model
HDPE	High-Density Polyethylene
HGJ	Har Ghar Jal
HR	Human Resource
HSC	House Service Connection
IA	Impact Assessment / Implementing Agency
IDSP	Integrated Disease Surveillance Programme

IEC	Information, Education and Communication
IFMIS	Integrated Financial Management Information System
IIT	Indian Institute of Technology
IMD	India Meteorological Department
IMIS	Integrated Management Information System
IoT	Internet of Things
IS	Indian Standard
ISO	International Organisation for Standardisation
ITI	Industrial Training Institute
IVR	Interactive Voice Response
IWMP	Integrated Watershed Management Programme
JJM	Jal Jeevan Mission
JSA	Jal Seva Aankalan
JSIC	Jal Shakti Innovation Centre
LWE	Left Wing Extremism
MAHA	Mission for Advancement in High-impact Areas
MBR	Master Balancing Reservoir
MGNREGA	Mahatma Gandhi National Rural Employment Guarantee Act
MGNREGS	Mahatma Gandhi National Rural Employment Guarantee Scheme
MIS	Management Information System
ML	Machine Learning
MoHFW	Ministry of Health and Family Welfare
MoHUA	Ministry of Housing and Urban Affairs
MoJS	Ministry of Jal Shakti
MoU	Memorandum of Understanding
MoWCD	Ministry of Women and Child Development
MPLAD	Members of Parliament Local Area Development
MTTR	Mean Time To Repair
MVS	Multi Village Scheme
NCVET	National Council for Vocational Education and Training
NDWM	National Drinking Water Mission

NER	North Eastern Region
NFHS	National Family Health Survey
NGO	Non-Governmental Organisation
NIRD&PR	National Institute of Rural Development and Panchayati RajAayogPanchayati Raj
NIT	National Institute of Technology
NITI Aayog	National Institution for Transforming India Aayog
NJJM	National Jal Jeevan Mission
NJMP	Nal Jal Mitra Programme
NJM	Nal Jal Mitra
NOC	No Objection Certificate
NPMU	National Project Management Unit
NRDWP	National Rural Drinking Water Programme
NRLM	National Rural Livelihood Mission
NRSC	National Remote Sensing Centre
NRW	Non-Revenue Water
NSQF	National Skills Qualifications Framework
NSS	National Service Scheme
NTC	National Technical Committee
NWE	National WASH Expert
NWIC	National Water Informatics Centre
O&M	Operation and Maintenance
OBF	Outcome-Based Financing
OHSR	Overhead Service Reservoir
OOMF	Output-Outcome Monitoring Framework
OTP	One-Time Password
PACS	Primary Agricultural Credit Societies
PFMS	Public Financial Management System
PHE	Public Health Engineering
PHED	Public Health Engineering Department
PIA	Project Implementing Agency
PIB	Press Information Bureau
PMKK	Pradhan Mantri Kaushal Kendra
PMKSY	Pradhan Mantri Krishi Sinchayee Yojana
PMKVY	Pradhan Mantri Kaushal Vikas Yojana

PMMVY	Pradhan Mantri Matru Vandana Yojana
PM-JANMAN	Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan
PPP	Public-Private Partnership
PR	Panchayati Raj
PRA	Participatory Rural Appraisal
PRI	Panchayati Raj Institution
PSU	Public Sector Undertaking
PVTG	Particularly Vulnerable Tribal Group
PWS	Piped Water Supply
QA	Quality Assurance
QC	Quality Control
QCI	Quality Council of India
RBI	Reserve Bank of India
RCC	Reinforced Cement Concrete
RD	Rural Development
RDWSD	Rural Drinking Water Supply Department
REDA	Renewable Energy Development Agency
RITES	Rail India Technical and Economic Services
RJJK	Rashtriya Jal Jeevan Kosh
RLB	Rural Local Body
RPL	Recognition of Prior Learning
RPWSS	Rural Piped Water Supply Scheme
RWPF	Rural WASH Partners' Forum
RWS	Rural Water Supply
SBM-G	Swachh Bharat Mission (Grameen)
SCADA	Supervisory Control and Data Acquisition
SCSC	State Command and Support Centre
SDG	Sustainable Development Goal
SHG	Self Help Group
SIDH	Skill India Digital Hub
SIP	Service Improvement Plan
SIRD	State Institute of Rural Development
SLA	Service Level Agreement
SLB	Service Level Benchmark

SNA	Single Nodal Account
SOP	Standard Operating Procedure
SPM-NIWAS	Dr. Syama Prasad Mookerjee National Institute of Water and Sanitation
SRLM	State Rural Livelihood Mission
SSC	Source Sustenance Committee
SSDM	State Skill Development Mission
STC	State Technical Committee
STT	Short-Term Training
SVS	Single Village Scheme
SWSM	State Water and Sanitation Mission
TOR	Terms of Reference
TPIA	Third Party Inspection Agency
TRL	Technology Readiness Level
UIDAI	Unique Identification Authority of India
UT	Union Territory
VAP	Village Action Plan
VGF	Viability Gap Funding
VHND	Village Health and Nutrition Day
VHSNC	Village Health, Sanitation and Nutrition Committee
VSSC	Village Source Sustenance Committee
VWSC	Village Water and Sanitation Committee
WASH	Water, Sanitation and Hygiene
WDC	Watershed Development Component
WMPSC	Water Management & Plumbing Skill Council
WQM&S	Water Quality Monitoring and Surveillance
WQMIS	Water Quality Management Information System
WSO	Water Supply Operator
WTP	Water Treatment Plant
ZBR	Zonal Balancing Reservoir
ZP	Zila Parishad

Definitions

Functional Household Tap Connection (FHTC): A household tap connection that provides potable water in adequate quantity (minimum 55 litres per capita per day), of prescribed quality (conforming to BIS IS:10500), on a regular basis at the delivery point.

Service Level Norm (55 LPCD): The prescribed minimum quantity of water, i.e. at least 55 litres per capita per day (lpcd), of prescribed quality, i.e. conforming to BIS:10500 standards, on regular basis. The prescribed quantity includes water for drinking, cooking, personal hygiene and domestic sanitation needs etc.

Design Life of Structures: The period for which water supply scheme structure or asset or component is designed to perform its intended function without major rehabilitation.

Population Projection: The methodology for estimating the future population of a service area used in the design of water supply schemes. Suggestive permissible limits for population growth projections under JJM are 18% at 15 years and 32% at 30 years from the base year. Population projections shall be prepared as per the methodology prescribed in CPHEEO Manual citing verified Govt. data/publication.

Domestic Demand: Water required for drinking, cooking, personal bathing, domestic sanitation, etc. Domestic demand does not include requirements of water for cattle, semi-commercial, commercial, institutional or industrial purposes.

Institutional Water Demand: Water demand of public institutions such as Gram Panchayat buildings, schools, Anganwadi Centres, Health Centres and Wellness Centres.

Non-Revenue Water (NRW): The difference between the volume of treated water put into the distribution system and the volume of water billed or accounted to consumers. NRW includes real losses (physical leakage from pipes, joints and storage structures), apparent losses (meter inaccuracies, unauthorized connections) and unbilled authorised consumption.

Master Balancing Reservoir (MBR) / Zonal Balancing Reservoir (ZBR): Large storage tanks or reservoirs used to balance water supply and demand within specific zones of a water distribution system, stabilizing pressure and flow and providing operational flexibility to match peak demand variations.

Elevated Service Reservoir (ESR) / Overhead Storage Reservoir (OHSR): Constructed storage tanks or reservoirs that provide treated water storage specifically for distributing water near to consumers, helping maintain hydraulic pressure and meeting immediate supply demands.

Conventional Water Treatment Plant (WTP): A typical water treatment facility employing standard processes such as coagulation, flocculation, sedimentation, filtration and disinfection to purify raw water for safe public use.

Life Cycle Cost (LCC) Assessment: An economic evaluation methodology that considers all costs associated with a water supply component over its entire design life, including capital expenditure (CAPEX), operation and maintenance expenditure (OPEX), rehabilitation and replacement costs and residual value.

SCADA (Supervisory Control and Data Acquisition): An automated control and monitoring system used to supervise, control and acquire operational data from distributed water supply infrastructure components such as pumps, valves, flow meter, pressure transmitters and treatment units.

Design Capacity: Maximum hydraulic or treatment capacity for which a Water Treatment Plant, pipeline, or reservoir is designed, usually expressed in million litres per day (MLD) or similar units to meet expected demand.

In-Village Infrastructure: In-Village Infrastructure refers to the distribution of treated water from the Overhead Storage Reservoir (OHSR) to individual households within the village. This includes the internal distribution network, service connections and all components facilitating last-mile delivery of potable water to end users.



Out-Village Infrastructure: Out-Village Infrastructure includes the transfer of treated water from the Water Treatment Plant (WTP) to the Master Balancing Reservoir (MBR) and from the MBR to the Overhead Tank (OHT), including all associated transmission lines. (OHSR not included)

Service Area: Geographical area or administrative boundary within which the water supply system operates and provides services to consumers.

Treated Water: Water that has successfully undergone all required treatment processes in a Water Treatment Plant and meets accepted potable water quality standards suitable for human consumption

Water Treatment Plant: An engineered facility designed to receive raw water and subject it to various physical, chemical and biological processes to remove contaminants, pathogens and undesirable substances, producing potable water that complies with health standards.

Single Village Scheme (SVS): Water supply scheme which includes the local source, storage with or without treatment under the administrative jurisdiction of a Gram Panchayat.

Multi-Village Scheme (MVS): Comprehensive water supply scheme providing infrastructure and services to multiple villages or Gram Panchayats collectively, commonly used for regional water resource optimization and economies of scale.

Bulk Water Scheme (BWS): It refers to the transfer of raw water from the point of intake through transmission lines up to the WTP.

Non-Piped Water Supply (Non-PWS) Scheme: A water supply arrangement that provides safe drinking water through community access points (stand posts, hand pumps, community water purification plants) in situations where household piped connections are technically unfeasible due to extreme terrain, climatic conditions, or dispersed habitations. Such schemes provide a minimum of 8–10 lpcd for drinking and cooking.

Community Water Purification Plant (CWPP): A point-of-use or community-scale water treatment unit designed to remove specific chemical contaminants

(arsenic, fluoride, iron, nitrates) or microbial contamination from raw water to produce potable water. CWPPs are deployed as interim measures in quality-affected habitations providing 8–10 lpcd until a piped water supply system is established.

BIS IS:10500 (Drinking Water Quality Standard): The Bureau of Indian Standards specification IS:10500 (latest edition, read with latest amendments) that prescribes permissible limits for physical, chemical, bacteriological and radiological parameters of drinking water in India. All rural piped water supply schemes under JJM shall ensure that water supplied to households conforms to BIS publication IS:10500 standards at the delivery point.

Free Residual Chlorine (FRC): The concentration of dissolved chlorine remaining in treated and distributed water after a specified contact period, which provides ongoing disinfection protection against microbial re-contamination in the distribution system.

Field Test Kit (FTK): A portable, field-deployable kit used to conduct rapid qualitative and semi-quantitative tests for basic water quality parameters. FTK testing by community members forms the first tier of the Water Quality Monitoring & Surveillance (WQM&S) framework under JJM.

Sanitary Inspection / Sanitary Survey: A structured field assessment of potential contamination risks in the catchment, source, treatment, distribution and household handling stages of a water supply system. Sanitary surveys identify environmental, structural, operational and behavioural factors that may compromise water quality.

State Water and Sanitation Mission (SWSM): The State-level institutional body responsible for implementation oversight, sector policy, scheme approval, resource allocation and monitoring of rural water supply under JJM.

District Water and Sanitation Mission (DWSM): The district-level institutional body chaired by the District Collector/Deputy Commissioner, responsible for overall implementation, supervision and monitoring of JJM at the district level.

District Technical Unit (DTU): A technical committee



constituted under the DWSM to provide continuous technical support to Gram Panchayat and VWSC-level micro utilities. The DTU coordinates between micro utilities (in-village infrastructure) and macro utilities (bulk/out-village infrastructure), oversees scheme commissioning and handover, manages asset registries, conducts source sustainability planning and operates District Command and Control Centres.

Village Water & Sanitation Committee (VWSC) / Pani Samiti: The village-level standing committee of the Gram Panchayat constituted to plan, implement, manage and oversee in-village rural piped water supply systems.

Micro Utility: The village-level decentralised water service delivery unit constituted by designating the VWSC (standing committee of the Gram Panchayat) as the service provider responsible for distribution of drinking water within the village, collection of user charges, management of in-village infrastructure, minor repairs and maintenance, water quality monitoring and financial management.

Macro Utility: A State, Union Territory, or regional-level professional body (parastatal agency, PHED, Jal Nigam, or specialised utility) responsible for planning, development, management and supervision of out-village water supply assets, including surface water intake structures, water treatment plants, bulk transmission systems, master balancing reservoirs and Multi-Village / Bulk Water Supply scheme networks.

Third-Party Inspection Agency (TPIA): An independent technical agency empanelled to conduct quality assurance inspections of rural water supply scheme construction activities.

Source Sustenance Committee (SSC): A multi-tier committee constituted at the State, District and Gram Panchayat levels within the JJM institutional structure, responsible for planning, implementing, monitoring and coordinating source sustainability measures.

National Project Management Unit (NPMU): A technical and managerial support unit of the NJJM comprising engineering, management and IT experts providing scheme engineering reviews, planning, monitoring, evaluation, predictive modelling and management of JJM-IMIS.

Har Ghar Jal (HGJ) Certification: A formal certification accorded to a Gram Panchayat or village where every rural household has been provided with a Functional Household Tap Connection (FHTC) supplying potable water at a minimum of 55 lpcd, of quality conforming to BIS IS:10500, on a regular basis.

Functionality Assessment: A periodic structured assessment to determine whether a village water supply scheme is delivering services meeting prescribed service level parameters: adequacy of supply (≥ 55 lpcd), quality compliance (BIS IS:10500), regularity of supply.

Jal Seva Aankalan (JSA): A GP/VWSC-led digital self-assessment tool and mechanism for evaluating the functionality, adequacy, quality compliance, institutional readiness and financial sustainability of rural water supply systems at the village level.

Service Level Benchmarks (SLBs): Quantitative performance indicators defined by regulatory or programme authorities to measure the quality and efficiency of water supply services.

Service Improvement Plan (SIP): A structured remedial action plan prepared by DWSMs and SWSMs based on the findings of Jal Seva Aankalan, identifying service gaps (infrastructure, operational, financial, institutional) and defining time-bound corrective measures, responsible entities and resource requirements to restore or improve water supply service delivery

Service Level Agreement (SLA): An agreement between service provider and contracting authority, clearly defining service delivery standards, responsibilities, measurable performance indicators.

Key Performance Indicator (KPI): A measurable, time-bound metric used to evaluate the performance of a utility, scheme, or institutional body against defined service standards.

Digital Twin: A real-time virtual replica of a physical water supply system — integrating source, treatment, storage, transmission, distribution, metering and household connection data — that mirrors the behaviour of the actual system through sensor feeds (SCADA/IoT), GIS asset registries and operational

transaction data.

Comprehensive Implementation and Reform Plan (CIRP): A strategic, time-bound operational plan prepared by each State/UT outlining scheme-wise completion timelines, financial allocations, reform commitments and governance milestones for the JJM 2.0 mission period.

Annual Action Plan (AAP): A one-year operational programme prepared by States/UTs within the CIRP framework.

SNA-SPARSH: An integrated financial management and payment mechanism that connects JJM-IMIS with the Public Financial Management System (PFMS), State Integrated Financial Management Information System (IFMIS) and the Reserve Bank of India's e-Kuber platform to enable scheme-wise, just-in-time payments, near-zero unspent balances and automated reconciliation of central and State fund releases.

Mother Sanction: The mother sanction in SNA SPARSH is the initial cumulative financial approval by DDWS through the PFMS to a State/UT establishing the authorised expenditure limit for a given financial year.

Inadmissible Expenditure (under JJM Central Share): Categories of expenditure that cannot be charged to the Central share under JJM.

Finance Commission Grants (15th/16th FC): Grants devolved by the 15th Finance Commission (covering 2021–26) and the 16th Finance Commission to Panchayati Raj Institutions (PRIs) for basic services, with a tied component specifically earmarked for water supply, sanitation, ODF maintenance and water quality improvement. States must ensure these grants are used only for permitted activities under FC guidelines and are properly accounted for.

Service Delivery Charges: Fees levied on households and other consumers for in-village water supply services, intended to recover the part/ full Operation & Maintenance costs of the village water supply system. A graded user charge system shall be adopted, ensuring full O&M cost recovery for basic domestic services while providing targeted support for vulnerable households. Higher tariffs apply to commercial and non-domestic users.

Reimbursement Mode: A fund-release mechanism where Central share is released only after the State/UT has incurred actual expenditure from its own resources even for Central share on eligible scheme components and submitted validated expenditure claims with physical progress evidence through the Programme Monitoring Module on Sujalam Bharat. No upfront Central release is made.

Sujalam Bharat (Digital Public Infrastructure — DPI): The national-level federated Digital Public Infrastructure for India's rural drinking water sector, developed as the digital backbone for GIS-based asset mapping, monitoring and lifecycle management of Rural Piped Water Supply Schemes (RPWSS). It assigns unique Sujal Gaon IDs to each service area, integrates interoperable digital registries, transaction platforms and decision-support systems and provides democratised data access to multiple cross-sectoral stakeholders through open APIs.

Sujal Gaon ID: A unique system-generated alphanumeric identifier assigned to each service area (habitation or group of habitations) within the Sujalam Bharat DPI framework, enabling complete source-to-tap digital mapping of rural piped water supply systems.

JJM-IMIS (Integrated Management Information System): The national web-based information management platform of the Jal Jeevan Mission that integrates data on scheme sanction, financial releases, physical progress, households covered, water quality monitoring (WQMIS), functionality assessments, grievances and training activities across all States/UTs.

GIS-Based Asset Mapping: The geo-tagging or geo-referencing of all rural piped water supply infrastructure assets (sources, WTPs, pumping stations, MBRs, ZBRs, OHTs, distribution pipelines and household connections) using GPS-enabled mobile applications and satellite imagery, linked to their Asset IDs, Scheme IDs and Sujal Gaon IDs on the Sujalam Bharat platform.

Designated Implementation Agency (DIA): The State-authorized agency registered on the Sujalam Bharat application entrusted for GIS asset mapping

Decision Support System (DSS) — Source Sustainability: A digital planning and analytical platform integrated with the JJM-IMIS framework that



combines geo-spatial layers, hydro-geological parameters, CGWB groundwater status data, rainfall data, aquifer maps and drinking water source datasets to support evidence-based planning, monitoring and prioritisation of source sustainability measures from district to village level.

Sujal Gaon Utility App: The mobile application serving as the standardised transactional data-capture platform for village-level O&M under JJM, enabling digital recording of daily pumping operations, water quality FTK/lab testing results, chlorination logs, preventive maintenance activities, user charge collection and grievance complaints.

District Command and Support Centre (DCSC): The central operational hub established under the DWSM that provides near real-time monitoring of scheme functionality, water quality alerts, grievance tracking and coordinated corrective response for rural water supply across the district.

Source Sustainability: The ability to ensure the long-term reliability, adequacy and safety of drinking water sources throughout the full lifecycle of rural water supply schemes.

Aquifer-Based Planning: A hydrogeological approach to designing and managing groundwater-based water supply schemes using aquifer maps and tools (such as NAQUIM, India-WRIS), groundwater assessment data and aquifer characteristics (type, depth, seasonal behaviour, recharge potential, extraction stage) to inform siting of borewells, planning of recharge structures (check dams, percolation tanks, recharge shafts) and preparation of Sustainability Action Plans.

Water Budgeting: A structured, data-driven assessment & planning exercise to estimate total water availability (from rainfall, groundwater recharge, runoff, surface storage and greywater reuse) and total demand (domestic, livestock, irrigation and other uses) within a defined service area.

Greywater Management: The collection, treatment and safe reuse of household wastewater generated from domestic activities (kitchen, bathing, laundry) excluding toilet waste (blackwater).

Source Protection Zone: A defined geospatial area

surrounding a drinking water source (borewell, tube well, spring, or surface water intake) within which land use is regulated and certain activities are prohibited to prevent contamination of the drinking water source.

Managed Aquifer Recharge (MAR): The intentional and planned replenishment of groundwater aquifers through engineered recharge structures (check dams, percolation tanks, recharge shafts, injection wells, farm ponds) to enhance groundwater availability, improve source sustainability and reduce seasonal stress on drinking water sources.

Nal Jal Mitra (NJM) / Jal Vitaran Sanchaalak: A locally trained and NCVET-certified multi-skilled O&M personnel (NSQF Level-4, job role: 'Jal Vitaran Sanchaalak') responsible for in-village water supply scheme operation and maintenance, including pump operation, basic plumbing repairs, routine maintenance, network upkeep, water quality reporting and grievance management.

NSQF (National Skills Qualifications Framework): A competency-based quality assurance framework established by the Government of India that organises qualifications according to levels of knowledge, skills and aptitude.

Recognition of Prior Learning (RPL): A certification pathway through which individuals with prior informal work experience in O&M of water supply or related domains are assessed, upskilled through a structured 120-hour programme and awarded the nationally recognised NSQF Level-4 Jal Vitaran Sanchaalak certification without undergoing the full Short-Term Training (STT) of 510 hours.

Sujalam Shakti: Community-based water quality sentinels constituted from women's Self Help Groups (SHGs) under State Rural Livelihood Missions (SRLM), trained and certified at NSQF Level-1 in basic water quality monitoring and surveillance.

Jal Arpan: The formal commissioning and community handover ceremony/protocol under which a completed rural piped water supply scheme (after a successful 15-day minimum trial run) is officially handed over by the implementing agency to the Gram Panchayat/ VWSC for long-term ownership, operation and maintenance.





SECTION 1

Background



CHAPTER 1

Introduction

Jal Jeevan Mission has brought a transformative change in rural India by ensuring access to safe and reliable tap water in more than 80% of the households. It has significantly reduced the drudgery of women and children who earlier spent hours fetching water, improved public health by providing clean drinking water and enhanced dignity and quality of life in villages. Beyond infrastructure, JJM has fostered community participation, strengthened local governance through Gram Panchayats and promoted sustainable water management practices. This holistic approach has not only improved service delivery but has also contributed to socio-economic development across rural India.

1.1. Historical Perspective of Rural Water Supply in India

Access to safe and enough drinking water has always been an important development need in rural India because it affects health, daily life and livelihoods. For many years, people in villages depended on sources that were not reliable or safe, which created a need for planned and policy-based action in the water supply sector.

The below timeline presents the historical evolution of rural drinking water initiatives in India, reflecting a gradual transition from supply-driven public health interventions to community-based management and service-level delivery frameworks.

The first major step was the Accelerated Rural Water Supply Programme (ARWSP) launched in 1972. It aimed to supplement the efforts of State Governments, especially in areas of scarcity and those endemic to

Figure 1: Historical Evolution of Rural Drinking Water initiatives in India

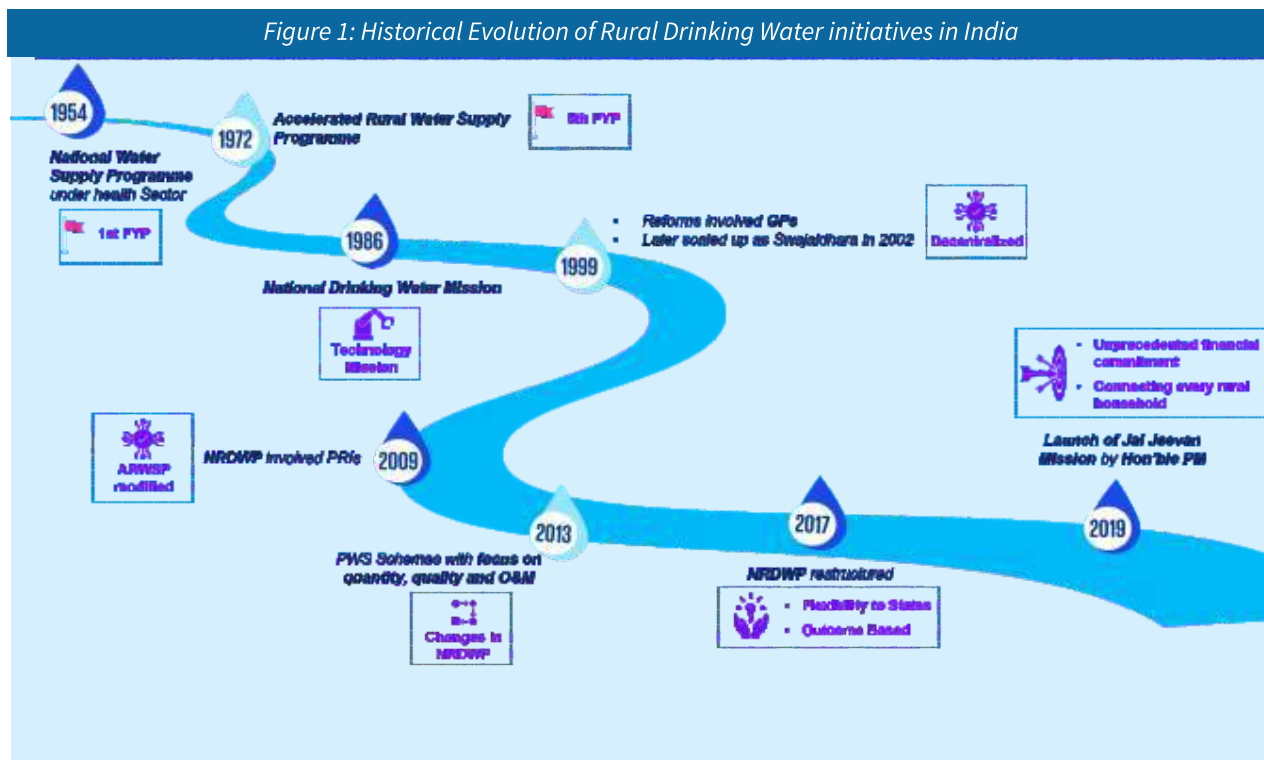
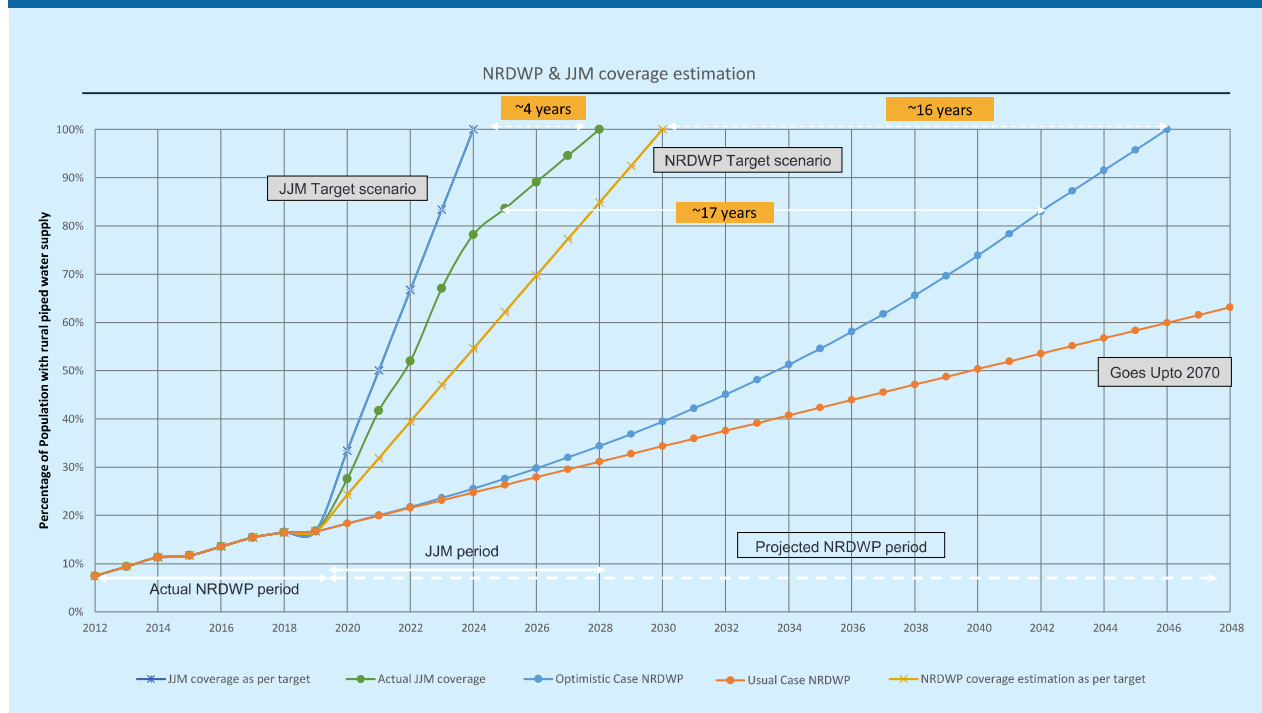


Figure 2: NRDWP and JJM PWS Coverage Estimation for Household Connections



The scenario assessment indicates that, had rural piped water supply expansion continued at the historical pace and investment levels of NRDWP, achieving universal coverage would have taken approximately 17 additional years compared to the progress under JJM. NRDWP followed a gradual, resource-constrained trajectory, whereas JJM represents a mission-mode national initiative characterized by a significantly higher scale of public investment.

water borne diseases. Although it improved access, the approach was mainly supply-driven and did not focus much on long-term source sustainability or community ownership. Later programmes, such as the Technology Mission (1986)/ Rajiv Gandhi National Drinking Water Mission, added better technology, water quality testing and institutional systems, but many challenges remained.

A major change came with the Swajaldhara Programme (2002), which promoted a community-led approach. It encouraged Gram Panchayats to take full responsibility for managing their schemes including operation and maintenance. While this improved ownership and reduced costs, many villages faced difficulties due to limited technical skills, financial issues and lack of long-term support systems.

The National Rural Drinking Water Programme (NRDWP, 2009) tried to bring together sustainability,

decentralised planning and service delivery. However, problems related to functionality, source security and service levels continued.

The Jal Jeevan Mission (2019) is the most ambitious reform in rural water supply sector. It focuses on providing every rural household with a Functional Household Tap Connection along with clear service standards, digital monitoring and stronger institutions. JJM builds on past lessons to ensure safe, reliable and sustainable drinking water for all rural households.

1.2. Jal Jeevan Mission - A Paradigm Shift

The Jal Jeevan Mission aims to provide Functional Household Tap Connections (FHTCs) to every rural household, ensuring safe and adequate drinking water.



Figure 3: JJM Solutions to NRDWP Challenges



The Mission marked a paradigm shift from infrastructure centric approaches to a service delivery framework with defined standards for quality, quantity, and reliability. Implemented in partnership with States/UTs, JJM has significantly expanded rural water supply coverage through enhanced investments, strengthened institutional frameworks, community participation and sustainability measures. This transformation has led to notable improvements in public health, dignity and ease of living, laying a strong foundation for the next phase of the Mission.

1.2.1 Key Outcomes under JJM

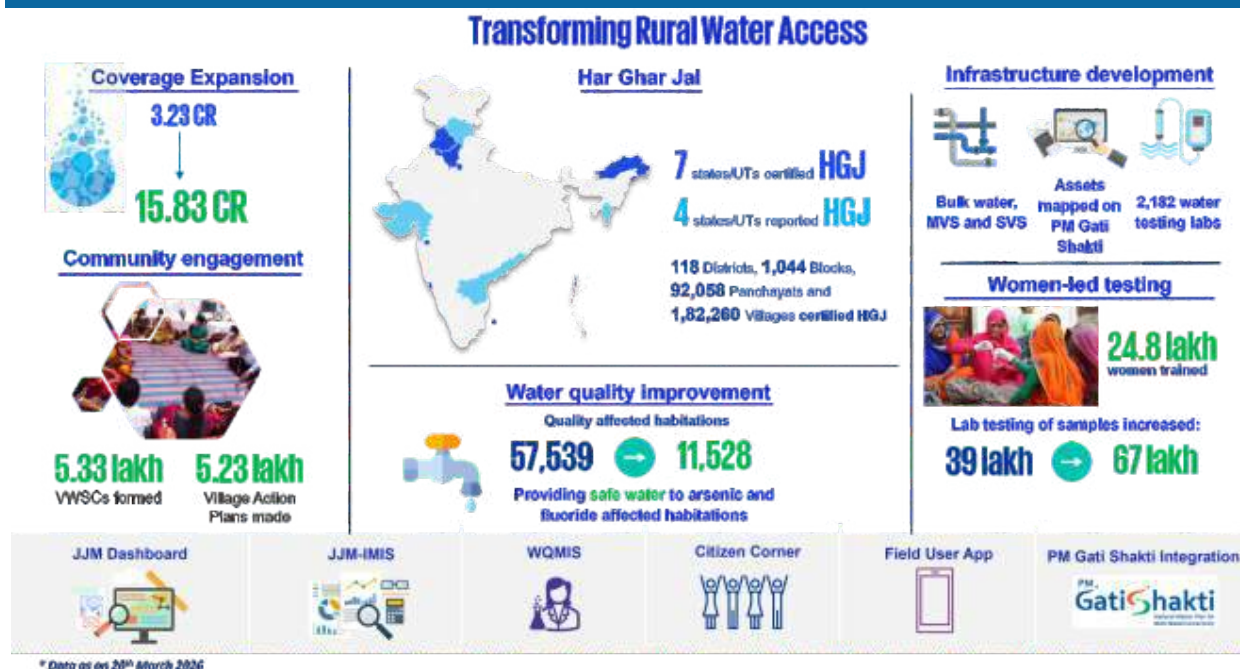
Following are key outcomes achieved under JJM, reflecting a significant transformation in rural drinking water service delivery across the country:

- i. **Universalisation of Household Tap Connection:** Shift from habitation-level access to household level service delivery has enabled provision of tap water to rural households, ensuring potable water of prescribed quality, in adequate quantity, on a regular and long-term basis, thereby enhancing convenience and dignity.
- ii. **Assurance of Minimum Service Level Standards:** Institutionalisation of a minimum service norm of 55 LPCD has ensured uniformity, adequacy and improved reliability in water supply, marking a transition towards

assured service delivery.

- iii. **Rapid Scale-Up of Coverage:** Tap water coverage has expanded from about 17% in 2019 to around 75% as on March 2024, with more than 11.39 crore additional households connected, reflecting strong Centre, State partnership and effective implementation.
- iv. **Strengthening of Water Quality Systems:** Reduction in water quality-affected habitations and universal coverage of Arsenic and Fluoride affected areas, supported by NABL accredited laboratories and community based surveillance, has enhanced public health safeguards and trust in tap water.
- v. **Women-Led Water Governance:** Ensuring at least 50% representation of women in VWSCs and their active role in water management has strengthened inclusivity, accountability and local ownership.
- vi. **Integration of Source Sustainability:** Mainstreaming of source sustainability, including groundwater recharge, has improved resilience of water sources and ensured long-term water security.
- vii. **Skilled Ecosystem for O&M:** Capacity building and initiatives such as Nal Jal Mitra Programme have created a local skilled

Figure 4: JJM: Transforming Rural Water Access



workforce for operation, maintenance, and minor repairs, ensuring continuity of services.

viii. **Expansion of WASH in Institutions:**

Increased tap water access in schools, anganwadi centres, and public institutions has improved hygiene, health outcomes, and dignity, particularly for children and women.

1.3 Jal Jeevan Mission: Implementation, Achievements and Socio-Economic Impacts

Under JJM, rural water supply schemes are designed to remain functional throughout the year, drawing lessons from the shortcomings of the erstwhile NRDWP, where the Performance Audit at the end of the 12th Plan period highlighted that 88% of piped water schemes were dependent on groundwater sources, affecting long term sustainability and safety and resulting in 4.76 lakh habitations slipping back during 2012–2017. Slip back occurred due to excessive groundwater extraction, inadequate attention to water quality aspects, poor sustainability of water sources and insufficient maintenance of existing systems, particularly in areas where agricultural withdrawal led to rapid depletion of groundwater, causing schemes to fail before their design period. Accordingly, States designed schemes under JJM selecting sustainable sources leading to a

paradigm shift from ground water to surface water-based schemes. Thus, surface water to groundwater ratio of 15:85 (as planned in initial stage of JJM) was changed to 48:52 (as approved during the mission period).

1.3.1 Achievements made during JJM

From a baseline of 3.23 crore rural households (~17%) with tap water connections in 2019, more than 12.58 crore additional rural households have been provided with tap water connections under JJM. Thus, at present, more than 15.81 crore rural households (81.73%) have tap water connections out of the total 19.36 crore rural households identified by the States/UTs.

Besides creating piped water supply infrastructure for new connections, additional households have also been benefitted through augmentation of source or infrastructure planned under JJM schemes. These household connections were previously being provided through central sponsored or state funded schemes but have been found to be non-compliant to the JJM service standards.

The designs of the schemes varied from region to region, reflecting the regional cost variances while ensuring uniform service delivery benchmarks at the household level across the country.



Figure 5: Ratio of HHs - Surface Water to Ground Water

Scheme	NRDWP (Actual)	JJM Planning 2019 (Estimation)	JJM (Actual)
Surface Water to Ground Water Ratio of HHs	12:88	15:85	48:52

The Jal Jeevan Mission has made remarkable progress across multiple dimensions, reflecting the scale, speed and systemic strengthening achieved under the programme. The Mission has considerably expanded household tap water coverage, strengthened rural water supply infrastructure and established community-based systems for service delivery. The use of digital platforms (JJM IMIS, PFMS) has improved transparency, streamlined fund flow and enabled regular monitoring and review of progress.

1.3.2. Scale of infrastructure planned

To realise the vision, States/ UTs undertook planning at an unprecedented scale, designing robust and sustainable rural drinking water supply systems. Given the diverse geographical challenges ranging from water scarcity and quality issues to difficult terrain, this involved the development of a wide spectrum of infrastructure, including single village schemes, multi-village schemes and large regional bulk water transfer systems.

These schemes incorporated extensive water transfer networks, treatment plants and zonal storage infrastructure to ensure long-term sustainability and functionality, even in areas where local water sources were inadequate or unreliable. Bulk water supply systems were designed to support and stabilise single village schemes, ensuring continuity and reliability of service delivery. This represents one of the largest infrastructure development efforts in the rural water supply sector globally.

For the first time, the rural water supply sector has witnessed an unprecedented scale of investment under Jal Jeevan Mission, with total allocations rising sharply to nearly ₹3.60 lakh crore in the period 2019–24. This steep escalation marks a transformative shift from historically modest funding levels, enabling large-scale infrastructure creation and sustained service delivery.

1.3.3. Water Supply Assets

A Piped Water Supply (PWS) scheme under the Jal Jeevan Mission is a comprehensive and integrated system designed to deliver safe, treated and reliable drinking water from source to households.

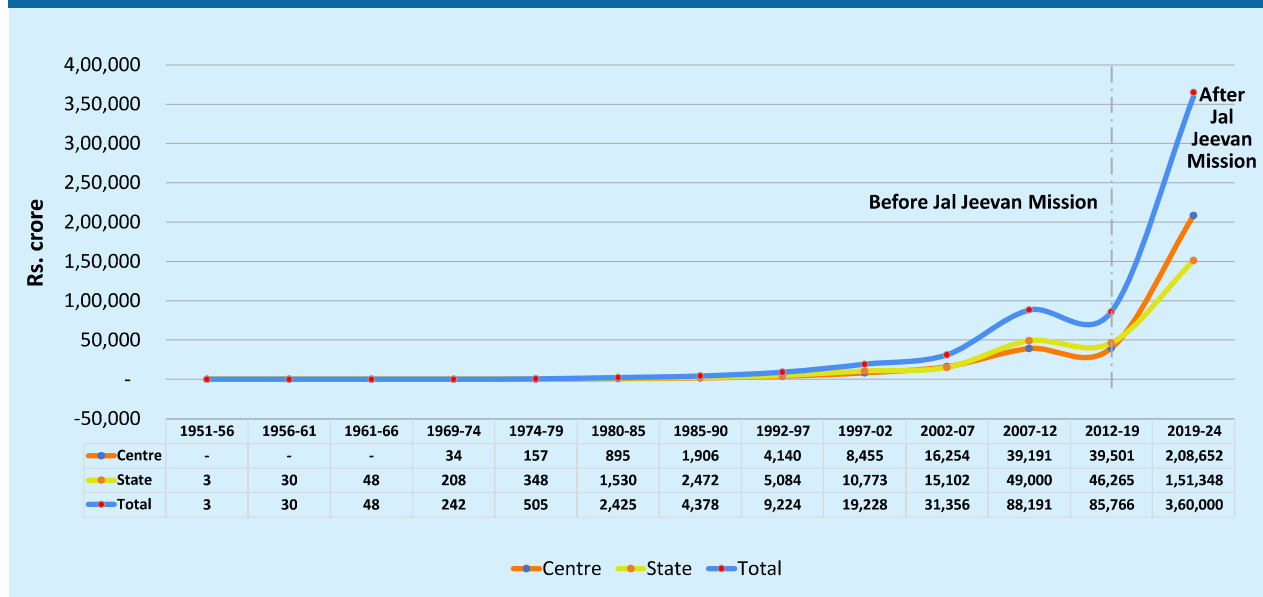
At a broad level, a typical PWS scheme comprises of a series of interconnected infrastructure components from source abstraction structures, pumping systems, transmission networks, water treatment facilities, storage reservoirs, to last-mile distribution infrastructure. These components are strategically designed to address diverse hydro-geological and topographical conditions, often involving long-distance water conveyance, multi-stage pumping and advanced treatment processes where required.

The architecture of a PWS scheme can be broadly categorised into out-village infrastructure, which includes bulk water abstraction, treatment & transmission systems and in-village infrastructure, which focuses on storage, distribution and household-level service delivery. Overall, a PWS scheme represents a complex engineering and service delivery framework, integrating physical infrastructure with operational systems to ensure sustainable and equitable access to potable water for rural households.

Implementation of the Mission has involved large-scale infrastructure development across rural India, including construction of extensive village distribution networks, water treatment plants (more than 15 folds increase in treatment capacity), water supply schemes (more than 9 folds increase in schemes under planned/ implementation).

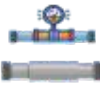
To strengthen water quality monitoring and surveillance 2,874 water testing laboratories (including 685 WTP located labs) out of which 1,709 are accredited/recognized. At the grassroots level, approximately 5.28 lakh Village Water & Sanitation Committees (VWSCs)/ Pani Samitis were constituted to institutionalize community participation in planning, implementation and Operation & Maintenance (O&M).

Figure 6: Investments made for Rural Water Supply



*Above are the estimated figures based on approved schemes subject to final assessment of the mission period of JJM2.0

Figure 7: Quantum Jump in Water Supply Assets

Assets	Unit of Measurement	Pre-JJM	Post-JJM*
 Water Quality Laboratories	Nos.	2,071 (619 accredited)	2,161 (1,709 accredited)
 Storage tanks & Reservoirs	MLD	1,200	34,000
 Pipelines for Water Distribution Network (transmission and distribution)	KM	~ 5 lakh	~ 70 lakh
 Water treatment	MLD	2,000	36,000
 PWS schemes	Nos	72,837	6,83,144
 Population covered by PWS	%	38%	100%

Post the first phase of JJM (2019-2024) implementation programme evaluation was undertaken to address implementation gaps and shift the focus from infrastructure creation to building sustainable systems for operations. With the creation of such an extensive rural water supply infrastructure, the need for a robust and sustainable operation and maintenance (O&M) ecosystem became critical to ensure long term functionality and service delivery. Recognizing this, the

National Jal Jeevan Mission (NJJM) placed focused emphasis during 2024-26 on building the institutional, governance and community ecosystem necessary for sustainable O&M through structural reforms. These reforms are aimed at strengthening service delivery systems, enhancing accountability, improving local institutional capacities and ensuring long term financial and operational sustainability of rural water supply arrangements through Jan Bhagidari.



1.4 Major Initiatives

The Department has undertaken many key initiatives to strengthen service delivery under Jal Jeevan Mission, focusing on technology integration, improved monitoring, quality assurance and institutional strengthening. These efforts aim to ensure transparency, accountability and sustainable operation and maintenance of rural water supply systems. Following are the major initiatives:

- i. **Development of Sujalam Bharat Module as Digital Public Infrastructure (DPI):** A national digital framework has been created wherein every service area of rural piped water supply schemes is assigned a unique Sujal Gaon ID, enabling complete end-to-end digital mapping of the water supply system, from source to household tap connection.
- ii. **Launch of State, District and GP level Interactive Dashboards:** Interactive dashboards have been enabled at the State, District and Gram Panchayat levels to support real time monitoring, promote active local involvement and establish Command and Control Centres for data driven oversight.
- iii. **Decision Support Systems for Source Sustainability Launched:** A Decision Support System (DSS) has been launched integrated with groundwater status data from the Central Ground Water Board (CGWB) to facilitate scientific planning and strengthening of source sustainability measures, especially for groundwater based schemes at the district level.
- iv. **Digital “Jal Seva Aankalan” was launched for Functionality Assessment:** Jal Seva Aankalan has been introduced as a digital tool to enable Gram Sabhas to record service delivery responses, supporting objective assessment of functionality in Har Ghar Jal villages.
- v. **Standardized Commissioning and Handover Protocols issued:** Protocols for commissioning and formal handover have been instituted through the Jal Arpan process, including a mandatory 15-day trial run with active participation of Gram Panchayats and Village Water & Sanitation Committees (VWSCs).
- vi. **Independent Assessment through Jal Seva Adhyayan introduced:** Assessment of JJM infrastructure and its implementation is being undertaken through “Jal Seva Adhyayan,” involving engineering students to provide independent insights and strengthen transparency and accountability.
- vii. **Monthly Joint Inspections:** Monthly joint inspections by administrative and technical officials have been institutionalised.
- viii. **Revised Terms of Reference for TPIAs issued:** Revised Terms of Reference for Third Party Inspection Agencies (TPIAs) were issued, introducing mandatory technical tests and enhanced administrative oversight by District Collectors/District Magistrates to ensure higher quality assurance in scheme execution.
- ix. **Regular Programme Evaluation through National WASH Experts and Central Prabhari Officers (CPOs):** A structured mechanism for regular programme evaluation has been instituted through National WASH Experts and the Central
- x. **Programme Evaluation through Central Nodal Officers (CNOs):** A structured programme evaluation mechanism has been instituted through reviews conducted by Central Nodal Officers (CNOs). The Department analysed CNO reports and has issued directives to States to address critical observations and implementation gaps identified during these assessments, thereby strengthening accountability and improving field-level performance.
- xi. **Monthly Review Meetings with States:** The Department holds one on one monthly meetings with State Mission Directors, to review progress and resolve implementation issues flagged by the CNOs.
- xii. **Advisories to District Collectors/ Magistrates:** Detailed advisories have been issued to all Dcs/ DMs in States/ UTs to prepare source sustainability plans, protect water supply infrastructure, monitor quality of works and ensure PM GatiShakti-based asset mapping for all rural water supply schemes.



- xiii. **Integration with e-Gram Swaraj Portal:** The JJM portal has been integrated with the e-Gram Swaraj platform and States/UTs have been directed to upload as-built drawings and adhere to standardized commissioning and handover protocols, mandating the presence of Gram Panchayats during the process.
- xiv. **Strengthening Local Institutions through VWSCs and District Technical Units (DTUs):** VWSCs are to be integrated as part of the Standing Committees of Gram Panchayats to ensure their statutory role in planning, implementation and oversight of rural water supply systems. Further, District Technical Units (DTUs) are being established to provide technical support to Gram Panchayats through DWSM, enhance engineering capacities and strengthen field-level implementation and monitoring.
- xv. **Community-Led Celebrations under 'Jan Bhagidari se Har Ghar Jal':** Standard Operating Procedures (SoPs) have been issued for community-led events to reinforce ownership and long-term service delivery outcomes. These include Jal Arpan (formal handing over of schemes), Jal Utsav (local celebrations to promote community ownership), Jal Bandhan (community commitment for asset maintenance) and Jal Sankalp (collective pledge for sustaining service levels).
- xvi. **Community Radio for Behavioural Change and Awareness:** A dedicated community radio initiative has been launched in 13 languages and 34 dialects across 25 States, 89 Districts and 100 Radio Stations (Pilot Phase) to enhance awareness, improve community participation and promote contextual communication on water service delivery and sustainability.
- xvii. **Direct Outreach with District Collectors:** The Department has conducted 29 one-on-one interactions with more than 700 District Collectors to strengthen last-mile implementation, resolve bottlenecks and ensure effective field-level monitoring of rural water supply systems.
- xviii. **Peyjal Samvad – Monthly Engagement with States/ Districts:** Regular Peyjal Samvad sessions are held every month with District Collectors and State Mission Directors aimed at strengthening district-led governance, promoting innovation, cross-sharing and peer learning among districts.
- xix. **Sarpanch Samvad through QCI Platform:** A nationwide engagement platform hosted by the Quality Council of India has been launched

Figure 8: O&M Policy Essential Components





under *Sarpanch Samvad*, enabling more than 1 lakh *Sarpanches* to participate in discussions on improving water service delivery, governance and community oversight.

- xx. **Sujal Gram Samvad – Direct Interaction with Beneficiaries:** The Department facilitates *Sujal Gram Samvad*, direct interactions with Gram Panchayat beneficiaries in their regional languages and local dialects, to encourage community ownership, gather feedback and strengthen transparency in scheme implementation.
- xxi. **Inclusion of Water, Sanitation and Hygiene (WASH) in School Curriculum:** Water, Sanitation and Hygiene components have been incorporated into the NCERT curriculum for Classes 1 to 8, fostering early awareness and behavioural change among school children as part of long-term sustainability efforts.
- xxii. **High Level Policy Dialogue on Sustainable O&M:** Policy dialogue was conducted with State Ministers and senior officials of PHE/RWS & Panchayati Raj Departments to strengthen understanding and commitment towards sustainable operation and maintenance of rural drinking water services. These engagements have reinforced the importance of institutional preparedness, long term financing and governance reforms required for reliable service delivery.
- xxiii. **O&M Policy Framework Based on 19 Essential Components:** A model framework with 19 essential components of a comprehensive O&M Policy has been shared with all States/UTs to assess their readiness for sustaining rural piped water supply systems. The draft policy framework and diagnostic matrix have been provided to guide States in aligning their systems with long-term sustainability requirements.

Agency in 2021, 2022 & 2024. These assessments examined key service level indicators, including adequacy of supply, quality compliance, regularity of service and sustainability of the source and system. The scope of the surveys has progressively expanded over the time to improve representativeness and geographic spread.

NITI Aayog also conducted an independent survey (2025) across 34 districts of 12 States/ UTs. The findings of the assessments revealed that 87% of households have adequate access to drinking water with 82% of households satisfied with tap water quality, where connections have been provided.

The outcomes of the assessments have been utilized for identification of service gaps, prioritization of remedial actions and strengthening of operation and maintenance frameworks under the Mission.

1.4.1. Functionality assessment 1.3.2. Scale of infrastructure planned

To assess compliance with the parameters, National Jal Jeevan Mission (NJJM) has undertaken functionality assessments through sample surveys by a Third-Party

Table 1: Coverage of Functionality Assessment Surveys

Unit	2020-21	2022	2024*
State covered	31	33	34
Districts covered	704	712	761
No of villages covered	6,992	13,299	22,812
No. of HHs covered	87,123	3,01,389	2,73,295

*In Har Ghar Villages only

Table 2: Results of Functionality Assessments

Parameters	2021	2022*	2024*
Working tap connections (Found working on the day of the survey)	94%	91%	87%
Adequate Quantity (i.e., 55 lpcd)	84%	88%	80%
Full Regular Supply (% HH) (as per declared supply schedule)	87%	84%	84%
Potable (Quality) (% HH)	61%	90%	76%
Overall Functionality	48%	69%	76%

*Survey outcome for Har Ghar Jal villages.





1.4.2. Potential socio-economic impacts of JJM

Beyond the expansion of household tap water coverage the mission is having a profound impact on the socio-economic development of rural communities, significantly enhancing their overall living standards. Access to safe and reliable drinking water has contributed to

measurable improvements in public health and hygiene, reduction in disease burden, enhancement of productivity and strengthening of local livelihoods. Evidence from research studies and sectoral assessments substantiates these outcomes and highlights the Mission's transformative impact on rural living standards.

Figure 9: Potential Impacts of JJM

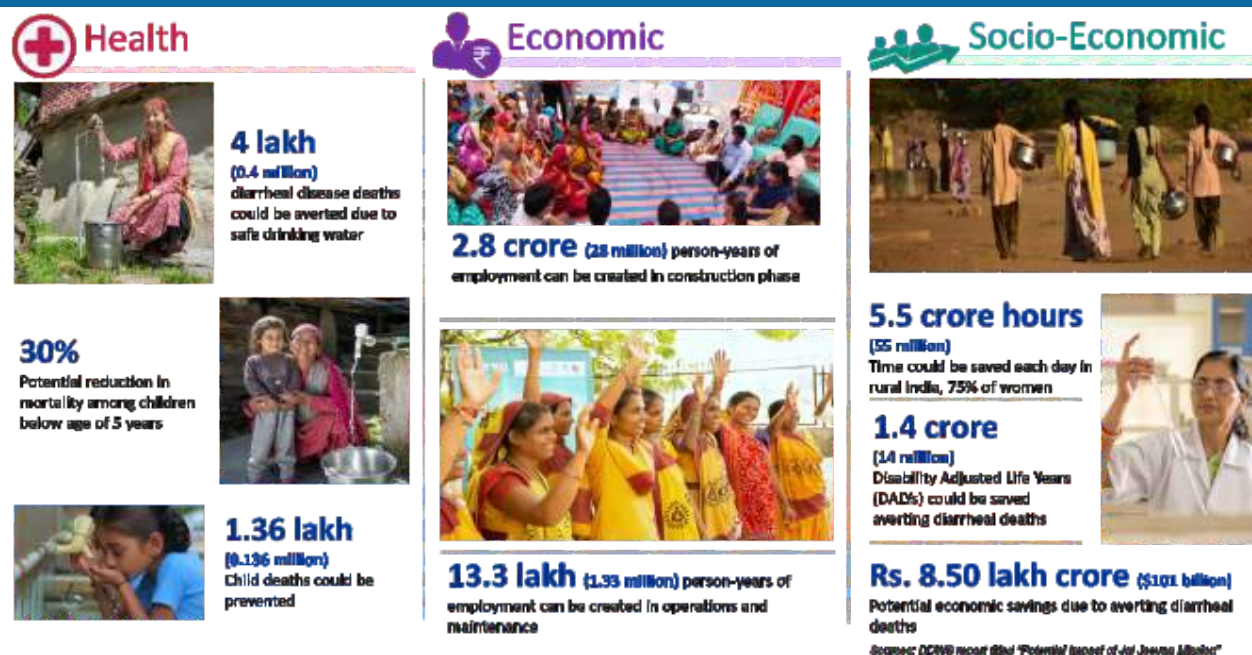


Table 3: National Initiatives and Envisaged Impacts

S.No.	National Initiatives	Impact
1	SBM G	Piped water availability enhances sanitation and wastewater management.
2	Poshan Abhiyan (National Nutrition Mission), ICDS (WCD)	Safe drinking water improves child nutrition and reduces malnutrition. Access to potable water at homes reduces the burden of pregnant and lactating mother and in Anganwadi centres impact child nutrition and hygiene practices.
3	National Health Mission (NHM)	Reduction in waterborne diseases lowers healthcare costs.
4	National Rural Livelihood Mission (NRLM) - SHGs	Engagement of SHGs in water testing, O&M etc. shall create entrepreneurship opportunities.
5	VB- G RAM G	JJM promotes water conservation, groundwater recharge, and source sustainability in convergence with VB- G RAM G.
6	National Water Mission & Atal Bhujal Yojana	Promote source sustainability and groundwater recharge.
7	Digital India	Digital monitoring of the operation and efficiency of infrastructure will enable rural India to connect with digital India.
8	Skill INDIA	Capacity building and training under Nal Jal Mitra Programme (NJMP).
9	PM Gatishakti	GIS mapping of all the pipe water supply lines and assets of piped water supply schemes.
10	Pradhan Mantri Awas Yojna (Grameen) - PMAY (G)	Providing water supply to all the households provided under PMAY (G).
11	PM JANMAN and DA JGUA	Making water accessible to most marginalized communities of the society such as PVTGs.

1.5.Vision

Every rural household has access to drinking water supply in adequate quantity of prescribed quality on regular basis through community participation (Jan Bhagidari) and bringing in structural reforms for sustainable O&M of rural water supply systems, thereby enhancing the living standards of rural communities.

1.6. Objectives

Building upon the objectives outlined in the previous Guidelines of JJM, the objectives for the extended phase of the Mission are as follows:

- i. **Universal and Inclusive Coverage:** To complete all ongoing and approved schemes to ensure tap water supply to every rural household by December 2028.
- ii. **Assured and Quality Service Delivery:** To ensure sustained supply of 55 LPCD potable water conforming to BIS 10500 standards through robust water quality monitoring and surveillance systems.
- iii. **Institutionalized Drinking Water Governance:** To establish a structured, accountable and utility-oriented governance framework at State (SWSM), District (DWSM/ DTU) and Village level (GP) and at the Community level (VWSC/

SHG/local youth).

- iv. **Community Ownership and Jan Bhagidari:** To promote “Har Ghar Jal” certification and participatory management of schemes through VWSCs, SHGs and community institutions for enhanced ownership and sustainability.
- v. **Sustainable Operation and Maintenance:** To ensure effective handover of in-village infrastructure to local bodies and institutionalize long-term O&M systems, including capacity building of local stakeholders and engagement of trained local youth.
- vi. **Technology-Driven Monitoring and Data-Based Decision Making:** To establish real-time monitoring systems, command-and-control mechanisms and leverage data analytics for improved service oversight and infrastructure performance.
- vii. **Service Utility Framework:** To transition from a departmental implementation model to a service delivery-oriented, utility-based approach through institutional reforms, performance-linked accountability and formalized Centre–State partnerships.



SECTION 2

Roadmap
to
Structural Reforms

CHAPTER 2.1

Establishing Institutional Architecture for Drinking Water Governance

Reform aims at:

- ◆ Establishing clear delineation of roles and responsibilities across governance, regulation and service delivery, replacing overlapping mandates.
- ◆ Transitioning from an infrastructure-centric approach to outcome-based service delivery
- ◆ Establishing greater accountability to citizens
- ◆ Strengthening regulatory oversight through a comprehensive legal framework
- ◆ Moving towards integrated, lifecycle-based water service systems in place of scheme-centric implementation.
- ◆ Enabling digital, data-driven governance and real-time monitoring, replacing fragmented and siloed information systems.

2.1.1. Drinking Water Governance

With the ongoing creation of massive rural water supply infrastructure, there is a need to bring in institutional reforms and establish administrative, regulatory and legal frameworks to ensure safe and reliable drinking water services in a citizen-centric manner.

Water is a State Subject under Entry 17 of the State List-II of Seventh Schedule to the Constitution of India placing responsibility for governance, management and distribution of rural drinking water on State Governments. The 73rd Constitutional Amendment Act (1992) devolved drinking water to Panchayati Raj Institutions (PRIs), empowering Gram Panchayats (GPs)

to plan, implement and manage local drinking water systems. This devolution strengthened community ownership, enabling GPs to undertake source conservation, infrastructure operation and distribution along with community involvement.

Jal Jeevan Mission (JJM) marked a sectoral shift toward a community-centric, service-oriented model of rural drinking water governance, mandating institutions such as VWSCs at the village level and DWSMs at the district level. However, gaps persist in areas such as source sustainability, O&M, water quality, financial sustainability and community participation. States/UTs are therefore required to reinforce institutional capacity, technical support systems and financial frameworks to enable GPs and VWSCs to manage in-village water supply systems effectively and deliver reliable household water services.

2.1.2. Institutional Mechanism

A. National Level

- i. The National Jal Jeevan Mission (NJJM), headed by Mission Director, will be responsible for steering and guiding the Mission at the national level. It will provide overall policy direction, strategic oversight and technical support to ensure effective implementation across States/UTs. NJJM will coordinate closely with States/ UTs for planning, implementation and monitoring, while promoting standardisation, capacity building and knowledge sharing.
- ii. The Mission will also operationalise performance-linked release of Central Assistance, linking funding to achievement of service delivery outcomes and progress on key reforms, thereby incentivising efficiency, accountability and sustainability. Additionally,



NJJM will support the development of regulatory frameworks, digital monitoring systems and performance assessment mechanisms to strengthen governance and ensure improved service delivery outcomes.

- iii. A National Command and Support Centre will be setup for monitoring progress, tracking reform milestones and processing State-level information for policy decisions. It will be integrated with Command & Support Centres at State & District levels.
- iv. National Project Management Unit (NPMU) with engineering, management and IT experts, will assist NJJM and the Mission Director. The NPMU will support programme management, technical support, programme evaluation and development of IT monitoring architecture.
- v. SPM-NIWAS will be responsible for building capacity, facilitating research and development related to low cost and efficient WASH technologies, enabling different Institutions to enhance their technical, financial and managerial skill in partnership with States/

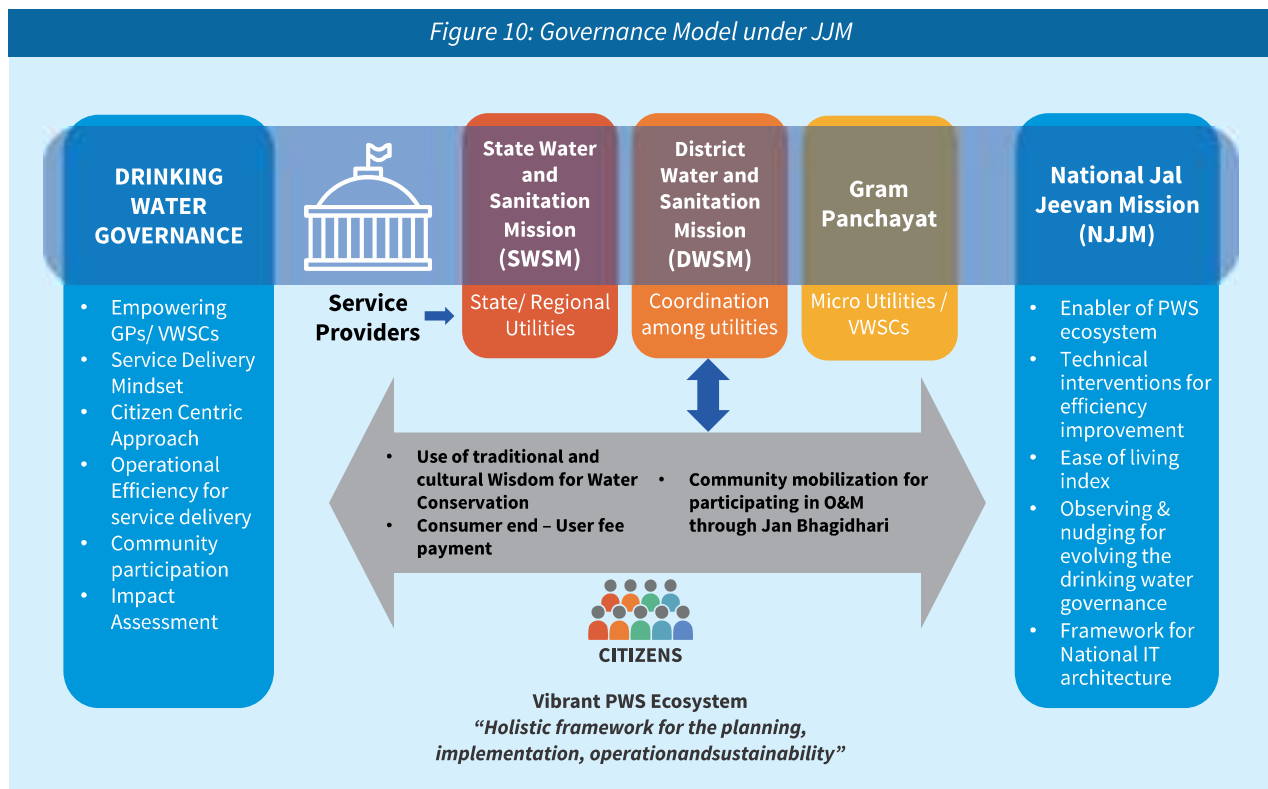
UTs, academia, NGOs and other partner agencies.

B. State Level (SWSM, Apex and Executive Committees)

The State Water and Sanitation Mission (SWSM) comprise of Apex Committee and Executive Committee:

- i. The Apex Committee, headed by the Chief Secretary and will comprise of Secretaries of PHED/RWS, Rural Development, Panchayati Raj, Health, Finance, Planning, Education and other related departments; Representative of the Government of India and State Mission Director as Member Secretary
 - a. It will guide policy, approve key decisions and support the functioning of water supply utilities under relevant State Acts and policies. It will also ensure coordination between departments. The Committee will issue directions on water security, regulatory measures, utility reforms and alignment with State priorities.

Figure 10: Governance Model under JJM





- b. The SLSSC shall be merged into Apex committee.
 - c. The committee shall meet on quarterly basis.
 - d. It will also review the progress of JJM implementation with DC/DMs periodically.
- ii. The Executive Committee will be headed by the State Mission Director who will be responsible for overall planning, implementation and management of the Mission at the State level. The State Mission Director shall be equivalent to Joint Secretary to the Government of India or above with at least 2 years of experience as DC/DM, ensuring adequate authority for inter-departmental convergence and effective decision-making. The composition of the SWSM executive committee may be determined by the States/UTs in line with the convergence requirements outlined in the guidelines, to enable effective coordination across relevant departments with mandatory participation from rural development and panchayati raj departments.
- a. The Executive Committee will ensure alignment with national guidelines, oversee the functioning of DWSMs and support planning, implementation and monitoring of schemes through effective operationalisation of District Technical Units. It will also facilitate engagement with district and Gram Panchayat institutions for timely execution of schemes and adherence to service delivery standards
 - b. Coordination with NJJM will be institutionalised through regular interactions between the Mission Director, NJJM and State Mission Directors. Monthly review meetings, joint monitoring and digital platforms for real-time data sharing will support aligned planning, reform implementation and data-driven governance
 - c. The Executive Committee will be supported by State Programme Management Unit (SPMU) to provide technical, managerial and administrative support for planning, implementation, monitoring and evaluation

tion of the Mission. The SPMU will also support data-driven decision-making, inter-departmental coordination and adherence to service delivery standards

C. District Level (DWSM with DTU & DCSC)

- i. The District Water and Sanitation Mission (DWSM), chaired by the District Collector/ Deputy Commissioner, will be responsible for overall implementation, supervision and monitoring of the Mission at the district level. It will ensure convergence across departments and effective coordination with Gram Panchayat institutions for timely and quality execution of schemes
- ii. The DWSM shall comprise key district-level officials, including the Chief Executive Officer (Zila Parishad)/ District Development Officer/Chief Development Officer, Divisional Forest Officer, Project Director (in ITDA/ ITDP districts), District Medical Officer, District Education Officer, Executive Engineers from Water Resources/Ground Water/Irrigation, District Agriculture Officer, District Information and Public Relations Officer, representatives of relevant line departments, nominated technical experts and the Executive Engineer of PHED/RWS Department as Member Secretary.
- iii. The DWSM shall conduct monthly review meetings with all stakeholders and upload the Minutes of Meeting (MoM) on the DWSM Dashboard through the JJM-IMIS portal
- iv. The DWSM will operationalise the District Command and Support Centre (DCSC) as the central hub for planning, monitoring and oversight of Mission activities, supporting Jal Seva Aankalan, third-party inspections, grievance escalation and real-time data-driven decision-making.
- v. Through the District Technical Unit (DTU) and District Programme Management Unit (DPMU), the DWSM will provide technical and managerial support for O&M, asset management, source sustainability, disaster preparedness and capacity building.

D. Gram Panchayat & VWSC



The Gram Panchayat (GP), as an institution of local self-government under the 73rd Constitutional Amendment, shall be responsible for ensuring delivery of safe and reliable drinking water services to all households within its jurisdiction. The GP, itself or through its standing committee i.e. Village Water and Sanitation Committee (VWSC) shall own, manage and be accountable for all in-village water supply infrastructure and lead the planning, implementation, operation and maintenance of such systems in accordance with defined service delivery standards for quality, quantity and reliability.

i. **Village Water and Sanitation Committee (VWSC)**

A standing committee for water and sanitation, constituted under the respective State Panchayati Raj Acts, shall function as the executive arm of the Gram Panchayat for water and sanitation services. The committee, which may be designated as the Village Water and Sanitation Committee (VWSC), shall be headed by an elected member of the Panchayat and remain accountable to the Gram Panchayat. The appropriate provisions shall be made under Panchayat Acts or rules to grant statutory status to such standing committees.

ii. **Composition and Financial Provisions**

The composition of the standing committee shall ensure inclusivity and community representation. It may consist of 10-15 members comprising elected members of Panchayat up to 25% of the composition; at least 50% women members; remaining 25% may consist of representatives of weaker sections of the village (SC/ST) proportional to their population; and participation from Anganwadi Workers (AWW), ANM/ASHA, Self-Help Groups (SHGs), Community-Based Organisations (CBOs) and other relevant stakeholders, as appropriate

In cases where a Gram Panchayat comprises more than one village, the GP may constitute more than one such standing committee to ensure effective service delivery. States/UTs may authorise separate bank accounts for these committees in accordance with applicable financial rules, enabling them to collect user/service charges and receive grants from

State Governments and Finance Commissions for WASH activities

iii. **Roles and Responsibilities**

The standing committee shall support the Gram Panchayat in preparing budgets, managing finances and strengthening service delivery systems. It shall be responsible for day-to-day service delivery functions, including; Collection of user/service charges and maintenance of O&M accounts; Management and upkeep of village-level water supply assets; Leak detection and preventive maintenance; Water quality monitoring and surveillance through Field Test Kits (FTKs) and laboratory testing; Implementation of source sustainability measures; Grievance redressal and citizen engagement; Conducting IEC/BCC activities; Facilitating community participation in operations and maintenance.

iv. **Service Delivery and Outsourcing**

The standing committee may undertake service delivery directly or engage external agencies for operations and maintenance. In such cases, a Service Level Agreement (SLA) shall be executed between the Gram Panchayat/ standing committee and the selected operator, clearly defining service standards, roles, responsibilities and performance conditions. A tripartite agreement among the Gram Panchayat/standing committee, State Government and the O&M agency may also be executed, as required.

v. **Monitoring, Reporting and Coordination**

The Gram Panchayat, supported by the standing committee, shall maintain asset and service records, support periodic Jal Seva Aankalan and ensure accurate reporting of service delivery and system functionality. It shall coordinate with the District Water and Sanitation Mission (DWSM) and District Technical Unit (DTU) for technical support, capacity building and compliance with service delivery standards.

The Gram Panchayat shall also promote water conservation, undertake source protection measures and ensure safeguarding of drinking water infrastructure within its jurisdiction.

2.1.3. Regulatory framework of rural drinking water

JJM 2.0 recognises household piped drinking water as the core element of “water for life”. States/ UTs shall ensure service delivery standards covering adequacy, regularity and BIS-10500 prescribed water quality. To support this, the following key regulatory measures shall be prioritised:

A. Source Protection Framework

States/ UTs shall establish a statutory framework for drinking water source protection. This shall include:

- i. Notification of Source Protection Zones around key drinking water sources such as borewells, springs and surface water intakes;
- ii. Regulation of land use, extraction and potentially contaminating activities within these zones;
- iii. Mandatory safeguards to prevent pollution and over-exploitation of sources;
- iv. Empowerment of district authorities for enforcement and compliance.

B. Domestic Water Allocation Priority

- i. States/UTs shall adopt a domestic water allocation priority principle, ensuring that drinking water supply to households is accorded the highest priority in allocation of both surface water and groundwater resources.
- ii. In surface water-based schemes, especially those dependent on reservoirs and bulk transfer systems, dedicated allocations for rural drinking water supply shall be ensured and protected during periods of scarcity, in line with public health priorities and principles of equity.

C. Protection of PWS Infrastructure

States/UTs shall establish regulatory provisions for protection of Piped Water Supply (PWS) infrastructure from damage, encroachment and unauthorised use. This shall include:

- i. Notification of protective zones around

critical infrastructure such as intake structures, treatment plants, transmission lines and storage reservoirs;

- ii. Provisions to prevent physical damage, tampering, or illegal connections;
- iii. Enforcement mechanisms at district and local levels to safeguard assets;
- iv. Clear accountability for maintenance and protection of infrastructure.

D. Sustainability and Source Security

Rural water supply planning shall be based on aquifer and watershed principles, ensuring long-term sustainability of sources. All piped water supply schemes shall incorporate backup or alternate sources to maintain continuity of service during source failure or seasonal variability.

E. Climate Resilience and Continuity Planning

States/UTs shall integrate climate-resilient measures, including source diversification, aquifer recharge and local storage systems. Each scheme shall have an emergency continuity plan to ensure uninterrupted water supply during droughts, floods and other disruptions.

F. Operations and Maintenance (O&M) Policy

States/ UTs shall notify a comprehensive Operations and Maintenance (O&M) policy for rural drinking water supply systems, clearly defining roles and responsibilities across all tiers-State, district and Gram Panchayat/ standing committees. The policy shall establish accountability for service delivery, outline standard operating procedures for system upkeep and ensure sustainable financing mechanisms, including user charges and targeted support, to maintain functionality, reliability and long-term sustainability of water supply systems.

G. Public Service Guarantee and Grievance Redressal

- I. States/UTs shall bring drinking water services under a Public Service Guarantee framework, prescribing defined timelines for provision of new household tap connec-



tions, extension of infrastructure and restoration of services. These service standards shall be notified and enforceable to ensure time-bound and accountable delivery.

- ii. A robust digital grievance redressal mechanism shall be established with defined

timelines for resolution, supported by digital platforms for complaint registration, tracking and escalation. Provision for escalation at district and State levels shall be ensured to address unresolved grievances, thereby strengthening accountability, transparency and citizen trust.

CHAPTER 2.2

Utility based approach for drinking water service delivery: Dual Utility Framework

Reform aims at:

- ◆ Transition towards a utility-based approach for sustainable and service-oriented drinking water delivery
- ◆ Institutionalise and operationalise Micro Utilities for in-village drinking water supply operations and maintenance
- ◆ Formation and operationalisation of State/regional-level macro utility constituted for out-village drinking water supply operations and maintenance
- ◆ Performance Benchmarking and periodic assessment of utilities for service level improvement

2.2.1. Background

India's rural water supply sector is now moving towards a more organised and professional "utility-based" model. This means that rural drinking water services should operate in a reliable, efficient and predictable manner—similar to how electricity or urban water utilities operate. As Piped Water Supply (PWS) systems expand across villages, there is a growing need to strengthen institutions, improve operational practices and develop skilled human resources for effective management of water supply systems. This strengthening includes improved planning processes, clearly defined procedures for operation and maintenance, capacity building and use of robust digital platforms for asset management, service monitoring, billing and performance tracking. States

and UTs will play a key role in supporting this transition by providing technical assistance, financial stability during the shift and guidance for sustainable water management including source strengthening and climate-resilient practices.

Over the coming years, rural water service providers are expected to gradually improve their systems and move towards stable, professional service delivery. This improvement includes setting clear responsibilities for operation and maintenance, adopting metering to encourage efficient use, reducing losses, improving energy and water efficiency and ensuring transparent service standards through citizen charters. Strengthening financial management and ensuring reasonable, affordable user charges will also help maintain the systems in the long run. As these improvements accumulate, the rural water sector will evolve into a sustainable service delivery ecosystem capable of meeting defined standards and providing safe drinking water throughout the system's lifecycle.

2.2.2. Utility¹ based approach for drinking water service delivery: Dual Utility Framework

JJM 2.0 envisages to bring in structural reforms in the rural water supply sector through development of a dual utility framework comprising (a) micro utilities at the village level and (b) macro utilities at the State/regional level.

- i. **Micro Utility:** The in-village water supply infrastructure shall be owned and managed by the GP, which shall designate its standing committee, i.e. the VWSC, to function as the micro service utility responsible for

¹ "...public utility service means... any industry which supplies power, light or water to the public; ...any system of public conservancy or sanitation..." Industrial Disputes Act, 1947; Furthermore, "...Utility means any person or entity engaged in the generation, transmission, sale, distribution or supply, as the case may be..." Electricity Act 2003; Again, "Public utility service includes transport, postal/ telegraph/ telephone service, supply of power, light or water to the public by any establishment..." Legal Services Authorities Act, 1987- Public Utility Service (Section 22A);



distribution, O&M and service delivery within the village. VWSCs may function as service providing utilities or may outsource the function with specific pre-conditions. Such pre-conditions for contracting out the functions of utilities must have service level agreement (SLA) between GP/ VWSC and the vendor. In any case, VWSCs shall be fully aware of the drinking water supply service in the GP/ VWSC managed area and the VWSC will facilitate Jal Seva Aankalan at regular intervals.

- ii. **Macro utility:** The out-village water supply infrastructure and bulk water transfers shall be managed by the State concerned departments and will operate as a professional body responsible for bulk water transfer, system optimisation and maintaining service standards across multiple service areas. Macro utilities shall be created only in States and UTs where bulk water transfer systems are available.

2.2.3. Village-level Micro Utilities

Functions of micro utilities

- i. The Micro Utility shall function as the primary village-level/ GP-level service provider, responsible for continuous, safe and equitable drinking water supply to all households and institutions within its jurisdiction.
- ii. Undertake daily O&M activities of the in-village water supply scheme, including pumping, local disinfection/ treatment, distribution, valve operations and network maintenance, as per approved Standard Operating Procedures, released from time to time by respective State/ UT.
- iii. Conduct routine pre-defined inspections for identification of leakages, illegal connections and network losses. Ensure timely rectification to maintain system integrity and service continuity.
- iv. Prepare under the guidance of DTU and implement a preventive maintenance calendar for pumps, motors, valves, reservoirs, disinfection/ treatment stocks/ units, meters and pipelines etc. to maximise asset life and reduce breakdown incidence.

- v. Carry out regular water testing using Field Test Kits (FTKs) and coordinate with NABL-accredited labs for periodic sampling. Furthermore, ensure compliance with prescribed drinking water quality parameters and initiate corrective measures where needed with the help of DWSM.
- vi. Facilitate community-level greywater collection, treatment and reuse systems and ensure proper O&M of soak pits, drainage structures and nature-based solutions for environmental safeguarding.
- vii. Maintain close coordination with designated Macro Utility, PHED/ Jal Nigam, DWSM and DTU for technical support, major repairs, capacity building and adherence to service quality norms.
- viii. Prepare the annual water service budget covering O&M costs, minor repairs, energy charges, water quality testing, community engagement and administrative expenses to ensure viable financial management.
- ix. Implement transparent and equitable user charge mechanisms approved by Gram Sabha. Ensure regular billing, revenue collection and accounting to maintain financial sustainability of village water services.
- x. Maintain cashbooks, vouchers, ledgers and audit-ready accounts. Submit financial utilisation reports to the Gram Panchayat and DWSM/ State authorities as per State/ UT guidelines.
- xi. Engage skilled human resources such as pump operators, plumbers, water quality testers and billing/ collection assistants as required. It shall ensure their training, certification and performance monitoring. A trained and certified operator, designated as “Jal Vitaran Sanchalak / Nal Jal Mitra” and aligned to NQSF Level 4 competency standards, is engaged for pump operation and routine maintenance. It shall establish a linkage between service delivery performance and financial discipline within the village water utility framework.
- xii. Establish protocols for handling water supply



disruptions, motor failures, contamination events and disaster impacts, ensuring timely response and community communication.

- xiii. Operate a village-level grievance register and provide time-bound resolution of consumer complaints related to water quality, supply timing, billing, leakages, or connection issues. Micro Utilities shall be the first point of contact for the beneficiaries for any grievance and must ensure local resolution whenever possible before any escalation further.
- xiv. Display water supply schedules, Service delivery charge information, contact details of O&M personnel, water quality results and financial statements on village noticeboards and digital platforms where available and before Gram Sabha periodically.
- xv. Conduct awareness campaigns on water conservation, household plumbing maintenance, source protection, metering and rational water use.
- xvi. Facilitate Gram Sabha deliberations on service delivery charge, O&M plans, water safety plans and performance reporting, ensuring participatory governance.

2.2.4. State/ Regional-level Macro Utilities

Functional Responsibilities of Macro Utilities

The designated Macro Utilities shall be responsible for the planning, development, management and supervision of regional, multi-village and bulk water supply assets. Their key functions will include:

- i. Identification, development, augmentation and long-term sustainability management of surface and groundwater sources used for regional supply systems.
- ii. O&M of water treatment plants, ensuring compliance with national drinking water quality standards. This includes monitoring raw water quality, process optimisation and ensuring safe bulk water delivery to village boundaries.
- iii. Operation, maintenance and monitoring of

rising mains, trunk pipelines, pressure management systems, bulk flow meters and master reservoirs to ensure uninterrupted and efficient conveyance of treated water to downstream village networks.

- iv. Execution of all major repairs, renewals and rehabilitation works associated with regional/ bulk/ macro water infrastructure, ensuring asset reliability and minimising service disruptions.
- v. Preparation of technical designs, DPRs, engineering standards and technical guidelines for regional water supply systems to provide technical support to DWSM/ DTUs, Micro Utilities and GPs/ PRIs for downstream service delivery.
- vi. Operational responsibility for MVS/ BWS systems upto GP/ village entry points. This includes but not limited to system balancing, pressure zone management and coordinated supply planning across multiple GPs.
- vii. Coordination with DWSMs, GPs/ PRIs, Micro Utilities alongside State/ UT departments/ agencies to ensure seamless bulk-to-distribution service interface and timely grievance resolution involving bulk supply disruptions.
- viii. Implementation of service standards, regulatory directives from NJJM/ State/ UT and performance benchmarks related to bulk water delivery, energy efficiency, minimization of water losses and operational reliability. The Macro Utility shall maintain transparent reporting and disclosure.
- ix. Prepare the annual bulk water supply service budget covering O&M costs, repairs, raw water and energy charges, treatment costs and organisational expenses to ensure viable utility-level financial management. The Macro Utilities must assess the viability gaps, if applicable, through convergence of all sources including but not limited to bulk water charges, government grants etc. Furthermore, technical and financial performances of Macro Utilities shall be reviewed by the respective SWSM/ Executive Committee of the SWSM every year.



2.2.5. Coordination between utilities

Under JJM 2.0, States and UTs shall establish an coordination mechanism among the utilities to ensure smooth planning and management of rural water supply services. Micro utilities (GP/VWSC-level utilities) and macro utilities (regional/bulk supply utilities) shall work in a coordinated manner to ensure efficient and reliable service delivery across villages. If any dispute arises between a micro utility and a macro utility, the matter shall be referred to the District Water and Sanitation Mission (DWSM) or the District Technical Unit (DTU).

The State/UT shall oversee the service delivery charge/bulk water transfer charge framework, particularly the tariff for bulk water transfer between macro and micro utilities. The DTU will support micro utilities by providing technical assistance, monitoring service delivery and strengthening capacities. The DWSM will guide district-level planning, ensure convergence of programmes, monitor the performance of micro utilities and handle grievance redressal to ensure smooth and uninterrupted service delivery. At the State level, the State Water and Sanitation Mission (SWSM) will provide policy direction, approve plans and monitor utility performance against set benchmarks, including O&M standards.

Benchmarking, Efficiency and 24x7 Service by 2047

Benchmarking helps rural water utilities measure how well they are performing by comparing their service levels, efficiency and financial health against defined standards. It uses clear indicators such as water supply regularity, non-revenue water, energy use, cost

recovery and customer service to assess performance. By regularly tracking these indicators, utilities can identify gaps, set improvement targets and adopt better practices. Benchmarking encourages transparency, motivates continuous improvement and helps move utilities toward reliable, efficient and accountable rural water service delivery.

For example, Denmark's long-term benchmarking system demonstrates that continuous measurement, universal metering, leak detection and public reporting can significantly improve performance. Danish utilities are able to maintain water loss at around 7%, use remote-read meters extensively and provide nearly 99.99% water availability with very limited annual service interruptions. These examples provide useful insights for India's rural water utilities as they grow and improve over time.

The following indicative SMART (*Specific, Measurable, Attainable, Realistic and Time bound*) Benchmarking points may be used by State/ Districts to review the performance of utilities. An indicative benchmarking matrix is as follows:

Furthermore, States/ Districts missions may use these benchmarks for macro and micro utilities respectively:

- Annual utility performance reviews.
- Provide improvement targets for utilities with technical assistance. Link improvements to incentives and capacity-building support.
- Publish utility performance ratings to promote transparency.

Table 4: SMART Benchmarking Indicators of Utilities

S.No.	Indicative Benchmarking Indicators	Basic Level	Improvement Level	Advanced Level	International Reference (Denmark)
1	Percentage of NRW or Unaccounted for Water (UFW)	>25%	15 to 25%	<15%	Around 7% water loss
2	Metering Coverage (%)	<20%	20 to 50%	>50%	66% remote read meters
3	Service Disruption (days of interruption/ month)	>3days/ month	1-3 days	<1 day	Around 46 min/ year interruption
4	Water Quality Compliance (%)*	<97%	97 to 99%	>99%	99.4% samples compliant



Table 4: SMART Benchmarking Indicators of Utilities

S.No.	Indicative Benchmarking Indicators	Basic Level	Improvement Level	Advanced Level	International Reference (Denmark)
5	Energy Use (kWh/ m ³)	>0.75	0.55–0.75	<0.55	Around 0.44 kWh/m ³ average
6	Number of Leakages in service	>1 per month	Once in three month	Once in 6 months	-
7	Financial Efficiency (Cost recovery)	<30%	30-50%	>50%	Managed by utilities
8	Grievance Redressal (Turn-around time)	>3 day	<3 days	<1 day	-

*Free Residual Chlorine 100%



CHAPTER 2.3

Establishing Technical Support Framework for Gram Panchayats

Reform aims at:

- ◆ Improved technical quality and standardisation of water supply schemes through strengthened adherence to CPHEEO norms.
- ◆ Rigorous and structured commissioning protocols ensuring system readiness.
- ◆ Enhanced sustainability of assets driven by systematic handing-over processes and strengthened community ownership.
- ◆ Strengthened technical capacity at District level through institutionalisation of District Technical Units as continuous support systems.
- ◆ Improved operational efficiency and service reliability through better design validation, monitoring and lifecycle-based implementation.

2.3.1. District Technical Unit (DTU) – Institutional Framework

States/UTs shall undertake constitution of DTU¹ – Institutional Framework.

2.3.2. Background and Rationale

To ensure long-term functionality, sustainability and reliable service delivery of the created rural water supply infrastructure, JJM's extended phase necessitates a shift from a construction-led, department-driven model to a utility-based, service-oriented governance framework. Accordingly, a DTU shall be established in every district as the technical backbone of the DWSM.

2.3.3. Composition and Role

The DTU shall comprise PHED officials at the district level, water quality officials, RD&PR representatives, Central/ State GWB officials, Irrigation/ Water resources

department officials and technical consultants, as required. The DTU shall function as the extended technical arm of the PHED for rural water operations and shall report to DWSM on all technical and operational matters. The DTU shall coordinate between micro utilities (GP/ VWSC-managed in-village infrastructure) and macro utilities (PHED/State Department-managed out-village and bulk water infrastructure), in accordance with the dual-utility system adopted under JJM.

2.3.4. Key Functions

The DTU shall be responsible for inter-alia the following functions:

- Scheme gap analysis, monitoring, commissioning and handover support.
- Technical review, performance assessment and functionality reporting support for functionality assessments (Jal Seva Aankalan) and SIPs preparation to DWSM.
- Water Quality Monitoring and Surveillance: The DTU should ensure the water quality monitoring and surveillance as per “Concise Hand Book for Monitoring Water Quality of Piped Drinking Water Supply to Rural Household”.
- Digital-first governance – all records on Sujalam Bharat Database, Sujal Gaon ID, GIS-linked digital asset registries, Meri Panchayat, JJM-IMIS, District/ GP dashboard etc. The DTU will consolidate all rural water supply schemes – including JJM, State, NRDWP and other schemes – into the Sujalam Bharat Database, creating a single unified digital registry for the district anchored to unique Sujal Gaon Ids.
- Preventive & Predictive Maintenance: DTUs will shift from breakdown-based repair to

¹ ‘District Technical Unit (DTU) – Institutional Framework’ as detailed in the letter Ref. DDWS letter No. WQ-11031/14/2025-WQ-DDWS-08 dated 12.01.2026



planned maintenance using Sujalam Bharat asset data. AI-based analytics will help predict failures, plan timely replacements and manage spare parts efficiently, reducing downtime.

- vi. Digital Ticketing for Technical Support: A digital helpdesk system will enable Gps/ VWSCs to raise and track technical issues quickly. Predictive tools will support faster diagnosis, timely resolution and better planning of repairs and asset replacement.
- vii. Source sustainability planning using Decision Support System (DSS).
- viii. Skilling, training, and capacity building through NSQF Level-4 training for Nal Jal Mitra programme, and NSQF Level-1 training for Sujalam Shakti group through Govt ITIs/ Polytechnics/ Govt. institutions.
- ix. The DTU will play a key facilitative role in convergence of available funding streams — including JJM allocations, Finance Commission grants to GPs and other State or scheme-specific funds
- x. Contract management for O&M – Service-level based, performance-linked: The DTU will guide districts and GPs in preparing O&M contracts - whether executed directly by GPs/ VWSCs or through engaged technical agencies - covering scope of work, staffing norms, spares and inventory requirements and safety provisions.
- xi. Standardisation of O&M practices, SOPs and district-specific O&M manuals: To ensure consistency and quality in operations across the district, the DTU will identify common and recurring O&M issues and issue SOPs for troubleshooting, preventive maintenance and emergency response. A District-specific O&M Manual, tailored to local technical and financial conditions, will also be prepared to serve as an operational reference for field functionaries and GP-level operators.
- xii. Operation of District Command & Support Centres (DCSC) for performance monitoring: DTU will establish and operate DCSC that serve as the nerve centre for real-time monitoring of

rural water supply scheme performance across the district.

2.3.5. Commissioning and Handing Over Protocol

States/ UTs shall undertake commissioning of water supply schemes, including comprehensive system testing and a successful trial run, for a minimum period of 15 days, in accordance with 'Guidelines for Commissioning and Handing Over Protocol & Community Ownership'². No scheme shall be declared complete, or funds released for its completion, without a duly certified Commissioning Report. The Commissioning Protocol follows a four-phase commissioning framework including Pre-Commissioning Documentation, System Testing, Trial Runs and Documentation and Compliance. Important activities includes hydrostatic pressure testing, flushing and disinfection, flow capacity testing, stabilisation of residual chlorine level, equipment, monitoring devices and electronic components, trial operations, engagement with the community, capacity building, skilling, transect walk and formal processing for system handover.

The objectives of Handing over protocol are to ensure a smooth, transparent and sustainable transfer of RPWSS from implementing agencies to GP, VWSCs, or Micro-Utilities. The outcomes are expected to include improved local-level governance, enhanced capacity for O&M and strengthened accountability mechanisms. Communities will be empowered to manage resources efficiently, resulting in a consistent and reliable water supply service. This process will also encourage a culture of preventive maintenance, timely reporting of faults and active monitoring by stakeholders.

Key action points:

- ◆ Include block-level technical officers within the DTU framework.
- ◆ States/UTs shall notify key provisions of commissioning and handover protocols.
- ◆ States/UTs shall establish asset management and inventory systems for preventive maintenance
- ◆ DTU shall train field functionaries and conduct periodic refresher programs for stakeholders.

² 'Guidelines for Commissioning and Handing Over Protocol & Community Ownership' circulated vide DDWS letter No. WQ-11031/14/2025-WQ-DDWS dated 19.11.2025



CHAPTER 2.4

Establishing Citizen-Centric Water Quality Governance

Reforms Points :

- ◆ Shift from reactive to preventive water quality management, with early detection and timely response to contamination risks.
- ◆ Empowered community-led monitoring, strengthening local ownership and ensuring last-mile vigilance through trained Sujalam Shakti Groups.
- ◆ Real-time, data-driven decision-making, enabled by digital platforms, geo-tagged reporting and integrated dashboards.
- ◆ Strengthened laboratory ecosystem and technical response systems, ensuring accuracy, standardisation and credibility of water quality data.
- ◆ Enhanced transparency and citizen trust, through public dashboards, grievance responsiveness and accessible water quality information.

Strengthening Water Quality Monitoring & Surveillance (WQM&S) demands targeted investments to upgrade infrastructure in alignment with BIS 10500 standards. While substantial progress has been made in expanding rural piped water supply systems, the long-term sustainability and public confidence in these systems depend heavily on the reliability, transparency and responsiveness of water quality monitoring mechanisms. By integrating laboratory-based testing with community-led surveillance through Field Test Kits (FTKs), the program reinforces shared responsibility and “Jan Bhagidari.”

Under JJM 2.0, it is aimed that States/UTs shall progressively transition from periodic testing frameworks to a continuous monitoring and response-based water quality governance system.

2.4.1. Principles of Water Quality Monitoring and Surveillance (WQM&S)

- Citizen-centric monitoring:** Active participation of village-level community groups in water quality testing and surveillance.
- Technology-enabled transparency:** Real-time reporting through digital platforms integrated with Sujalam Bharat and Sujal Gaon Ids.
- Institutional accountability:** Clearly defined monitoring responsibilities across Gram Panchayat, District and State levels.
- Scientific validation:** Strengthened laboratory network for confirmatory testing and technical advisory.
- Rapid response and remedial action:** Timely identification and resolution of water quality issues.

2.4.2. Key Features of the WQM&S Framework

- A. Community-Based Water Quality Surveillance:** Village-level monitoring shall be undertaken through trained women groups designated as Sujalam Shakti, preferably linked with Self Help Groups under State Rural Livelihood Missions. Their responsibilities shall include:
- Periodic water quality testing using Field Test Kits (FTKs) conforming to the 'Concise Handbook for Monitoring Water Quality of Piped Drinking Water Supply to Rural Households' released in December 2024.
 - Monitoring of free residual chlorine and disinfection practices.
 - Reporting of anomalies or contamination risks by conducting sanitary surveys.



- iv. Supporting community awareness and source protection activities.

B. Digital Monitoring and Data Integration

All water quality observations recorded by Sujalam Shakti groups shall be entered through the Sujal Gaon Utility App, which shall serve as the transaction layer linked to the Sujalam Bharat Digital Public Infrastructure. Key features include:

- i. Geo-tagged WQ testing records linked to Sujal Gaon ID.
- ii. Real-time dashboards for DTU, DWSM and SWSM.
- iii. Automated alerts for water quality deviations.
- iv. Integration with existing WQM&S/ WQMIS and laboratory reporting systems.
- v. States may further customise cluster-level applications or digital modules to strengthen surveillance and response mechanisms as per local requirements.

C. Setting up of and Strengthening of Laboratory Network

States to ensure compliance with the quality requirement as per the quality protocol issued by NJJM. States shall establish a multi-tier laboratory network consisting of:

- i. State/ regional-level reference laboratories.
- ii. District laboratories.
- iii. Cluster/ block/ Sub-divisional level laboratories where required.

These laboratories may be State-owned laboratories, or established through collaboration with accredited institutions, universities, or private laboratories. The laboratory network shall provide:

- i. Confirmatory testing.
- ii. Technical support for remedial measures.
- iii. Analytical reports for regulatory and public health response.

D. Institutional Response Mechanism

- i. Source protection measures and related advisories: At the local level, Gram Panchayats may issue orders regarding the protection of water sources and measures to prevent contamination. Similarly, for surface water

sources and other common community sources, the DWSM should issue appropriate protection measures and advisories.

- ii. DTU to analyse water quality monitoring data and coordinate technical interventions by mandatory issuance of Disinfection protocols to HGJ Villages and ensuring the training and orientation prior to Jal Arpan and HGJ declaration.
- iii. DTU to analyse the feedback from GPs through Jal Seva Aankalan regarding Infrastructure repair or treatment system upgrades.

2.4.3. Monitoring and Evaluation Framework

Monitoring and evaluation of the WQM&S framework under JJM 2.0 shall be undertaken through a combination of digital reporting systems, community-based oversight and institutional monitoring mechanisms across the four governance tiers. The monitoring framework shall utilise data captured through the Sujal Gaon Utility App, laboratory reporting systems and the JJM-IMIS platform, enabling real-time analysis and timely technical response.

A. Village Level Monitoring

- i. Sujalam Shakti groups shall conduct routine water quality testing using Field Test Kits (FTKs), including monitoring of residual chlorine levels and basic water quality parameters. Test results shall be recorded and uploaded through the Sujal Gaon Utility App, linked to the respective Sujal Gaon service area ID.
- ii. The VWSCs shall review the water quality testing results reported by Sujalam Shakti groups and ensure appropriate actions such as disinfection, pipeline inspection, or source protection and cleaning measures where required.
- iii. Water quality monitoring outcomes and related actions shall be periodically reviewed in the Gram Sabha, ensuring community awareness, transparency and collective oversight of drinking water safety. Village-level monitoring indicators shall include:
 - a. Number of FTK tests conducted per month



- b. Compliance with prescribed testing frequency
- c. Residual chlorine monitoring compliance
- d. Number of water quality alerts reported
- e. Timely reporting through Sujal Gaon Utility App

B. District Level Monitoring

At the district level, monitoring shall be undertaken by the DTU in coordination with the DWSM to ensure timely technical response and system reliability.

The DTU and DWSM shall monitor the following parameters:

- i. Monitoring of the completeness, timeliness and accuracy of water quality data reported through the Sujal Gaon Utility App and laboratory reporting systems.
- ii. Verification of compliance with prescribed testing frequency for different water sources and supply systems, as specified in 'Concise Handbook for Monitoring Water Quality of Piped Drinking Water Supply to Rural Households' released in December 2024.
- iii. Monitoring the time taken for technical intervention and corrective action following detection of contamination or water quality deviations.
- iv. Ensuring timely submission of water samples for confirmatory testing and monitoring laboratory turnaround time for reporting results.
- v. Providing technical guidance to Gram Panchayats and VWSCs for addressing water quality issues through disinfection, infrastructure repair, treatment solutions, or source protection measures.

District-level monitoring dashboards shall enable DTUs and DWSMs to identify high-risk locations, recurring contamination issues and operational gaps in water quality management.

C. State Level Monitoring

The SWSM shall undertake periodic review of water quality monitoring performance across districts to ensure effectiveness and institutional accountability.

The State-level monitoring framework shall include the

following indicators:

- i. Laboratory Performance Indicators
 - a. Turnaround time for laboratory testing
 - b. Availability of trained laboratory personnel
 - c. Compliance with testing protocols and reporting standards
 - d. Accreditation status of laboratories
- ii. Community Monitoring Coverage
 - a. Number of operational Sujalam Shakti groups
 - b. Percentage of villages undertaking routine water quality testing
 - c. Number of persons trained under Sujalam Shakti groups
- iii. State-Level Response Mechanisms
 - a. Effectiveness of district technical response to water quality alerts
 - b. Timely communication of advisories to citizens
 - c. Integration of water quality monitoring data with State dashboards

SWSM shall also review trends in contamination events and develop corrective policy and operational measures where systemic issues are identified.

D. National Level Monitoring

The National Jal Jeevan Mission (NJJM) shall periodically review the overall performance of the Water Quality Monitoring and Surveillance ecosystem across States and Union Territories.

The national-level review shall focus on the following parameters:

- i. State Performance Indicators
 - a. Progress of water quality monitoring across rural habitations
 - b. Digital reporting compliance through Sujal Gaon systems
 - c. Laboratory network functionality
- ii. Effectiveness of WQM&S Ecosystem
 - a. Responsiveness of district-level technical support systems



- b. Integration of community surveillance with laboratory confirmation
- c. Performance of digital monitoring platforms
- iii. Public Health Outcomes
 - a. Trends in water quality contamination cases
 - b. Reduction in water-borne disease risks associated with improved monitoring and response mechanisms
 - c. Citizen confidence in drinking water safety

The insights generated through national monitoring shall inform policy guidance, technical advisories and capacity-building initiatives for strengthening water quality governance under JJM 2.0.

2.4.4. Utilisation of Water Quality Monitoring and Surveillance (WQM&S) Funds

The WQM&S component under Jal Jeevan Mission 2.0 is intended to support the establishment of a robust, citizen-centric water quality governance framework across rural drinking water supply systems. States shall utilise these funds to build an integrated ecosystem comprising field-level surveillance, institutional monitoring, laboratory support systems and responsive remedial mechanisms, ensuring long-term safety and reliability of drinking water services.

A. Establishment of Community-Based Water Quality Surveillance

States shall prioritise the establishment of field-level surveillance mechanisms through community institutions. For this purpose, women-led Self Help Groups (SHGs) shall be organised as Sujalam Shakti groups, which will support Gram Panchayats and Village Water and Sanitation Committees (VWSCs) in routine water quality monitoring, community awareness and coordination for remedial actions.

The Sujalam Shakti groups shall undertake activities such as:

- i. periodic water quality testing using Field Test Kits (FTKs);
- ii. monitoring of residual chlorine levels and disinfection practices;

- iii. reporting potential contamination risks through the Sujal Gaon Utility App;
- iv. assisting VWSCs in community surveillance and awareness activities; and
- v. supporting local response actions where water quality deviations are observed.

B. Capacity Building and Skill Development

To strengthen technical capacity at the community level, one to two members of each Sujalam Shakti group shall undergo structured training leading to certification under NSQF Level-1 in basic water quality monitoring and surveillance.

The training programme shall include:

- i. use of Field Test Kits (FTKs);
- ii. understanding of basic water quality parameters and contamination risks;
- iii. safe sampling practices;
- iv. residual chlorine monitoring and disinfection awareness; and
- v. digital reporting through the Sujal Gaon Utility App.

The training shall be supported under the WQM&S component of JJM 2.0, with convergence encouraged with State Rural Livelihood Missions (SRLMs) and other Govt. training institutions to ensure scale and sustainability.

C. Incentive-Based Fund Release Linked to Governance Milestones

After supporting the establishment of community surveillance and training of field functionaries, the remaining funds under the WQM&S component shall be released to States in tranches linked to achievement of defined governance milestones.

The milestone-based incentive structure is intended to support States in systematically establishing the institutional components required for effective water quality governance.

Indicative milestones may include:

- i. operationalisation of Sujalam Shakti groups and digital reporting through the Sujal Gaon Utility App;
- ii. establishment of district, regional and cluster/Sub-divisional level laboratory monitoring



systems;

- iii. adoption of standard water quality testing protocols and remedial response mechanisms;
- iv. strengthening of laboratory networks with adequate technical manpower;
- v. accreditation of laboratories and compliance with national testing standards;
- vi. development of State-level protocols for water quality surveillance, preventive maintenance and response mechanisms; and
- vii. establishment of sustainable State budgetary provisions for long-term water quality monitoring and surveillance.

D. Strengthening Institutional Water Quality Governance

The remaining funds under the WQM&S component shall be released to States in tranches linked to achievement of defined governance milestones. The milestone-based structure is designed to ensure that States progressively establish field-level surveillance, laboratory monitoring systems, technical response mechanisms and sustainable institutional arrangements for water quality governance.

States shall utilise WQM&S funds to ensure that water quality monitoring systems are supported by clear institutional arrangements, including:

- i. technical oversight by District Technical Units (DTUs);
- ii. laboratory support for confirmatory testing;
- iii. monitoring through State Water and Sanitation Mission (SWSM); and
- iv. digital reporting integrated with the Sujalam Bharat platform and JJM-IMIS systems.

This approach will strengthen public confidence in drinking water safety and ensure long-term sustainability of rural piped water supply systems under Jal Jeevan Mission 2.0.

2.4.5. Transparency and Citizen Information

The Citizen Corner - JJM WQMIS will provide a platform to States/UTs to enable access to citizens regarding information on:

- i. Water quality testing results: Water quality data generated through community testing by

Sujalam Shakti groups, routine laboratory analysis and surveillance activities shall be digitally linked with the Sujal Gaon service-area architecture under the Sujalam Bharat platform. Relevant water quality information pertaining to supplied drinking water may be periodically displayed on the Citizen Corner to enable citizens to access the latest available testing results for their respective service areas.

- ii. Status of remedial actions: District Technical Units, DWSM and State Water and Sanitation Missions shall monitor the responsiveness of this system to ensure timely updating of test results, effective communication with citizens and prompt remedial action in cases where water quality deviations are reported.
- iii. Functionality of local water supply systems: This will strengthen public confidence in drinking water safety and promote community participation in water governance.

2.4.6. Roadmap for strengthening WQM&S:



Figure 11: Roadmap for Strengthening WQM&S





CHAPTER 2.5

Source Sustainability & Water Security Framework

Reform aims at:

- ◆ Decision Support System (DSS) introduced as a digital platform for source monitoring and planning, integrating multi-source datasets under the Water as One Data approach.
- ◆ Aquifer-based planning institutionalised to guide scientifically suitable recharge and conservation measures.
- ◆ GIS-based identification of water-stressed villages and vulnerable sources for prioritised sustainability measures.
- ◆ District Source Sustainability Action Plans (DSSAP) introduced for systematic identification of critical sources and phased sustainability measures.
- ◆ Strengthened institutional and convergence framework, including Source Sustenance Committees and alignment with National and State programmes for recharge and conservation works.

In the service-delivery phase of JJM, long-term drinking water security is to be ensured by sustaining the reliability and quality of all drinking water sources throughout the scheme life cycle. The focus shall progressively shift from infrastructure creation to scientific assessment, aquifer-based planning, climate-risk screening and proactive management of withdrawals and recharge.

Source sustainability refers to ensuring the long-term reliability, adequacy and safety of drinking water sources of rural water supply schemes. It requires maintaining both the quantity and quality of the source at levels sufficient to meet present and future demand, while preventing ecological stress and avoiding disruptions in scheme functionality.

2.5.1. Key Features of Source Sustainability

- i. Ensures adequate and reliable quantity & quality of drinking water through sustained source yield, seasonal stability, regular testing and timely corrective measures.
- ii. Supports scheme functionality by reducing breakdowns, supply interruptions and dependence on temporary water arrangements.
- iii. Requires annual and seasonal assessments of groundwater levels, yield, quality and recharge behaviour for all drinking water sources.
- iv. Enables predictive identification of scheme criticality through continuous and real-time monitoring, helping detect declining yields, seasonal stress or emerging quality risks.
- v. Strengthen implementation through convergence with different ministries and departments, such as VB G RAM -G, Watershed Development Component-Pradhan Mantri Krishi Sinchayee Yojana (WDC-PMKSY) 2.0, etc., for recharge, watershed development and catchment improvement.
- vi. Promotes multi-department coordination for recharge planning, source protection, water budgeting, demand management and climate-resilient interventions.
- vii. Integrates water budgeting and community participation, enabling Gram Panchayats to monitor water availability, demand and sustainability measures.
- viii. Utilises GIS-based mapping and annual risk categorisation to identify water-stressed villages and prioritise interventions.



- ix. Relies on continuous monitoring of groundwater and surface water sources throughout the scheme life cycle.
- x. Achieved through a combination of technical measures (recharge structures, aquifer-based planning, source protection) and institutional mechanisms (SSC, DWSM reviews and governance systems).

2.5.2. Institutional Framework

Source sustainability shall be supported through clearly defined roles at the State, District and Gram Panchayat levels, ensuring that technical review, coordination and monitoring are part of regular planning and implementation processes.

Source Sustenance Committee

As envisaged in the Operational Guidelines for the implementation of Jal Jeevan Mission, para 5.2, Source Finding Committees were constituted at the planning stage of each scheme to identify and recommend the availability of potable water in adequate quantity in prescribed quality for the scheme design period. In Operation & Maintenance phase of Jal Jeevan Mission, the Source Sustenance Committee (SSC) shall be constituted at Gram Panchayat, District and State level in alignment with the JJM institutional structure to ensure periodic assessment, protection and long-term sustainability of drinking water sources. At the Gram Panchayat level, the VWSC shall act as the Source Sustenance Committee and shall be responsible for source-level monitoring, protection and reporting activities. At the District level, the SSC shall function under the District Water and Sanitation Mission (DWSM) supported by the District Technical Unit (DTU) to plan, implement and monitoring of source sustainability measures. At the State level, the SSC shall function under the Executive Committee of the State Water and Sanitation Mission (SWSM) to provide policy guidance, technical oversight and coordination with relevant State departments.

A. Gram Panchayat Source Sustenance Committee

The Gram Panchayat Source Sustenance Committee shall periodically assess and support:

- i. Maintenance of basic source records and seasonal registers for drinking water sources within the Gram Panchayat, including providing inputs for digital monitoring.

- ii. Ensure protection and upkeep of drinking-water sources and supports water-quality testing (labs/ FTK) with prompt reporting of any quality issues or supply distress.
- iii. Contribute to GP/ village water budgeting and assist in preparation of GPDP.
- iv. Assist in the upkeep of minor recharge or conservation works approved by the district and escalate emerging risks (drying, turbidity, leakage).
- v. Plan and oversee recharge and source protection measures at GP level, including structures implemented through convergence with VB G RAM G, WDC-PMKSY 2.0 and other schemes.

B. District Source Sustenance Committee

The district Source Sustenance Committee shall periodically assess and support:

- i. Review the seasonal performance of all drinking water sources and assess sustainability, especially during lean periods.
- ii. Plan and oversee recharge and source-protection measures, including structures implemented through convergence with VB-G RAM G, WDC-PMKSY 2.0 and other schemes.
- iii. Identify at-risk sources and GP/villages facing depletion or quality deterioration and recommend appropriate corrective actions.
- iv. Facilitate water budgeting, demand-management measures and inter-departmental coordination for recharge, conservation and catchment protection.
- v. Submit recommendations to the DWSM for approvals, convergence planning and required follow-up actions.

C. State Source Sustenance Committee

The State SSC shall:

- i. Review consolidated district reports on source sustainability and track overall progress.
- ii. Issue policy guidance and technical advisories to districts based on emerging challenges and priorities.
- iii. Provide support to districts facing recurring or widespread sustainability issues, including technical inputs and coordination assistance.
- vi. Facilitate convergence and coordination with

State-level departments responsible for water resources, groundwater, watershed management and environmental protection.

D. Digital monitoring and support system

Digital monitoring and support for source sustainability under Jal Jeevan Mission shall be enabled through a Decision Support System (DSS) developed on the JJM-IMIS platform. The DSS shall act as an enabling digital platform under the “Water as One Data” approach, integrating geo-spatial layers, hydro-geological parameters, rainfall information, aquifer data and drinking water source datasets from various national and departmental platforms. This includes integration of CGWB groundwater status data, Sujalam Bharat source mapping and digitised inspection inputs from districts. Accessible through dashboards for District Water and Sanitation Missions (DWSMs) and Gram Panchayats, the system shall support evidence-based planning, monitoring and prioritisation of source sustainability measures from district to village level.

The DSS shall generate composite risk indicators, early warning alerts and possible intervention options to help identify vulnerable sources and prioritise recharge and conservation measures. By bringing together data from multiple departments in a single platform, the system shall support coordinated planning, improve transparency and strengthen proactive management of drinking water sources to ensure sustained functionality of rural water supply systems under Jal Jeevan Mission.

2.5.3. Aquifer-based planning

States/UTs shall adopt aquifer-based planning to ensure that drinking water sources are developed and managed in accordance with local hydro-geological conditions. Districts shall use NAQUIM aquifer maps and groundwater assessment data from CGWB/ INGRES to understand aquifer types, seasonal groundwater behaviour and long-term trends, particularly in semi-critical, critical and over-exploited areas. Recharge and conservation measures such as check dams, percolation tanks, recharge shafts and spring-shed interventions shall be selected based on aquifer characteristics, depth to water table and recharge potential. The Decision Support System (DSS) shall integrate these datasets to support informed siting of drinking water sources, planning of recharge structures and preparation of Sustainability Action Plans under JJM.

2.5.4. Sustainability Action Plans

District Water and Sanitation Missions (DWSMs) shall prepare a District Source Sustainability Action Plan (DSSAP), aligned with the Comprehensive Implementation Reform Plan (CIRP) of the State/UT, to ensure the long-term reliability and functionality of drinking water sources under Jal Jeevan Mission. The Action Plan shall identify vulnerable or critical sources and assess hydro-geological conditions, groundwater trends, seasonal source performance and water quality risks and outline appropriate corrective and preventive measures along with responsible departments and implementation timelines. Preparation of the Action Plan shall also incorporate local knowledge and traditional water management practices to support locally appropriate and community-responsive sustainability measures. Districts shall utilise datasets and analytical outputs available through the Decision Support System (DSS) to support evidence-based prioritisation of sustainability interventions. Information on drinking water sources mapped under the Sujalam Bharat platform shall also be reviewed to assess source performance and criticality, in order to guide timely sustainability planning.

The Action Plan shall adopt a phased approach covering near-term stabilisation of vulnerable sources, short-term implementation of recharge and conservation measures and medium-term aquifer-scale and watershed interventions, while promoting convergence with relevant programmes and integration with GPDPs and District Action Plans to strengthen long-term water security. States and districts shall also promote Jal Utsav/ Jal Arpan Diwas as a periodic village-level initiative for inspection, cleaning and preventive maintenance of drinking water sources, recharge structures and related infrastructure to support sustained source functionality.

2.5.5. Commissioning, Handover and Annual Source Maintenance

Source sustainability planning shall be aligned with the commissioning and handover processes under Jal Arpan. Prior to commissioning of a drinking water scheme, a successful trial run shall be undertaken to verify the performance of the source, treatment system and distribution network. Upon commissioning, water supply assets shall be formally handed over to the Gram Panchayat/ VWSC for operation and maintenance, including responsibilities for source protection,



monitoring and upkeep of associated recharge and conservation structures.

To support sustained functionality of drinking water sources, Jal Utsav shall be organised periodically as a village-level initiative for inspection, cleaning and maintenance of water sources, recharge structures and related infrastructure. Such activities shall reinforce community participation, preventive maintenance and long-term sustainability of rural drinking water sources under Jal Jeevan Mission.

2.5.6. Source Protection and Contamination Measures

To ensure the safety and sustainability of groundwater sources used for drinking water supply, States/UTs shall implement source protection and contamination prevention measures in accordance with applicable guidelines of the Central Public Health and Environmental Engineering Organisation (CPHEEO), Central Ground Water Board (CGWB) and Bureau of Indian Standards (BIS).

Appropriate minimum sanitary protection distances shall be maintained around drinking water sources such as tube wells and borewells to minimize the risk of contamination from anthropogenic activities.

District administrations and Gram Panchayats shall utilize geospatial information and datasets available through the Decision Support System (DSS) and other digital platforms to:

- i. Map drinking water source locations
- ii. Monitor surrounding land use patterns
- iii. Identify potential contamination risks
- iv. Support planning and implementation of appropriate source protection measures

The surroundings of drinking water sources shall be maintained in a clean and well-kept condition and may be suitably developed or beautified to promote community ownership, responsible behaviour and protection of the source.

To prevent contamination of drinking water sources, the following safeguards shall be ensured within the prescribed protection zone:

- i. Prohibition of landfill sites or solid waste disposal facilities in proximity to drinking water sources.

- ii. Prevention of disposal or storage of hazardous or polluting substances near tube wells and borewells.
- iii. Avoid construction of soak pits, leach pits, septic tanks, magic pits or other waste disposal structures within the sanitary protection radius of drinking water sources. Where such structures cannot be avoided due to site/area constraints, appropriate protective and mitigation measures shall be adopted to prevent contamination of the drinking water source.

Further, to prevent over-extraction of groundwater and hydraulic interference between wells, installation of additional tube wells or borewells within the prescribed safe distance from an existing drinking water source shall be reviewed in accordance with CPHEEO, CGWB and BIS technical standards.

These measures are essential to maintain source integrity, prevent contamination and ensure long-term sustainability of groundwater-based rural drinking water supply systems.

2.5.7. Water Budgeting

Water budgeting at the Gram Panchayat (GP) level shall be undertaken using a structured, data-driven approach to assess water availability and demand across domestic, livestock, irrigation and other uses. Priority shall be given to allocation of water for drinking water supply, ensuring that domestic water requirements under Jal Jeevan Mission are safeguarded before planning for other uses. GPs shall use relevant datasets available through the Decision Support System (DSS) such as rainfall, groundwater trends, land use and aquifer information along with local data on population, livestock, crop patterns and water storage structures to estimate water availability from rainfall, groundwater recharge, runoff, surface storage and greywater reuse. The resulting water balance can support evidence-based decision-making, prioritisation of interventions and preparation of Village Water Security Plans under JJM.

2.5.8. Greywater management

Effective management of household greywater shall be an important component of source sustainability under Jal Jeevan Mission. Gram Panchayats shall promote safe collection, treatment and reuse of greywater through appropriate systems such as soak pits, leach



pits, drainage channels, kitchen garden and nature-based treatment options, in convergence with programmes such as SBM(G) Phase II. Proper greywater management helps prevent contamination of groundwater and drinking water sources, reduces stagnation and local pollution around water supply infrastructure and enables beneficial reuse for activities such as irrigation of plantations and community green spaces. These measures contribute to improved village sanitation, protection of drinking water sources and long-term water security.

2.5.9. Convergence with Government Schemes/ Grants

States/ Districts/ Gps shall plan and implement source sustainability measures through convergence with Central and State programmes supporting groundwater recharge, catchment protection, greywater man-

agement, water conservation and demand regulation, ensuring that investments are directed to priority locations. Programmes such as VB-G RAM-G, WDC-PMKSY 2.0 and Finance Commission Grants may be leveraged for this purpose. As per recent central government notification provisions, a minimum share of expenditure on water-related works shall be prioritised in groundwater assessment units (blocks) based on their groundwater status 65% in over-exploited and critical blocks, 40% in semi-critical blocks and 30% in safe blocks thereby directing recharge and conservation investments towards water-stressed areas and strengthening long-term sustainability of drinking water sources under JJM.

An indicative list of schemes under which activity/ fund convergence can be prioritized to improve the sustainability of villages is attached as **Annexure-I**.



CHAPTER 2.6

Establishing Digital Data Governance in Rural Drinking Water Supply Systems

Reform aims at:

- ◆ Shift from fragmented data systems to an integrated digital ecosystem, enabling end-to-end visibility of rural water supply systems.
- ◆ Enhanced real-time monitoring and decision-making, through multi-level dashboards tracking service delivery and scheme performance.
- ◆ Complete asset and network visibility, with GIS-based registries and digital mapping of hydraulic connectivity across assets.
- ◆ Improved operational efficiency and accountability, driven by digital capture of operational, financial and water quality transactions at the service provider level.
- ◆ Strengthened citizen engagement and transparency, through integrated platforms for feedback, grievance redressal and participatory monitoring.

The Sujalam Bharat Framework provides a national-level digital governance architecture aligned with the vision of Viksit Bharat @2047 to ensure assured and professionally managed rural drinking water supply services. The framework mandates the integration of digitised GIS-based physical asset infrastructure layers created in the *Sujalam Bharat* Digital Public Infrastructure (DPI) with stakeholder transaction layers enabled through the Sujal Gaon Utility Application and integration of data from SCADA/ Digital monitoring tools thereby institutionalising a unified, data-driven governance mechanism for Rural Piped Water Supply Schemes (RPWSS). This digital ecosystem establishes standards for accountability, transparency, accuracy and sustained service delivery across the asset

lifecycle, from creation to operation and maintenance.

Objectives of Sujalam Bharat

- i. Provision of integral and interoperable digital data layers of PWS asset infrastructure and transaction layers to the stakeholders through open API scalable systems.
- ii. Enable assured drinking water service delivery, which includes enabling of future expansion of supply to remote and underserved rural areas
- iii. Facilitate transparent access to RPWSS infrastructure information for citizens, industries, investors and public institutions, thereby supporting informed decision-making and availability of PWS as a professionally managed service.
- iv. Promote ease of living and investment readiness through modular and standards-based DPI-enabled data on water quality, infrastructure and service delivery.
- v. Establish uniform protocols for the operation, maintenance and benchmarking of rural PWS systems, enabling the development of national standards and data-supported decision-making mechanisms.

Reforms for Establishing Digital Data Governance in PWS Operations:

Digital Data Governance under the *Sujalam Bharat* Framework is anchored on four core principles:

- i. *Sujalam Bharat* foundational as a Digital Public Infrastructure (DPI)
- ii. GIS-Enabled Asset Mapping for Transparent Physical & Financial Monitoring
- iii. Near real time Data entries for Transaction mapping to enable Digital Twin systems in RPWSS at Service provider level

- iv. Empowering citizens to be a part of decision-making process.

2.6.1. Sujalam Bharat foundational DPI

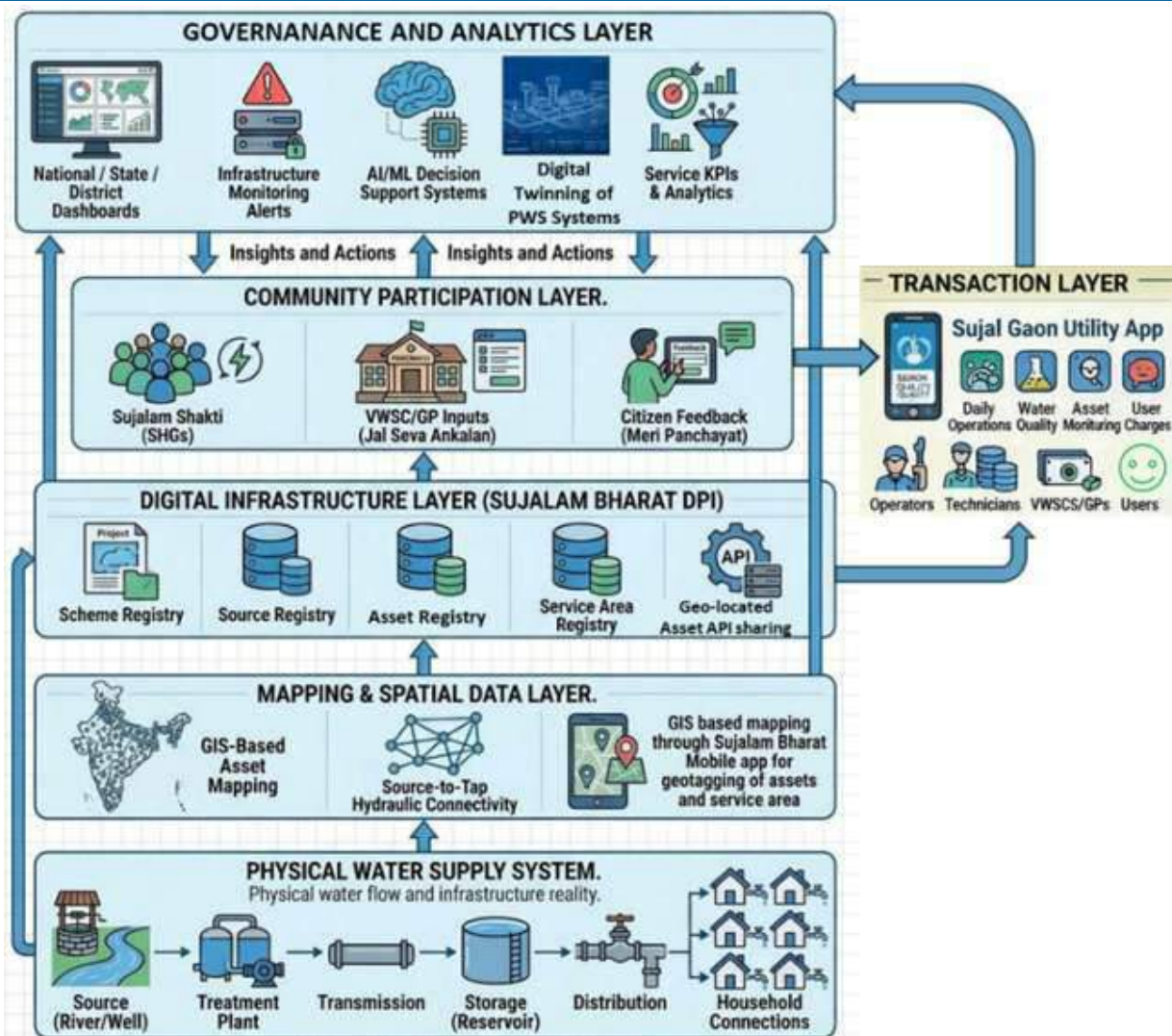
Sujalam Bharat is being developed as a federated DPI for India's drinking water sector for creation of interactive digital asset layers to the stakeholders as per the data entries created by the State and District users for a complete digital hydraulic connectivity of assets in rural piped water supply schemes (RPWSS).

- i. The DPI provides Interoperable database to multiple cross sectoral stakeholders and enables exchange of modular and scalable asset data registries through open API.

- ii. Designed to move beyond infrastructure creation-based monitoring towards comprehensive service delivery assurance. It integrates interoperable digital registries, geo-spatial asset mapping, transaction platforms and decision-support systems to create a unified yet flexible national digital backbone for water service delivery from source to tap.

- iii. Sujalam Bharat creates an architecture that enables seamless coordination across Centre, State, District and local communities, while maintaining operational autonomy of States/UTs and establishes a trusted, population-scale digital infrastructure to support sustainability, transparency, financial accountability and long-term management of

Figure 12: Functional Framework for Sujalam Bharat





drinking water systems.

- iv. NPMU-IT/ NJJM shall function as the National Drinking Water Data Governance & Intelligence Centre, responsible for setting and enabling data interoperability standards, validating data quality as entered by States, generating national insights and providing decision-support to states.

Role of State/UT

- i. Ensure accurate and complete data entry in the Sujalam Bharat (RPWSS) Module as per approved SOPs, enabling system-generated Sujal Gaon IDs and maintaining uniformity across all scheme entries.
- ii. Adhere to the *Sujalam Bharat* DPI IT architecture to ensure consistent end-to-end mapping of source-to-habitation water supply systems, including correct linkage of schemes with the Asset IDs.
- iii. Ensure correct and transparent classification of Out-Village and In-Village infrastructure, based on functional role and O&M jurisdiction, not merely geographic location.
 - a. Assets serving multiple Gram Panchayats (e.g., WTPs, MBRs, upstream infra) shall be mapped as Out-Village, even if located within a village boundary.
 - b. Assets exclusively serving a single GP and operated locally may be mapped as In-Village.
- iv. Review and validate all In-Village entries created by District-level users to ensure alignment with hydraulic connectivity, operational logic and source-to-tap representation within the *Sujalam Bharat* (RPWSS) architecture.
- v. Ensure correct sequencing of infrastructure components in the *Sujalam Bharat* (RPWSS) module in accordance with actual scheme design. States shall map assets strictly as per real operational flow without altering physical sequencing.
- vi. Develop and implement digital literacy and capacity-building programs for GPs and

VWSCs, in coordination with NJJM, to develop digital maturity models and structure digital improvement plans using *Sujalam Bharat* DPI.

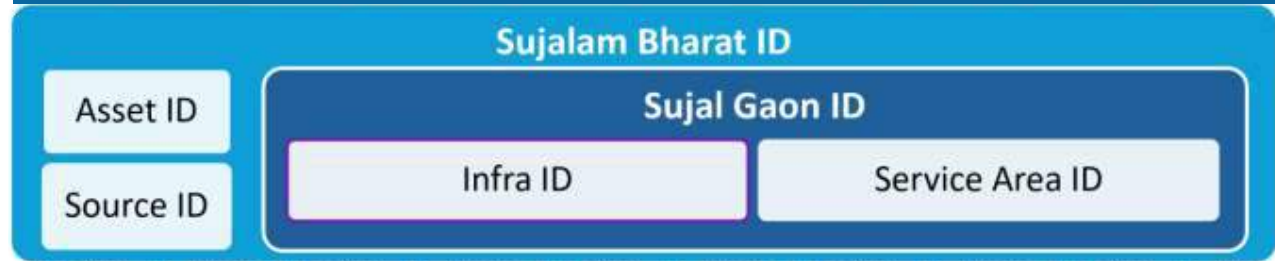
- vii. Leverage *Sujalam Bharat* DPI asset registries to develop contextualised digital applications for PWS management, improving local planning, monitoring and service delivery as use cases.

2.6.2. GIS-Enabled Asset Mapping for Transparent Physical & Financial Monitoring

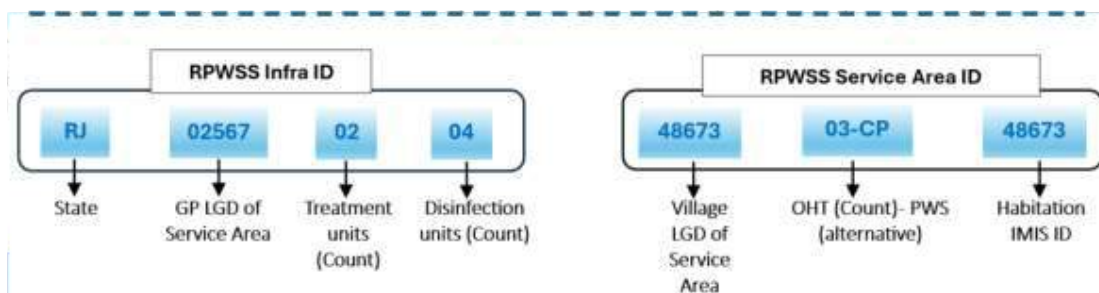
To ensure transparent, accurate and real-time monitoring of physical and financial progress of JJM schemes, NRDWP (legacy) schemes and State Schemes, all drinking water supply assets shall be mandatorily captured through a GIS-based spatial asset mapping mobile application integrated with the *Sujalam Bharat* Digital Public Infrastructure (DPI). NJJM, integrating PM GatiShakti has developed the *Sujalam Bharat* Mobile App as the national platform to map RPWSS geospatial layers, ensuring asset geo-tagging, pipeline mapping and generation of asset location identifiers. Each asset to be geotagged shall be linked to its Asset ID, Scheme ID and system-generated *Sujalam Bharat* ID as established in the *Sujalam Bharat* (RPWSS) module to enable end-to-end visibility of RPWSSs and maintain the coordination with *Sujalam Bharat* (RPWSS) module.

- i. Every water supply asset from source to last mile distribution at the habitation level must be geotagged or georeferenced as per the completion status defined in the *Sujalam Bharat* SOP.
- ii. Integration of financial data with the geospatial asset registry shall ensure parallel and verifiable alignment between expenditure and physical execution milestones.
- iii. GIS-based asset mapping shall enable creation of interactive dashboards at the State, District and Gram Panchayat levels, enabling transparent progress tracking and decision-making.
- iv. These dashboards which are composed of GIS based digitalized asset shall serve as the foundation for data integrity, audit trails, spatial validations and risk detection, supporting proactive governance and improved drinking water service delivery across all tiers.

Figure 13: Relationship between Sujalam Bharat ID, Source ID, Sujal Gaon ID, Service Area ID, Infra ID and Asset ID



Sujal Gaon ID: with Out-Village and In-Village Components



Sujal Gaon ID: with Only In-Village Components

As part of the RPWSS generation process, the system creates the Sujalam Bharat ID, which acts as a single unique identifier by mapping the Asset IDs, Source IDs. All PWS assets in a scheme from source to tap are registered and Geotagged against this unique system generated Sujalam Bharat ID

Role of State

- Onboarding and registration of Designated Implementation Agency (DIA) for Bulk Water, Out-Village and In-Village assets geotagging/georeferencing.
- Assign Asset IDs to corresponding Scheme IDs for geotagging/georeferencing to the registered DIA.
- Review, approve or reject geospatial submissions (geotagging/georeferencing) for all asset categories and ensure timely correction and resubmission by DIA.
- Ensure all assets follow mandatory geospatial mapping rules: GPS accuracy, administrative boundary checks, pipeline geometry validation and enforce maker-checker workflow integrity.
- Review, approve, or return submissions of In-Village assets mapped by Implementing Agencies registered under the District-level



users and verify asset locations as per assigned District boundaries.

- vi. Ensure that DIA follow mandatory spatial validations (buffer limits, geometry rules, accuracy requirements and critical points mapping).
- vii. Ensure that the DIA reviews the uploaded KMLs where required and review pipeline KMLs before geotagging.
- viii. DIA shall function as data makers and are not authorized to approve or finalize any asset geotagging; they must ensure timely submission of all mapped assets to the designated State or District checker for validation.

2.6.3. Near real time data entries for Transaction mapping to enable Digital Twin systems for Rural Piped Water Supply Schemes (RPWSS)

To institutionalise digital O&M and assured service delivery in RPWSS, all routine operational, water-quality, financial (O&M) and grievance transac-

tions shall be captured in near real time through the Sujal Gaon Utility App, integrated via APIs with National/State platforms (e.g., JJM-IMIS, WQMIS, e-Gram Swaraj, Sujalam Bharat RPWSS). This enables a unified, role-based transactional layer for daily operations (pumping, storage, distribution), chlorination and FTK/lab testing, preventive maintenance, user charges and grievance cycles, with automated alerts, escalations and GP-District-State dashboards for monitoring and response.

- i. Standardised digital workflows shall be used by VWSCs/GPs, Nal Jal Mitras/WSOs, SHGs and FTK-trained women to ensure uniform capture of operational and water-quality events, replacing fragmented/manual registers.
- ii. Closed-loop actioning (alert → action → escalation (in the event of non compliance) → outcome → evaluation) shall be enforced for no-supply events, RC/turbidity deviations, breakdowns, leakage and poor collections.
- iii. Dashboards at GP/DTU/DWSM/SWSM levels shall provide real-time service KPIs (supply hours/ volume, water-quality compliance, failure incidence, grievance Turnaround time(TAT), user-charge performance) and

Figure 14: Digital Twin for Rural Water Supply Systems



support preventive maintenance planning.

- iv. Interoperability with National/ State systems shall avoid duplicate reporting and maintain a single source of truth data integrity is ensured through role-based access, OTP login, audit logs and versioning.
- v. Continuous data flows from the Sujal Gaon Utility App for transaction mapping between stakeholders with disruption reports and grievance records. It shall form the dynamic behavioural layer of the Digital Twin.
- vi. With the integration of SCADA/ IoT systems, the Digital Twin shall enable simulation of flow dynamics, energy utilisation and functional adequacy to support preventive maintenance planning and resource prioritisation.
- vii. States/ UTs may also use this transactional layer database to develop Digital Twin visualisations for service Utilities to ensure efficient O&M.

Role of State

- i. Adopt and notify the Sujal Gaon App as the standard transactional platform for village-level O&M data capture.
- ii. Issue SOPs and data standards for daily/periodic entries (operations, water-quality, finance, grievance), thresholds for alerts/escalations and minimum reporting frequencies; enforce uniform use across all GPs.
- iii. Enable and oversee user onboarding/ authorisation (GP-anchored registration of VWSCs, Nal Jal Mitras/ WSOs, SHGs/ FTK users) and mandate OTP-based role access with audit trails.
- iv. Establish dashboard governance by defining State/ District/ DTU KPIs, review the responsibility for actioning alerts monitor scheme functionality, water-quality compliance, grievance resolution and user-charge performance.
- v. Mandate closed-loop response to system alerts with time-bound actions and escalation to DTU/DWSM/SWSM and track outcomes on

dashboards.

- vi. Direct DTUs to provide technical support (fault diagnosis, preventive maintenance guidance, major repair validation) based on Sujal Gaon App alerts and trends which will help in prioritising resource allocation accordingly.
- vii. Ensure financial monitoring by monthly capture of demand/collection/expenditure balances use analytics to flag persistent shortfalls and initiate corrective measures.
- viii. Undertake digital literacy and capacity building for VWSCs/GPs/field functionaries on standard workflows, quality sampling, disinfection logging and grievance handling within the App

2.6.4. Project Monitoring Module

The Project Monitoring module in the Sujalam Bharat App has been developed to effectively track and oversee the implementation of schemes during the infrastructure creation phase. It facilitates systematic capture of both physical and financial progress linked to the Asset IDs of Rural Piped Water Supply Schemes (RPWSS). This ensures transparent, accurate and real-time monitoring throughout the entire project implementation lifecycle.

- i. **Single backbone:** Uses Sujalam Bharat IDs to anchor every RPWSS, preserving hydraulic linkages from source to distribution and enabling scheme-wise, asset data mapping.
- ii. **Lifecycle tracking:** Records mergers/ demergers/ restructuring/ completion status against the same Sujalam Bharat IDs, maintaining continuity of technical, physical and administrative histories.
- iii. **Audit-ready trail:** Captures administrative actions and financial transitions for each restructuring event, creating an end-to-end digital audit trail.
- iv. **Progress sheet:** Dashboards summarise schemes (JJM/Non-JJM), schemes connected to Sujalam Bharat IDs with fully/partially connectivity and derived Total Scheme Cap Cost against the Sujalam Bharat IDs Cost.



2.6.5. Empowering citizens to be a part of decision-making process

Citizens shall be empowered as active participants in rural water-supply governance through transparent access to service-delivery information and an institutionalised feedback and grievance-redressal loop linked to Sujal Gaon IDs. All citizen inputs captured through e-Gram Swaraj, Jal Seva Aankalan and Meri-Panchayat App, shall be geo-tagged to the corresponding service area, enabling pinpointed issue resolution across the PWS value chain. Multi-level dashboards (NJJM, SWSM, DWSM, Panchayat) shall provide continuous visibility of feedback status, grievance resolution and service-delivery outcomes, supporting data-driven decision-making at all administrative levels.

- i. System-generated Sujal Gaon IDs shall serve as the anchor for linking all citizen feedback and grievances to the exact habitation and scheme infrastructure, enabling targeted action by GP, District and State authorities.
- ii. Jal Seva Aankalan shall function as the GP/VWSC-facing digital feedback mechanism, capturing functionality assessments and service issues and notifying DWSM and DTU for technical support.
- iii. The Jal Jeevan Mission (JJM) module on the Meri Panchayat App, developed in collaboration with the Ministry of Panchayati Raj (MoPR), serves as a dedicated citizen interface to strengthen participatory governance at the Gram Panchayat level. It provides village-level information on water supply schemes, water quality status and details of VWSC/Pani Samiti etc. for public access. It has a facility for citizens to provide feedback on Jal Seva Aankalan, which is a self-assessment of functionality of drinking water supply in their village, submitted by Gram Panchayat, thereby completing the feedback loop for improved accountability.
- iv. It is envisaged that the module on Meri Panchayat App will be further integrated with the Sujalam Bharat Project Monitoring Module to enable citizens to track the physical progress of ongoing water supply schemes in their village. Additionally, linkage with the Sujal

Gaon ID will facilitate visibility of the entire source-to-tap hydraulic connectivity, while integration with the Sujal Gaon Utility App will provide real-time information on water supply service delivery. It will also provide a mechanism for citizens to provide feedback on progress and service delivery which will enable community engagement, ensuring greater accountability and strengthening decentralized water governance systems.

- v. Multi-level dashboards (NJJM, SWSM, DWSM, Panchayat) shall display real-time grievance status, pendency, escalations and resolution timelines, strengthening transparency, accountability and evidence-based governance.

2.6.6. Multi-level Governance Dashboards

- i. **NJJM Dashboard:** Provides national-level monitoring, analytics and benchmarking of States/UTs.
- ii. **SWSM Dashboard:** Enables State-level oversight, advisory functions, data quality assurance and coordination with Districts and GPs.
- iii. **DWSM Dashboard:** Supports District-level supervision, scheme functionality monitoring, grievance management and technical facilitation through DTUs.
- iv. **Panchayat Dashboard:** Offers GPs real-time information on scheme functionality, water quality, supply schedules and VWSC actions, enabling local decision-making and community-led monitoring.

Role of State

- i. Establish dashboard governance at State and District levels to monitor grievance pendency, service-level compliance, monitor repeated issues and resolution performance. Trigger corrective actions where service failures persist.
- ii. Issue SOPs defining feedback categories, routing protocols, escalation timelines and

closure verification processes across GP, DWSM, DTU and SWSM levels.

- iii. Promote usage of citizen feedback ecosystem under Jal Seva Aankalan, Meri Panchayat App and e Gram Swaraj integrated with Sujal Gaon Ids
- iv. Ensure all grievances are traceable by mandat-

ing linkages to Sujal Gaon IDs, enabling precise localisation of service breakdowns and targeted response.

- v. Strengthen coordination between SWSM, DWSM, DTUs and GPs for time-bound resolution of community-raised issues, including provision of technical support through DTUs.





CHAPTER 2.7

Establishing Participatory Governance through ‘Jan-Bhagidari’

Reform aims at:

- ◆ Participatory governance through Jan Bhagidari.
- ◆ IEC as social infrastructure for community ownership.
- ◆ Structured communication framework for transparency and engagement.
- ◆ Jal Utsav as the national umbrella for community participation.
- ◆ Jal Arpan and Jal Seva Aankalan to strengthen ownership and accountability.
- ◆ Structured Samvad platforms for dialogue, peer learning and programme feedback.
- ◆ Use of digital platforms and AI-enabled localisation tools for scalable citizen outreach.

Under JJM 2.0, the focus shall shift from infrastructure creation to sustained service delivery. Household tap connections shall mark the beginning of a long-term service commitment, to be managed through continuous O&M by Gram Panchayats and VWSCs/ Pani Samitis, supported by community through Jan Bhagidari.

Under JJM 2.0, participatory governance through Jan Bhagidari shall be institutionalised to create an enabling environment where communities are actively engaged and take responsibility for managing their water schemes. Community engagement shall follow a structured progression from awareness to ownership, promoting changes in perception, behaviour, and responsibility. The approach shall recognise water as a shared natural and cultural resource, encouraging communities to value, celebrate and sustain it.

Community engagement shall be guided by a progres-

sion from awareness to ownership, wherein communities take responsibility for managing and sustaining their water systems.

2.7.1 IEC for Strengthening Community Participation

IEC under JJM 2.0 shall act as a key enabler for building a people owned water supply system. It shall focus on creating an environment where communities are informed, engaged and motivated to take collective responsibility for sustaining village water systems. Through sustained outreach and dialogue, IEC shall support the transition from awareness to active community ownership.

- IEC shall be treated as social infrastructure supporting the sustainability of rural water supply systems. It shall move beyond one-time awareness campaigns or event-based activities and function as a continuous process to enable informed, responsible and actively engaged communities.
- IEC shall strengthen Jan Bhagidari by promoting community participation in planning, management and oversight of rural water supply systems. Different efforts shall focus on encouraging responsible water use, timely payment of service delivery charges, preventive maintenance, water quality vigilance and protection of drinking water sources. These efforts shall aim to build a sense of ownership among communities towards village water supply systems.
- States/ UTs shall adopt participatory and dialogue based approaches and tools such as Participatory Rural Appraisal (PRA) and interpersonal communication (IPC). Platforms such as Gram Sabha, Jal Chaupal and other commu-

Figure 15: IEC as JJM's Social Infrastructure - Moving from Awareness to Governance



nity interactions shall be used regularly to facilitate two-way communication, enabling communities to participate in decision making, service monitoring and grievance redressal.

- vi. Communication efforts shall maintain a unified identity and consistent messaging across all levels, while allowing contextual adaptation to local socio-cultural conditions.

2.7.2 Structured Communication Framework for Transparency and Accountability

All communication, media engagement, community participation, and IEC activities under JJM 2.0 shall be aligned with the national umbrella framework of “Jal Utsav” and shall be planned as part of the Annual Action Plan (AAP). Jal Utsav shall serve as a structured platform to promote community participation by strengthening the cultural and emotional connect with water, while ensuring national coherence and flexibility at State, District, and Gram Panchayat levels to incorporate local cultural and regional elements. In addition, continuous IEC interventions to support operation, maintenance, and service delivery shall also be undertaken, provided that all such activities are

included in the approved AAP. All activities shall be planned in an integrated manner and shall not be undertaken in isolation. Accordingly, participatory governance under JJM 2.0 shall be operationalised through an integrated communication framework to strengthen Jan Bhagidari, covering:

- a. Public Disclosure for transparency and accountability.
 - b. Media Engagement for wider outreach and awareness.
 - c. Reporting & Documentation for monitoring and learning.
 - d. Strengthening Jan Bhagidari through structured community engagement.
 - e. Branding and Unified Identity for consistent messaging.
 - f. Activities aligning with long-term sustainability of piped water supply management.
- i. **Quarterly Public Disclosure Mechanism:** Every Gram Panchayat and District shall ensure public disclosure of key service delivery parameters once every quarter to promote transparency, accountability and informed community participation. The following



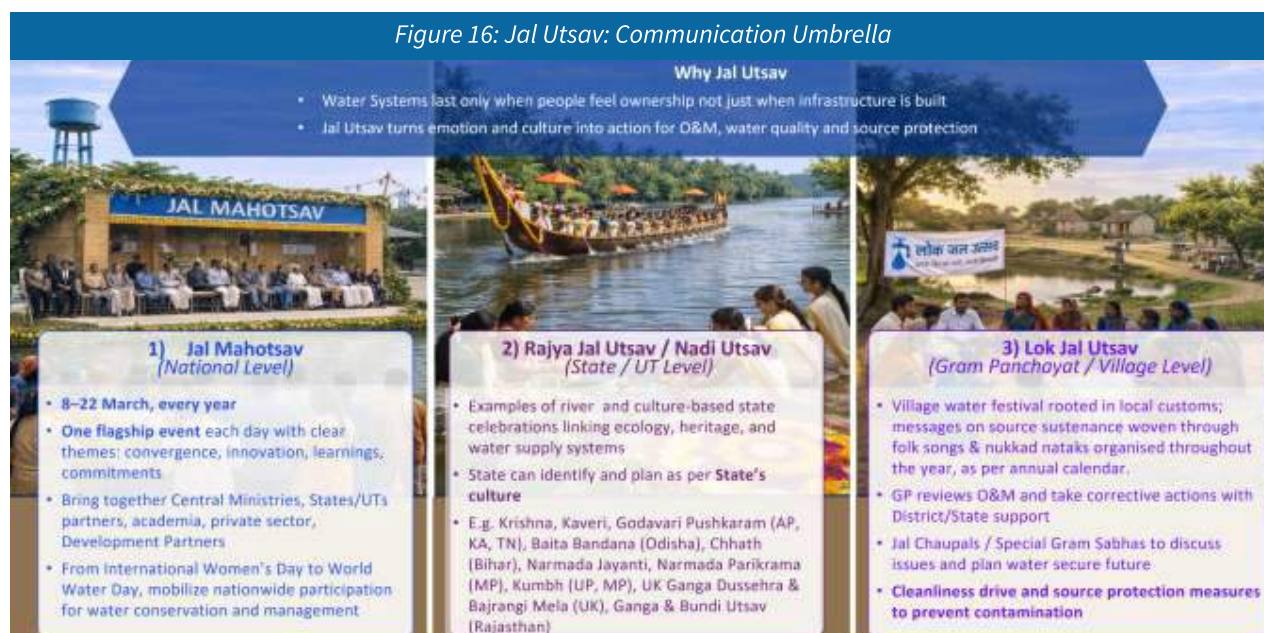
service parameters shall be disclosed:

- Water Quality Testing Status.
- Functionality Status (through Jal Aankalan or as specified from time to time)
- Grievance Redressal Data
- Collection of service delivery charge and other O&M Fund and Utilisation
- Source Sustainability Actions

Such disclosure shall be ensured through public display at the Gram Panchayat level, presentation in Gram Sabha or Jal Chaupal and reporting in standardised

organised under this framework to ensure coordinated, community-led participation across the country. Jal Mahotsav shall be positioned not merely as an annual event, but as a national platform to nudge States, Districts and Gram Panchayats to institutionalise community-led celebrations rooted in local customs, traditions and cultural associations with water. The emphasis shall be on fostering an enabling cultural and governance environment for sustained behavioural change, rather than organising isolated events.

Figure 16: Jal Utsav: Communication Umbrella



formats or as specified by NJJM from time to time.

- Jal Utsav Framework:** Jal Utsav shall serve as a national level framework to promote community participation and engagement through culturally rooted activities, while providing a unified direction with flexibility for contextual implementation at State, District, and Gram Panchayat levels.

- Jal Mahotsav – National Level:** At the national level, Jal Mahotsav shall be observed annually to create a collective national focus on water sustainability. Activities at the National, State/UT and Gram Panchayat/Village levels shall be

- Rajya Jal Utsav / Nadi Utsav at State/ UT level:** At the State/UT level, Rajya Jal Utsav / Nadi Utsav shall be organised to reflect regional water traditions, local river systems and culturally rooted conservation practices, including alignment with traditional festivals and community celebrations associated with water. State specific calendars shall be developed to promote convergence across departments and to ensure active participation of districts and local institutions.
- Gram Panchayat/ Village Level - Lok Jal Utsav:** At the Gram Panchayat/Village level, Lok Jal Utsav shall be organised as community-led water festivals rooted in local



traditions, folklore and cultural practices, reflecting the community's historical relationship with water and customary practices such as "Kuan Pujan". Gram Panchayats shall prepare annual Lok Jal Utsav calendars, ensuring regular community engagement through activities such as Gram Sabha/ Jal Chaupal, water quality demonstrations, asset awareness and collective water conservation actions.

- d. **Annual Calendar:** States/UTs shall prepare annual calendars for Rajya Jal Utsav/ Nadi Utsav and Lok Jal Utsav and incorporate them into their Annual Action Plan.
- e. **Periodical Review:** The framework of Jal Utsav shall also be utilised to periodically review the operation and maintenance of water supply systems, identify gaps and undertake corrective actions with active community participation.
- iv. **Integrated Governance Instruments:** Community institutions viz. Gram Panchayats, VWSCs will play a central role in planning, managing, operating and sustaining rural drinking water services. To translate this responsibility into day to day action at the village level, a set of integrated governance instruments will be implemented to promote community ownership, enable regular service assessment and institutionalise dialogue and peer learning. Together, these instruments will function as a unified accountability loop connecting communities, local governments and higher administrative levels.

A. Jal Arpan Linking Commissioning and Handing Over Protocols with Community Participation :

Commissioning and handing over of rural water supply systems under JJM 2.0 shall not be treated as purely technical milestones, but as community centric processes that establish ownership, accountability and readiness for sustained service delivery. Jal Arpan framework shall serve as the institutional mechanism to integrate community participation into these protocols, ensuring that Gram Panchayats and communities are actively involved in the transition from construction to operation and maintenance.

Accordingly, States/UTs shall ensure that commu-

nity participation is embedded in commissioning and handing over processes through the following:

- i. **Community Involvement during Trial Run:** Gram Panchayat/ VWSC shall participate in the 15–30-day trial run to verify service levels (supply hours, pressure, quality) and identify gaps.
- ii. **Transect Walk by Community:** Gram Panchayat/VWSC, along with community members, shall undertake a transect walk from source to storage and distribution points to physically verify system components, identify leakages or deficiencies, and build confidence in understanding and managing the scheme.
- iii. **Validation through Gram Sabha / Jal Chaupal:** Commissioning status, service parameters and readiness for O&M shall be presented and discussed in Gram Sabha or Jal Chaupal for community validation.
- vi. **Transparency of System Details:** Key documents such as as-built drawings, asset registers, water quality reports and O&M plans shall be shared with the community and made accessible at the Gram Panchayat level.
- v. **Orientation of Local Stakeholders:** VWSC members, operators (Nal Jal Mitra) and community representatives shall be oriented on their RPWSS, operation, maintenance protocols, water quality monitoring and grievance redressal mechanisms etc. associated with their Piped water scheme.
- vi. **Formal Handover through Jal Arpan:** Completed schemes shall be formally handed over to the Gram Panchayat/ community through Jal Arpan, reinforcing collective ownership and responsibility.
- vi. **O&M Readiness and Financial Preparedness:** Service delivery charge mechanisms, O&M fund arrangements and roles and responsibilities shall be clearly established prior to handover.
- vii. **Public Disclosure and Community Awareness:** Service levels, water quality status and grievance mechanisms shall be publicly disclosed as part of Jal Arpan activities.

Following completion and formal handing over of the scheme and the first observance of Jal Arpan



Diwas, it shall thereafter be observed annually as a community platform for reviewing system performance, reinforcing accountability, and undertaking corrective actions to ensure long-term sustainability of rural water supply systems. States/UTs shall integrate this observance into the Lok Jal Utsav Calendar, thereby institutionalising annual maintenance and community-led review processes.

Detailed Standard Operating Procedures (SOPs) on Jal Utsav, Jal Arpan and related activities are provided in the Handbook titled “Community Managed Piped Water Supply Systems in Rural India: Jan Bhagidari se Har Ghar Jal”¹ released by the Department in November 2025, which shall be referred to, along with other advisories issued by the Department, while planning and conducting such activities.

B. Jal Seva Aankalan as a Community-led Governance Tool:

Jal Seva Aankalan is a community led self-assessment process undertaken by Gram Panchayats/VWSCs to evaluate the performance and sustainability of rural water supply services. It enables communities to assess key aspects such as service functionality, water quality, operation and maintenance, financial management, and source sustainability, and to identify gaps for continuous improvement.

- i. Jal Seva Aankalan shall be institutionalised as a community led governance tool under JJM 2.0 to enable Gram Panchayats and VWSCs to undertake periodic self-assessment of rural drinking water service delivery. The process shall promote community participation, strengthen local accountability and support informed decision making for sustained operation and maintenance of piped water supply systems.
- ii. The assessment shall be carried out by the Gram Panchayat/VWSC with participation of community members and the findings shall be presented in Gram Sabha / Jal Chaupal and publicly disclosed at the Gram Panchayat level to ensure transparency, community validation and collective ownership.

- iii. The outcomes of Jal Seva Aankalan shall be used to identify gaps in service delivery and to prepare Service Improvement Plans (SIPs) or other corrective action plans at the Gram Panchayat, District and State levels, as applicable.
- iv. Aggregated findings of Jal Seva Aankalan shall support monitoring, review and planning at higher administrative levels and shall contribute to strengthening the overall accountability framework under JJM 2.0.

C. Pey Jal Samvad as a Dialogue and Learning Platform:

NJJM has recently initiated structured dialogue platforms such as Pey Jal Samvad with District Collectors and Sujal Gram Samvaad with Gram Panchayat and community, to strengthen participatory governance and continuous learning under JJM 2.0. These platforms shall be institutionalised to facilitate engagement, exchange of experiences and collective problem-solving across all levels.

- i. At the district level, Pey Jal Samvad shall provide a platform for District Collectors and officials to present successful practices, innovations and implementation experiences. At the national level, Sujal Gram Samvad shall enable Gram Panchayats and communities to share their achievements, challenges and learnings and directly engage with the Government of India.
- ii. These platforms shall promote peer learning, encourage replication of good practices and provide opportunities to address implementation challenges through collective dialogue. They shall also serve to recognise community efforts, build trust and motivate stakeholders towards improved performance and sustained service delivery.
- iii. States/ UTs shall actively motivate and facilitate Gram Panchayats to come forward and share their experiences, innovations and success stories on platforms such as Pey Jal Samvad and Sujal Gram Samvad, enabling wider learning, recognition and replication of

¹ Handbook: Community Managed Piped Water Supply Systems in Rural India: Jan Bhagidari se Har Ghar Jal vide D.O. Letter No. W-JJM-/23/2023-JJM-VI-DDWS dated 3rd Dec. 2025

good practices.

- D. Sarpanch Samvad:** Sarpanch Samvad, facilitated by the Quality Council of India (QCI), is a digital platform and mobile application aimed at connecting, training, and empowering Sarpanches across the country. It provides a structured platform for sharing best practices, accessing e-learning resources, and promoting peer learning to strengthen local governance and support improved service delivery outcomes. Under JJM, the Platform shall be leveraged to engage elected representatives, capture field-level insights, share experiences and strengthen feedback loops for improved service delivery.

2.7.3. Knowledge Management and Citizen Outreach

- i. Communication under JJM 2.0 shall be continuous, localised and participatory. Knowledge management efforts shall focus on translating policy provisions into simple, relevant and citizen centric narratives, tailored to diverse language, regional and socio-cultural contexts. Documentation and capacity building of IEC functionaries shall be strengthened through structured training programmes.
- ii. Digital media shall be used strategically to enhance outreach, while ensuring that content is contextualised and accessible. States and districts shall prioritise vernacular communication, engage local influencers and promote community-generated content to build trust and strengthen ownership.

2.7.4. Use of Digital Tools and Artificial Intelligence

- i. Digital tools, including artificial intelligence, may be leveraged to support planning, content development, translation, dissemination and feedback analysis of IEC activities. These tools shall be used to enhance efficiency, scalability and responsiveness, while ensuring that outputs remain contextualised, accurate and accessible. Appropriate human oversight and quality assurance mechanisms shall be ensured in the use of such tools to maintain

accuracy, reliability and contextual relevance of outputs.

- ii. States/ UTs shall promote the creation and dissemination of short videos, including reels, testimonials to document and showcase community-led initiatives, success stories and good practices under JJM. Such content shall be used to enhance outreach, encourage peer learning and motivate wider community participation.

2.7.5. Convergence and Unified Messaging

- i. JM 2.0 shall adopt a whole of government approach to communication by strengthening coordination across Ministries and Departments. A unified WASH communication framework shall be followed to ensure consistent messaging, coordinated outreach and effective dissemination.
- ii. Impact assessment shall be undertaken in convergence with relevant programmes, schemes, and Ministries/ Departments such as WCD, MoHFW and MoRD, as part of a whole of government approach.
- iii. States and UTs shall integrate WASH into school curriculum in coordination with SCERT. Engagement with Schools shall be made to promote awareness and responsible behaviour through outreach activities strengthening Jan Bhagidari.
- iv. Aligned communication calendars, common branding and messaging, and coordinated media engagement shall be ensured to enhance clarity and impact, while integrating water-related messaging with nutrition, health, education, and related schemes.
- v. RWPF shall serve as a partnership platform to support participatory governance and IEC through knowledge sharing, capacity building, demonstrating good practices and coordinated efforts to strengthen Jan Bhagidari.

All IEC activities at State, District and GP levels shall be planned and included in the CIRP/Annual Action Plan.



CHAPTER 2.8

Capacity Building Framework for continuous service improvement

Reform aims at:

- ◆ Shift from generic training to function- and competency-based capacity building, ensuring role clarity and measurable outcomes.
- ◆ Professionalisation of service delivery systems, with utility-oriented, citizen-centric skill development for both micro and macro utilities.
- ◆ Creation of a scalable and standardised training ecosystem, through a structured ToT-based hub-and-spoke model.
- ◆ Strengthened district-level technical capacity, enabled by data-driven assessments and targeted interventions using Jal Seva Aankalan.
- ◆ Improved accountability and tracking of capacity outcomes, through time-bound training targets supported by digital monitoring tools.

2.8.1. Introduction

With the substantial expansion of rural piped water supply infrastructure under the Jal Jeevan Mission across the country, the focus is now increasingly shifting towards ensuring sustained and efficient service delivery. As the mission approaches its completion phase in many States / UTs, the priority is to ensure that systems continue to function efficiently throughout their design life. This requires systematic capacity building and competency enhancement of all stakeholders involved at utility and governance level. Strengthening governance orientation and institutions is also critical so that institutions can build and sustain the necessary support systems beyond the mission period. States will therefore need to continue skill and

competency-based capacity building of stakeholders to ensure that rural piped water supply systems are effectively managed and remain functional in the long term.

2.8.2. Vision

Rural piped water supply systems range from simple village-level operations to complex, large-scale service delivery arrangements involving multiple stakeholders and interconnected processes. Therefore, while strengthening their capacities, it is equally important to ensure that linkages between stakeholders are clearly defined and consistently maintained. Stakeholders must be oriented not only to their individual responsibilities but also to the institutional coordination and collaboration required for effective service delivery.

Governance institutions must lead this transformation by driving change management efforts and ensuring that service utilities operate effectively and transparently. This includes continuous training, orientation and capacity building of all stakeholders so that institutions can perform their roles efficiently while maintaining proper oversight, support and coordination across the entire service delivery chain.

2.8.3. Key Stakeholders

Stakeholders are structured across following core levels, with each level carrying specific responsibilities that collectively ensure sustainable rural piped water service delivery:

- Gram Panchayat Level (Community level): Community and Community based Organizations (SHG/PACS, NJM, Community level workers/Volunteers, functionaries, AWWs, ASHAs) who manage and support day to day village water supply activities directly within the community,
- Service Utility Level-

- a. Macro Utilities that have dedicated managerial and technical wings and operate at the block/district/ regional level to manage and oversee large or multiple water supply schemes effectively
- b. Micro Utility: GPs /VWSCs or any other entity working as service provider at village level
- iii. Governance Level-Institutions that provide policy direction, oversight, regulation and capacity support for effective water service delivery to states
 - a. **Local governance:** Gram Panchayat, PRI Members
 - b. **District level:** DWSM (Administrative) and DTU (Technical) including Block level target groups
 - c. **State level:** SWSM, Water Resource Dept/ PHED/RWS/RDWS/PR&RD.

- d. **National Level:** Institutions that establish policy, mission guidelines, Advisories, allocate funding and monitor progress through program level dashboards during the mission period.

2.8.4. Capacity Building Approach

This capacity-building approach uses a structured training framework that aligns skill development with the specific functional roles of stakeholders in rural water supply service. It assigns clear responsibilities, identifies competencies required to perform those responsibilities effectively and outlines targeted training areas to strengthen performance. The approach will also promote convergence with relevant sectoral institutions and programmes, including Panchayati Raj Institutions, rural development systems, livelihood missions and technical training institutions, to leverage existing capacities and

Table 5: Roles and Responsibilities of Key Stakeholders

S.No.	Level	Key Stakeholders	Roles & Responsibilities
1	Community Level	- GPs/ VWSCs - Community-based organizations such as SHGs/ PACS etc. - Local Functionaries such as AWWs/ ASHAs - NJM/ Local operators & technicians - Volunteers	- Daily operation of village water systems - Minor repairs & maintenance - Water quality testing - User charge collection - Community engagement & support - Contract/ vendor management
2	Service Utility Level	Micro level - GPs/ VWSC/ Village-level micro-utilities Macro level - PHED/ Other similar bodies in states - DTU - Professional water service agencies	- Daily operation of village water systems - Minor repairs & maintenance - Contract/ vendor management - Operation & maintenance of infrastructure - Asset management & preventive care - Contract/ vendor management - Ensuring reliable service delivery
3	Governance Level	- Gram Panchayat - DWSM - SWSM	- Policy implementation & oversight - Monitoring and evaluation - Capacity building and change management support - Financial, technical facilitation
4	National Level	National Jal Jeevan Mission	- National policy formulation and strategic planning - Mission guidelines, reforms, and standards - Funding allocation - Monitoring progress through national dashboards



strengthen local support systems.

States and UTs will institutionalize the technical, managerial, financial, governance and digital competencies needed for effective planning, implementation, operation and management of rural water supply systems.

RWPF may also be utilized as a partnership platform to support capacity building through knowledge sharing, technical assistance, and demonstration and scaling of innovative practices. The Forum shall facilitate collaboration among partners to strengthen competencies for sustainable rural water supply management

2.8.5. Core Competency Areas -GPs /Village Level

Gram Panchayat will function as the village-level governance authority for drinking water services while Village Water & Sanitation Committees (VWSCs) will function as local service delivery institutions responsible for operation, maintenance and management of rural piped water supply systems. To effectively perform these functions, capacity-building interventions will focus on strengthening competencies across technical, institutional, financial, digital and governance domains. Accordingly, competency and capacity building areas may be prioritized and an indicative list thereof is attached as **Annexure-II**.

2.8.6. Core Competency Areas-District level

The District Water and Sanitation Mission (DWSM) will focus primarily on program administration, coordination and monitoring, while the District Technical Unit (DTU) will provide continuous technical support and hand-holding to Gram Panchayats and VWSCs. The DTU will also support the preparation of District-level Service Improvement Plans using data from Jal Seva Aankalan and other relevant data available locally. District-level capacity building will strengthen the competencies required for effective oversight, technical guidance and facilitation of decentralized, community-led water service delivery. An indicative list thereof is attached as **Annexure-III**.

2.8.7. Core Competency Areas – State level

At the State level, the State Water and Sanitation

Mission (SWSM), in coordination with the Departments of Panchayati Raj (PR) and Rural Development (RD), shall lead the transition from infrastructure creation to a sustainable drinking water service delivery framework. The SWSM shall be responsible for sector policy formulation, strategic planning, regulatory systems and ensuring that institutional, financial and technical arrangements are in place to support the long-term operation and maintenance of rural water supply systems.

Given the central role of Gram Panchayats and community institutions in managing rural piped drinking water systems, close coordination with PR and RD departments will be essential for strengthening decentralized governance, promoting community participation and enabling convergence with rural development programmes.

Accordingly, capacity building at the State level will focus on strengthening leadership and institutional competencies for service assurance, financial planning, performance monitoring and enabling districts and Gram Panchayats to function effectively as decentralized water service delivery institutions. The specific competency levels and areas of focus at the State level will be determined based on the state-specific gap assessment and priorities identified by the State/ Uts.

2.8.8. Training Content

The National Jal Jeevan Mission (NJJM) will provide the overall training content framework with support from SPM-NIWAS, Premier Institutions, Professor Chairs, Institutes of Excellence and NIRD&PR. States will then customize this training content through their State Institutes of Rural Development (SIRDs), working alongside any other technical and management institutions. The training materials will be adapted to suit regional needs, including local languages, geography and socio-cultural contexts with additional modules addressing specific challenges such as disaster management, coastal conditions, hilly terrains and other locally relevant issues. Wherever required, digital content shall also be incorporated to support effective and accessible training delivery. Establishment of a digital Learning Management System (LMS) shall also be explored to strengthen and scale up capacity building efforts.

2.8.9. Training Delivery

Training will follow a Hub and Spoke Model, using a

structured Training of Trainers (ToT) approach. Master Trainers will be trained at the hub level and will cascade training down through successive levels, ultimately reaching cluster-level functionaries. Under NJJM, NCVET approved NJMP and Sujalam Shakti Drinking Water Quality certificate course, will be offered through government approved institutions - ITIs, Polytechnics, RSETIs, and other government training centres. States may also design their own certificate courses wherever required, with the decision to conduct certificate-based or routine training programmes resting entirely with the State governments.

Within this model, trainers will deliver training to field level functionaries, especially those associated with VWSCs and Gram Panchayats. Community-level orientation activities will be undertaken through Jal Arpan, as detailed in the Chapter “Establishing Participatory Governance through Jan Bhagidari” and Handbook on Community Managed Piped Water System in Rural India.

At the district governance level, capacity building activities will focus on development of managerial, operational and technical aspects. Training will be delivered primarily at the institutional level for District Water and Sanitation Missions (DWSMs) and other relevant parastatal bodies. Within the hub and spoke framework, capacity building will extend to District Technical Units (DTUs), Micro utilities, Gram Panchayat governance structures and community level stakeholders. At the macro utility level, training will similarly be structured across managerial and operational levels.

2.8.10. Training Cycle

Each field level functionary will undergo a minimum of two training cycles to ensure reinforcement of knowledge and practical capacity. Under the hub-and-spoke model will include:

- i. Orientation training
(Capacity gap assessment before Refresher training through different ways)
- ii. Refresher training

To support systematic monitoring, NJJM will explore leveraging the existing digital platform of e-Gram Swaraj under the Ministry of Panchayati Raj for tracking field-level capacity-building activities related to rural

drinking water supply. Necessary provisions or customized modules may be integrated within the platform, in coordination with the Ministry of Panchayati Raj, to enable elected and non-elected representatives to monitor the status of trainings and capacity-building initiatives.

Where required, complementary features may also be developed within the Jal Jeevan Mission digital systems to ensure seamless data capture and reporting.

Further, for HGJ certification of any Gram Panchayat or village, all designated field-level functionaries shall complete the required trainings either prior to certification or within six months from the date of HGJ certification, to ensure readiness for sustained operation and management of rural drinking water supply systems.

For non-HGJ villages, States/ UTs complete the training of all the field functionaries in a time bound manner before their certification. Also, States/ UTs will be responsible for the orientation of contractors and vendors using their own resources.

2.8.11. Key Monitoring Points

To strengthen the overall effectiveness of the Comprehensive Implementation Reform Plan (CIRP) and ensure sustainable service delivery under the mission, it is essential to prioritize capacity building across all levels of field functionaries. Hence following milestones should be prioritized:

- i. All field functionaries in Har Ghar Jal (HGJ) Gram Panchayats shall be trained, and training completion records shall be maintained on e-Gram Swaraj/GP Dashboard.
- ii. At least one round of training shall be completed in all HGJ GPs by June 2027. Refresher training shall be conducted after six months or as decided by the State/District



CHAPTER 2.9

Establishing Human Resource Skilling Ecosystem

Reform aims at:

- ◆ Availability of certified Human Resources with all the RLBs/ GPs for operations of water supply schemes
- ◆ NSQF-aligned certification through NCVET-Jal Vitaran Sanchalak as a nationally recognised job role at NSQF Level-4 approved by NCVET, giving trained personnel with a valid and portable qualification.
- ◆ Recognition of Prior Learning (RPL)-a structured pathway to certify the existing informal O&M workforce already working in villages.
- ◆ Engineering Elective and Operator Diploma in formal academic institutions - water supply O&M embedded into AICTE-affiliated engineering colleges and polytechnic diploma programmes, creating a long-term institutional pipeline.
- ◆ End-to-end digital tracking-lifecycle monitoring from enrolment through training, certification, and GP-wise deployment through SIDH and the DWSM dashboard.

2.9.1. Introduction and Objective

Under JJM, substantial progress has been achieved in creation of rural drinking water infrastructure. With the transition to JJM 2.0, the focus is on ensuring sustained functionality, service delivery, and long-term operability of assets created. In this context, establishment of a robust human resource skilling ecosystem is critical to ensure that adequately trained personnel are available at all levels for O&M of rural water supply systems. The objectives of this ecosystem are to:

- i. Ensure availability of trained and certified human resources at Gram Panchayat level

- ii. Promote community ownership and decentralised O&M through VWSCs/Paani Samitis
- iii. Enable standardisation of service delivery through NSQF-aligned training and certification
- iv. Facilitate local employment and skill-based livelihoods in rural areas

2.9.2. Multi-tier Human Resource Framework

States/UTs shall adopt a multi-tier workforce structure aligned with infrastructure typology.

Table 6: Multi-tier Human Resource Framework for O&M

Level	Role	Qualification	Scope of Work
In-Village	Nal Jal Mitra (Jal Vitaran Sanchalak)	NSQF Level-4	In-village SVS: pumps, pipelines, OHTs, water quality reporting
Out-Village	Multi-skill Technician/ Operator	ITI/ Diploma Polytechnic/ equivalent NSQF level course	WTPs, BWS, MVS, MBRs - routine maintenance, monitoring and minor repairs; Day-to-day WTP and source operations - pump, valve, routine checks
Supervision	Engineer	Engineering Elective in Degree College	SCADA operations, quality control, scheme supervision at WTPs
Village (Cross-cutting)	Sujalam Shakti Groups - FTK Testing & Monitoring	Micro-skilling/ equivalent NSQF Level 1 course	Water quality testing using FTK – identified members of Sujalam Shakti Groups



2.9.3. Nal Jal Mitra Programme (NJMP)

Nal Jal Mitra Programme (NJMP) aims to provide skill-based trainings to local persons to equip them with a comprehensive set of skills, so that they can function as scheme operators and are able to carry out minor repairs and maintenance, including preventive maintenance, of the piped water supply scheme(s) in their village.

The Nal Jal Mitra Programme is to be implemented as an NSQF Level-4 multi-skilling programme under the job role 'Jal Vitaran Sanchalak', approved by NCVET, to build a cadre of locally available multi-skilled personnel for in-village O&M. The Nal Jal Mitra Programme (NJMP) shall be implemented as the primary intervention for village-level skilling. Two types of training are provisioned under NJMP:

- i. Short Term Training (STT)¹ of 510 hours for fresh candidates and
- ii. Recognition of Prior Learning (RPL) & Upskilling² of 120 hours for experienced candidates and those who are already performing minor repair & maintenance tasks under O&M of water supply schemes.

2.9.4. Multi-skill Technician Training/ Operator Diploma Programme

Multi-skill Technicians for O&M of out-village components including Bulk Water Schemes (BWS), Multi Village Schemes (MVS), Water Treatment Plants (WTPs), Elevated Storage Reservoirs (ESR), and Master Balancing Reservoirs (MBRs). As per the guidelines the out-village infrastructure is envisaged to be maintained by technical agency such as PHED/RWS/ Jal Nigam/Boards etc. The State Governments may include such specialised courses in ITIs/ polytechnic and ensure utilisation of trained personnel for O&M of out-village infrastructure. The existing engineering colleges may also be encouraged by States/ UTs to offer such courses

2.9.5. Engineering Elective on Water Supply Infrastructure Management

Autonomous Technical universities, Engineering colleges or AICTE-affiliated engineering colleges are to

be engaged for adoption of the elective course focused on rural water supply infrastructure management to be integrated into undergraduate engineering programmes. These electives cover design principles, treatment technologies, distribution network management, automation and SCADA systems, water quality management, and regulatory frameworks etc.

2.9.6 Micro-skilling of Sujalam Shakti Groups for WQM&S

The identified members of Sujalam Shakti groups in village will be trained for WQM&S using Field Test Kits, especially to monitor bacteriological contaminants and residual chlorine. Training is to be delivered through water supply departments/ Lab networks. This programme enables women at the grassroots level to participate in water quality surveillance thereby strengthening community-based monitoring and public confidence in water supply services. The NJJM will also endeavour to develop an NSQF Level-1 certified course in collaboration with relevant sector skill council. The States may utilise their Water Quality laboratories to provide training on testing especially residual chlorine along with other parameters.

2.9.7 Institutional Arrangements

- A. **National Level:** DDWS in convergence with the MSDE and other relevant ministries shall ensure effective implementation of skilling of human resources required for O&M of rural drinking water supply system through policy formulation, financial support, and progress monitoring.
- B. **State Level:** State Water and Sanitation Missions (SWSMs) shall identify the State Skill Development Missions (SSDMs) or State Rural Livelihood Mission (SRLM) as Programme Implementing Agencies (PIA) and coordinate with them for planning and delivery of all training. Annual district-wise training plans are to be prepared covering all workforce categories and submitted to NJJM as part of Annual Action Plans.
- C. **District Level:** District Water and Sanitation Missions (DWSMs) are responsible for implemen-

¹ Refer NJMP guidelines circulated through email on 16.08.2023

² Refer RPL guidelines circulated vide letter No. W-11011/6/2023-JJM-III-DDWS-Part(1) dated 28.02.2024



tation, coordination, and monitoring of skilling activities at the district level. Each DWSM is to designate DTU as a nodal for skilling. The DTU has to coordinate with SSDMs, ITIs, Polytechnics, RSETIs, and other implementing agencies for preparation of proposal and training delivery. For this, at least one person from DTU may be identified for analysis, planning, coordination, implementation and monitoring of the skill requirement of the districts. All skilling activities are to be entered and reflected on the DWSM dashboard.

- D. Gram Panchayat/ Village Water and Sanitation Committee:** Gram Panchayat/ Village Water and Sanitation Committee shall mobilize and recommend candidates for the NJMP/ Women - FTK Testing & Monitoring from Sujalam Shakti Group. GP may deploy the candidates post successful certification.

2.9.8 Training Infrastructure

All training is to be delivered through government institutions - ITIs, Polytechnics, RSETIs, and other government training centres. All training centres are to obtain necessary accreditation and meet infrastructure standards prescribed under guideline before commencement of training.

2.9.9 Financing

Common Cost Norms notified by MSDE from time to time, shall be applicable for training of human resources, wherever not explicitly indicated in this Guidelines, including for difficult geographies.

The financial support for skilling programme under JJM 2.0 will be met from below-mentioned funding sources:

- i. Finance Commission tied grants
- ii. JJM Support component³
- iii. Convergence through PMKVY, RSETIs, other skilling programmes of government

Further, detailed SOP along with schedule of implementation will be issued separately.

³ A detailed SoP on utilisation of funds under JJM Support component for skilling activities shall be issued by DDWS.

2.9.10. Course Adoption and Certification

NSQF aligned courses approved by NCVET shall be adopted for training programme. Assessment is to be carried out by Water Management & Plumbing Skill Council or any other body approved by NCVET. Certification is to be issued upon successful completion of assessment. The Skill India Digital Hub (SIDH) platform shall be utilised for lifecycle tracking of training activities – from candidate registration through training, assessment, certification, and deployment. The Operator Diploma Programme and Engineering Elective course may be adopted by universities, polytechnic, engineering colleges etc. in coordination with AICTE or respective regulatory boards/bodies.

2.9.11. Monitoring and Reporting

A. Monitoring

- i. **Central Level:** Overall monitoring responsibility shall lie with the NJJM at the Centre. NJJM shall review implementation progress, provide policy direction, and ensure alignment with national programme objectives.
- ii. **State Level:** At the State level, SWSM under the leadership of ACS/PS/Secretary, PHE/RWS, shall periodically monitor (preferably on a quarterly basis) the State Action Plans. The State shall ensure timely data reporting, review district performance, and facilitate coordination with training institutions and other skilling programmes.
- iii. **District Level:** At the district level, DWSM under the leadership of the District Collector/Deputy Commissioner/CEO-ZP, shall be responsible for implementation monitoring, validation of progress, and ensuring convergence with relevant government skilling initiatives.

- B. Reporting:** All skilling-related activities shall be reported through the SIDH portal and reflected on the DWSM dashboard to ensure real-time monitoring and transparency. At the district level, digital monitoring shall be undertaken through the DWSM dashboard, with skilling-related data reported through the Skill India Digital Hub (SIDH) portal. The following data



points shall be maintained and updated regularly:

- i. Number of O&M personnel assessed and requirement mapped (category-wise and scheme-wise, Village/GP-wise)
- ii. Candidates enrolled, trained, assessed, and certified (category-wise)
- iii. Training institutions engaged, including accreditation status and batch utilisation
- iii. Deployment status of Nal Jal Mitras and Multi-skill Technicians (GP-wise and scheme-wise)
- iv. Women trained for FTK testing and monitoring (village-wise) and
- v. Convergence achieved with other government skilling programmes

Districts shall ensure completeness, accuracy, and timeliness of data entry for effective review at State and Central levels.

2.9.12. Key action points for States/UTs

- i. Ensure NJMP training in all Har Ghar Jal villages with at least one trained and certified Nal Jal Mitra per Gram Panchayat
- ii. Undertake assessment of human resource

requirement at all levels for O&M of schemes and compute district-wise training gaps

- iii. Adopt and rollout the training courses across identified training institutions in all States/UTs
- iv. Formulate policy for engagement of human resources at rural water service utility levels
- v. Appoint Programme Implementing Agencies (PIA) for planning, coordination and delivery of all training
- vi. Review state-wise progress against NJMP enrolment, training, and certification targets - in alignment with Annual Action Plans
- vii. Monitor deployment status of certified Nal Jal Mitras and Multi-skill Technicians - mapped against scheme-wise O&M requirements
- viii. Ensure convergence with other skilling programmes of government
- ix. Monitor Quality of training delivery through assessment pass rates, dropout rates, and feedback from certified personnel and Gram Panchayats
- x. Ensure adequacy of training infrastructure including accredited institutions, operational batches, and geographic coverage across blocks and districts



CHAPTER 2.10

Operational and Financial Sustainability of Water Supply Schemes

Reforms aims at:

- ◆ Shift from reactive repairs to preventive maintenance-driven operations, reducing breakdowns and extending asset life.
- ◆ Reduction in O&M costs through efficient system management, including pressure control, leak detection and lowering Non-Revenue Water (NRW).
- ◆ Transition from ad-hoc financing to structured and sustainable O&M funding, ensuring uninterrupted and reliable service delivery.
- ◆ Improved scheme functionality and reduced slippages, through regular monitoring, planned operations and lifecycle-based asset management.
- ◆ Strengthened community-led oversight and accountability, with Gram Panchayats actively engaged in supervision, management and sustainability of systems.

As drinking water is a State subject, the primary responsibility for ensuring the effective Operations and Maintenance (O&M) of rural water supply systems rests with the State Government and the respective local bodies. These institutions shall ensure sustained functionality and reliable service delivery by undertaking robust financial planning and securing adequate, timely and dedicated O&M financing as an essential and mandated element of the sector's overarching fiscal framework

Financial and operational efficiency is vital to keep water supply schemes reliable and affordable. Inefficient operation increases O&M costs because fluctuating pressures in Pipe Water Supply systems

may damage pipes, cause unplanned repairs and disrupt services. Poor planning also leads to higher NonRevenue Water (NRW), where treated water is lost without revenue. When schemes are not operated regularly or preventive maintenance is ignored, minor issues become major faults, significantly increasing repair costs. Therefore, steady operation, pressure control, leak management and timely preventive maintenance are essential to reduce O&M burden and ensure efficient service delivery.

Earlier, many rural waters supplied schemes under NRDWP remained nonfunctional, had limited piped coverage and frequently slipped back due to the absence of regular monitoring, preventive maintenance planning and community ownership. Financing for O&M was weak, with no robust mechanism for dedicated funds and schemes operated inefficiently with high O&M costs and little focus on reducing operational burdens. Infrastructure sustainability received less attention, causing assets to deteriorate quickly, while poor community involvement further weakened system reliability. Under the reform-based approach of JJM, States now ensure structured O&M financing for uninterrupted services, mandatory O&M budgeting for every scheme and systematic planning to control O&M costs. Schemes are operated using preventive maintenance schedules, with strong emphasis on infrastructure sustainability and long-term asset care. Community engagement has been strengthened, making Gram Panchayats responsible and involved in vigilance, supervision and day-to-day management. JJM also focuses on preventing slip backs, promoting conjunctive use of water sources and strengthening water quality monitoring. These improvements show why sector reforms are essential to shift from a fragmented, unreliable system to a sustainable, well-planned, community led drinking water service that ensures safe water for every household over the long term.

States/UTs shall focus on key intervention areas that strengthen Operation and Maintenance (O&M) systems in a sustainable manner, ensuring that the overall functioning remains efficient and outcome-oriented. In this context, emphasis shall be placed on:

- A. Efficient Financial Management**
- B. Efficient Operation Management**

2.10.1. Efficient Financial Management

Financial management in the drinking water sector refers to the ability of a water supply system to generate, retain and efficiently utilize financial resources to meet its full operation, maintenance, lifecycle replacement and service improvement costs on a long-term basis, without depending on unpredictable external subsidies or ad-hoc fiscal support.

A. Revenue Adequacy

States/UTs must ensure that water services generate predictable and sufficient revenue to cover all Operation & Maintenance (O&M) expenses, including electricity, staff, water quality testing, routine repairs and emergency needs. Revenue planning must also include provisioning funds for unexpected breakdowns and periodic overhauls. Collection of Water Distribution Charges/User charges for in village schemes and Service Delivery Charges for Bulk Water Transfer should be steady and timely, with strong collection systems, digital payments and incentives for on-time payment.

B. Development of Financial Tools for effective O&M

The State should develop a financial planning tool that is directly linked to the SUJALAM Bharat ID used under the Jal Jeevan Mission for capturing component wise village-level data. By connecting this data to a financial tool, the State can generate accurate estimates of the annual O&M needs for each village and for the State as a whole. This tool can automatically fetch technical and operational information from SUJALAM Bharat ID and convert it into yearly budget requirements for power, manpower, water quality and repairs. It will help the State prepare consistent, data-based O&M plans, identify villages with higher needs and support transparent budgeting.

C. State Budget Linkage for Rural Water Supply

For rural water supply to run without interruption, it is important that all major operation and maintenance

expenses are linked directly to the State budget. This means the State should create clear and budgetary allocation for electricity, major repairing like replacement of high HP pumps and capital item replacement. When these costs are included in the State's yearly financial planning, funds become stable and predictable. This prevents villagers from struggling to keep systems running during times of low community collection.

D. Proper Use of Finance Commission Funds

The State must ensure that tied grants from the Finance Commission are utilised strictly for water and sanitation purposes as per prescribed guidelines. The next cycle, the 16th Finance Commission, also asks for clear financial records, regular audits and proper functioning of Panchayati Raj institutions. The State should help districts and Panchayats meet these rules so that funds continue without stoppage.

E. Ring-Fencing of Funds in Asset Management

A financially sustainable drinking water system depends on predictable, ring-fenced revenue and efficient spending to ensure uninterrupted, high-quality service and timely repair or replacement of assets. Under JJM, ring-fencing O&M funds in a dedicated, transparent account is a key reform that prevents diversion of money and ensures that financing is always available for essential activities like preventive maintenance, electricity payments, water quality testing and emergency repairs. By securing funds specifically for pumps, pipelines, reservoirs, treatment units and source structures, States and UTs strengthen asset management, improve financial discipline and reduce system failures. Through digital monitoring and clear processes, ring-fencing helps every water supply scheme remain reliable, well-maintained and financially secure year after year.

F. Operation and Maintenance (O&M) Budgeting Process

O&M budgeting begins at the village level, where each VWSC prepares a simple, need-based annual estimate for every scheme using SWSM formats. The DWSM then compiles and verifies these budgets, adds the O&M needs of bulk systems and submits a consolidated district requirement to the State. The SWSM aggregates



district figures, assesses available resources from user charges, Finance Commission grants, State allocations, CSR and convergence schemes and identifies the O&M funding gap. This bottom-up process ensures realistic field-based planning, clear gap identification and structured resource convergence, while mandatory digital entry through the Sujal Gaon ID financial tool ensures transparency, uniformity and strong financial governance.

G. Cross-Subsidized Financing for Financial Sustainability

A cross-subsidized model can significantly strengthen the financial sustainability of rural and peri-urban water supply schemes by using revenue from peri-urban, commercial and industrial users to support schemes that cannot fully recover their O&M costs. This approach works effectively only when planning is consolidated at the State, district and scheme levels, allowing authorities to clearly identify which schemes generate surplus revenue and which operate at a deficit. Through systematic assessment of low-revenue or high-cost schemes, States and UTs can allocate additional income from higher-tariff user categories to ensure stable funding for essential services in underserved areas. This model not only helps bridge O&M gaps but also promotes equity, service reliability and long-term financial viability across the entire water supply system.

H. Key Principles of Efficient Financial Management

Following are the key features of Efficient Financial Management:

- i. **Transparency in Accounts:** All financial records, receipts and payments must be clear, traceable and maintained in an open manner for easy verification.
- ii. **Accurate and Timely Record-Keeping:** Revenue, expenditure and balances must be recorded regularly without delays to support correct decision-making.
- iii. **Digital Billing and Payments:** Use digital tools for billing, fee collection and vendor payments to reduce errors, improve traceability and increase collection efficiency.
- iv. **Cost Control and Efficiency:** Minimize unnecessary expenditure through leak

reduction, energy efficiency, preventive maintenance and optimized manpower deployment.

- v. **Regular Monitoring and Audits:** Conduct routine financial reviews and conduct annual audits to identify gaps, ensure compliance and strengthen accountability.
- vi. **Planing of Funds:** Plan budgets based on actual needs and ensure continuous fund flow to support uninterrupted water supply services

Table 7: Key Financial Monitoring Indicators

Indicator	Description / Formula	Monitoring Level
O&M Cost Recovery Ratio	User charges collected ÷ Total O&M expenditure	GP / VWSC
Electricity Cost Ratio	Electricity expenditure ÷ Total O&M cost	GP
Collection Efficiency	Amount collected ÷ Amount billed	GP
O&M Budget Utilization	Actual expenditure ÷ Approved O&M budget	District
Non-Revenue Water (NRW)	Difference between water produced and water billed	GP / District
Timeliness of Financial Reporting	Compliance with monthly financial reporting	District / State

2.10.2. Efficient Operation Management

Operations Management focuses on ensuring that the water supply system functions smoothly every day without interruption. Skilled operators, properly trained technicians and functional VWSCs are essential for daily operations. Standard Operating Procedures (SOPs) must be followed for pump operation, chlorination, valve control, water quality testing and customer grievance handling. Real-time monitoring tools such as flow meters, pressure sensors, online water quality devices and SCADA help detect issues early and improve decision-making. Record-keeping must be systematic so that energy use, water production, service hours and breakdowns can be tracked regularly. Daily data entry of operational activities on Sujal Gaon App is essential for operational sustainability and ensuring consistent, safe and reliable water supply to households as per JJM service standards.

If operational efficiency is not maintained, the overall expenditure of the water supply system can increase very quickly. Uncontrolled pressure, irregular operations, neglected maintenance and frequent breakdowns can sharply raise O&M costs and reduce the life of the system. Therefore, for long-term financial sustainability, it is essential to follow efficient financial management practices together with strong operational discipline. When both finances and operations are managed efficiently, schemes run smoothly, water losses are minimized and resources are used wisely ensuring reliable service delivery at a lower cost.

For effective operation management, following measures are to be taken:

A. Community Vigilance and Surveillance

Community vigilance is an important part of effective operation and management of water supply schemes. When community members regularly observe and monitor the functioning of their scheme, disruptions, leakages, pressure changes, or breakdowns can be detected early and corrected quickly. This active involvement reduces the chances of major failures, prevents water loss and lowers overall O&M costs. Communities must be encouraged and engaged in routine checks, reporting issues and supporting timely repairs. Through strong community surveillance, schemes become more reliable, responsive and financially efficient.

B. Asset Management & Lifecycle Planning

Asset management under JJM requires maintaining a complete, updated record of all scheme assets pumps, motors, pipelines, storage tanks, treatment units, meters and source structures through a detailed asset register capturing installation date, condition, repair history and expected lifespan. Lifecycle planning must guide timely rehabilitation and replacement, with depreciation provisions to ensure future funding. Effective asset management prevents sudden failures, reduces long-term costs and supports sustainable O&M.

The DTU must maintain district-wide, updated asset and inventory records, verify assets on the ground, update condition data and ensure timely entry of new and replaced components. Efficient inventory management ensures availability of spares, faster repairs and better use of funds. DTU must also use the PM Gati Shakti platform for GIS-based mapping of all pipelines, pump houses, tanks, sources and treatment units, ensuring every asset is geo-tagged and visible on the SUJALAM Bharat Platform to support transparent planning and long-term sustainability.

C. Monitoring the Efficiency of Water Supply Schemes

Monitoring the efficiency of water supply schemes is essential to ensure that systems remain functional, financially sustainable and capable of delivering safe water every day. Regular monitoring helps detect issues early, plan timely maintenance, control O&M costs and keep service levels consistent with JJM standards. A structured approach supported by financial tools, preventive maintenance, budgeting discipline and community involvement strengthens long-term performance and reduces breakdowns.

2.10.3. Preventive Maintenance

Preventive maintenance is a key sustainability measure required under JJM. It involves planning and conducting regular servicing and inspection activities to avoid sudden breakdowns and costly emergency repairs. Activities include:

- i. Scheduled service of pumps and motors,
- ii. Leak detection,
- iii. Pipeline flushing,
- iv. Valve operation cycles,
- v. Tank cleaning,
- vi. Electrical inspection and



vii. Calibration of meters and sensors.

Predictive maintenance tools like vibration monitoring and pressure analysis help detect early warning signs of failure. A preventive maintenance schedule must be prepared by the DTU. Conducting preventive maintenance on time extends the life of assets, reduces O&M costs and ensures uninterrupted water supply.

**2.10.4. Key action points for the States/
UTs**

- i. State may adopt financial monitoring tools
- ii. Complete assessment of all post Har Ghar

Jal schemes

- iii. Ensure annual O&M budgeting is provisioned in the State budgets.
- iv. DTU to publish a preventive maintenance schedule for all schemes
- v. Engage the community in regular surveillance and vigilance to identify disruptions early.
- vi. States to adopt structural reforms alongside providing funds, function and functionaries for sustainable O&M of rural piped water supply infrastructure.

CHAPTER 2.11

Research, Innovation & Knowledge -Ecosystem

Reform aims at:

- ◆ **Institutionalized Technology Appraisal:** Establishment of a structured technology appraisal system through the National Technical Committee (NTC) and State Technical Committees (STCs) to guide systematic evaluation and adoption of technologies.
- ◆ **Structured Validation and Adoption Framework:** Introduction of processes that enable solutions to progress through structured development, assessment, piloting and field-based learning prior to wider deployment.
- ◆ **Collaborative Innovation and Knowledge Sharing:** Strengthening collaboration with academia, research institutions, startups and industry, along with systematic documentation and national dissemination of innovations to facilitate wider sectoral learning.
- ◆ **Outcome-Based Financing for R&D Proposals:** Introduction of a results-linked financing approach that aligns R&D funding with clearly defined milestones and measurable outcomes, ensuring that technologies progress through evidence-based development and systematic field validation.

- i. Development and deployment of low-cost, scalable and locally appropriate technologies
- ii. Strengthen indigenous innovation, design and manufacturing capabilities.
- iii. Institutionalize evidence-based validation, field testing and piloting of emerging technologies.
- iv. Foster partnerships among academia, industries, startups and implementation agencies.
- v. Facilitate systematic documentation and dissemination of best practices, operational learnings and knowledge resources across all levels.

This structured approach shall guide States/UTs, institutions and partners to establish sustainable, climate-resilient and cost-effective rural water service delivery systems in alignment with the vision of Atmanirbhar Bharat.

2.11.2. Institutional Framework for Innovations

In 2014, the then Ministry of Drinking Water and Sanitation constituted a Standing Committee for Examination of best Technologies concerning Water and Sanitation. The committee was chaired by Dr. Mashelkar, former Director General, Council for Scientific and Industrial Research. Many technologies on water and sanitation were considered by the Committee and approved for use in the sectors.

Earlier, a National Technical Committee was constituted under the chairmanship of PSA to Govt. of India to examine and review the R&D proposals. Presently, the National Technical Committee has been reconstituted and shall function as the apex body for technology appraisal and innovation promotion in WASH under the chairmanship of Member, NITI Aayog.

A. National Technical Committee (NTC)

The National Technical Committee has been reconsti-

2.11.1. Objective

Jal Jeevan Mission (JJM) shall establish and promote a structured ecosystem dedicated to research, innovation and knowledge management in the rural water supply sector.

The ecosystem shall facilitate the following outcomes:



tuted and shall function as the apex body for technology appraisal and innovation promotion in rural WASH sectors.

The NTC shall undertake functions, including but not limited to:

- i. Evaluate and recommend innovative technologies and R&D proposals.
- ii. Facilitate technology validation and adoption across States/ UTs.
- iii. Promote indigenous technology development.
- iv. Enable collaboration with national and international research institutions.
- v. Issue advisory notes on emerging technologies and sectoral innovations.
- vi. Review and evaluate applied research studies and reports submitted by JJM Professor Chairs as mentioned in the verification protocol of their Disbursement Linked Indicators..

B. State Technical Committee (STC)

States/ UTs shall constitute a State Technical Committee (STC) aligned with the national framework. The STC shall report to the State Water and Sanitation Mission (SWSM) to ensure alignment with state priorities and to facilitate coordinated review, piloting and adoption of appropriate technologies.

Roles and Responsibilities of State Technical Committee

The STC shall undertake functions, including but not limited to:

- i. Evaluate region-specific technologies and innovations.
- ii. Review proposals for pilot implementation of new technologies.
- iii. Assess suitability of solutions based on local hydro-geological conditions.
- iv. Facilitate adoption of innovations recommended by the NTC.
- v. Recommend scale-up of validated technologies.
- vi. STC may refer R&D proposals to the NTC for their consideration.

C. Role of SPM-NIWAS

SPM-NIWAS shall provide secretarial support for the

compilation, shortlisting and facilitation of the evaluation of innovative technologies and R&D proposals. It shall also function as a hub-and-spoke platform to enable collaboration among research institutions, industry, startups and implementation agencies, including facilitating international collaboration and knowledge exchange. It shall coordinate joint research initiatives, innovation challenges and support water technology innovators and startups.

D. Role of National Jal Jeevan Mission

National Jal Jeevan Mission (NJJM) shall function as a national platform to promote the development and adoption of innovative technologies in the rural water supply sector. NJJM shall provide strategic guidance to SPM-NIWAS for coordinating sectoral innovation through collaboration among research institutions, academia, startups and industry for the development, piloting and validation of solutions, integrating evidence-based knowledge and facilitating sectoral learning and the exchange of global and indigenous best practices. Where required, NJJM may also undertake or facilitate such initiatives to support convergence and technology adoption.

2.11.3. Low-Cost Water Treatment and Climate Resilient Technologies

States/UTs shall consider low- cost water treatment technologies to address local water quality challenges through affordable, easy to operate and community friendly technologies.

Indicative technologies include:

A. Low-Cost Water Treatment

- i. Decentralized filtration units: Point-of-use or community-level filtering systems.
- ii. Absorption/Adsorption or media- based treatment systems: Contaminant removal using natural or engineered filter media.
- iii. Solar powered disinfection technologies: Pathogen inactivation using solar energy-driven processes.
- iv. Catalytic or non-membrane treatment systems: Physicochemical treatment processes without pressure-driven membranes.
- v. Modular contaminant removal units for: Scalable units for targeted removal of
 - a. Iron



- b. Fluoride
- c. Arsenic
- d. Nitrate
- e. Microbial contamination
- f. Region specific emerging contaminants

B. Climate Resilient

- i. Atmospheric Water Generators (AWGs): To provide drinking water for sparse population.
- ii. Managed Aquifer Recharge technologies: To enhance ground water recharge.
- iii. Solar powered water pumping system: To ensure reliable water supply in remote areas.

C. Technology selection shall be based on:

- i. Hydro-geological suitability.
- ii. Cost effectiveness.
- iii. Ease of operation and maintenance.
- iv. Scalability and community acceptance.

2.11.4. Research & Development in Water Sector Technologies

A. Priority Areas for Research

To support technological advancement and strengthen service delivery in rural water supply systems, research and innovation efforts shall focus on the following thematic areas:

i. Water Source Security and Quality

- a. Source Sustainability and Aquifer Management: Strengthening understanding of groundwater systems and supporting long-term sustainability and security of drinking water sources.
- b. Emerging Contaminant Management: Improving detection, monitoring and mitigation of emerging contaminants affecting drinking water quality.
- c. Low-cost Advanced Treatment Technologies: Development of efficient, affordable and context-appropriate treatment solutions suitable for rural water supply systems.

ii. Water supply system efficiency and climate resilience

- a. Climate-Resilient Water Supply Systems:

Advancing approaches that enhance resilience of water supply systems to climatic variability.

- b. Energy-efficient Pumping and Treatment: Promoting solutions that reduce energy consumption and improve operational efficiency in water supply systems.

iii. Digital and Operational Innovation

- a. Predictive Maintenance and Digital Monitoring: Enabling digital tools for real-time monitoring, predictive diagnostics and improved reliability of water supply infrastructure.
- b. Cost-effective O&M Models: Supporting innovative approaches that strengthen operational sustainability and reduce lifecycle costs.

B. State-Driven R&D Initiatives

States/ UTs shall undertake R&D initiatives aligned with local hydro-geological conditions, operational requirements and sectoral priorities identified under the Mission. These efforts shall include identifying areas requiring research attention, initiating need-based R&D projects.

States/UTs shall take up various research-oriented solution streams, including:

- i. **Service centric solution:** Development and piloting of technologies that enhance the efficiency, reliability and operational management of rural water supply systems, including tools for monitoring, asset management and operational decision-support.
- ii. **Citizen centric solutions:** Innovations that strengthen user participation, transparency and accountability in water service delivery, behavior change interventions, community-based monitoring tools and strengthened grievance redress system.
- iii. **Sustainability Centric Solutions:** Research and pilots that improve long-term water security through improved source protection, aquifer understanding, water conservation approaches and resource-efficient technologies.
- iv. **Climate resilient solutions:** Development and assessment of approaches and technologies that enhance the resilience of rural water



supply systems to climate variability and extreme events, including tools for risk appraisal, drought-responsive planning, diversified sourcing and resilient infrastructure design.

2.11.5. Support for Water Technology Startups

To promote indigenous innovation, the Mission shall facilitate joint research initiatives, innovation challenges and mentorship activities to encourage water-tech startups in developing indigenous technologies.

A. Convergence and Collaboration with Government Innovation Platforms and Industry Bodies

- i. **Department for Promotion of Industry and Internal Trade (DPIIT):** To leverage the national startup ecosystem, incubation networks and innovation initiatives for development, piloting and scaling of emerging technologies relevant to rural water supply.
- ii. **Federation of Indian Chambers of Commerce and Industry (FICCI):** To facilitate industry engagement, technology partnerships and knowledge exchange for promoting innovative and scalable solutions in the water sector.
- iii. **Confederation of Indian Industry (CII):** To enable private sector participation, technology demonstrations and industry-led innovations that support improved efficiency, sustainability and operational performance of water supply systems.
- iv. **Jal Shakti Innovation Centre (JSIC):** The JSIC shall be leveraged for technology incubation, field-level demonstration, validation and the scaling of indigenous innovations.
- v. **Department of Science & Technology (DST):** National innovation and research platforms, including the Anusandhan National Research Foundation (ANRF) – Mission for Advancement in High impact Areas (MAHA) Water Program, shall be leveraged to support the development and deployment of field-validated solutions for water security. The MAHA Water Program facilitates multidisciplinary and multi-

institutional consortia to advance priority technologies from early-stage research to field-ready applications, including assessment, validation, prototyping and scalable deployment.

2.11.6. Piloting and Scaling of Innovative Technologies

Emerging technologies shall progress through structured pilot testing prior to large-scale adoption. The pilot framework shall include:

- i. Technical appraisal by the National and State Technical Committees.
- ii. Performance monitoring under field conditions.
- iii. Assessment of performance & reliability, cost-effectiveness, ease of O&M and scalability and replicability.

Following successful pilots, subsequent steps shall include:

- i. Development of standard implementation protocols.
- ii. Preparation of O&M guidelines.
- iii. Phased scaling in similar hydro-geological regions.
- iv. Dissemination through National and State knowledge platforms.
- v. Support for scaling through industry partnerships and manufacturing ecosystems.

Outcomes from pilot projects shall be documented and shared with the National Jal Jeevan Mission to support wider adoption, strengthen evidence-based decision-making and enhance national-state convergence.

2.11.7. Collaborative Research and Knowledge Platforms

A. Collaboration with Academia, Research Institutions and Line Departments

- i. To strengthen research, innovation and evidence-based decision making in the rural water supply sector, the Mission shall establish and institutionalize collaborations with IITs, NITs, CSIR laboratories, leading Central and State universities and international academic

institutions to support the development, field validation and standardization of affordable and low-cost technologies for the rural water sector.

- ii. The NJJM shall undertake collaborative research programmes in convergence with the Department of Science and Technology (DST), Indian Space Research Organization (ISRO), Ministry of Earth Sciences and Ministry of Housing and Urban Affairs (MoHUA) and other allied Ministries/ Departments to leverage their specialized domain expertise.
- iii. SPM NIWAS shall collaborate with industry bodies such as FICCI and CII to leverage their water sector platforms and initiatives, including the FICCI Water Mission, India Industry Water Conclave & awards and programs of the CII-Triveni Water Institute to promote technology innovation, technology transfer and knowledge sharing in the water sector.

B. JJM – Professor Chair

JJM Professor Chair established by the DDWS shall be

engaged to strengthen the research, innovation and knowledge ecosystem in the rural water sector.

Performance of DDWS-supported Professor Chairs shall be linked to measurable deliverables including applied research studies, technology validation pilots, policy inputs, State technical engagements and knowledge repository contributions. Disbursement and continuation of support shall be contingent upon achievement of annually defined Disbursement Linked Indicators (DLIs) verified by SPM-NIWAS

C. International Collaborations on Global exchange of Technologies

International collaboration under the Jal Jeevan Mission shall act as a strategic driver for improving service delivery, sustainability and innovation in the rural water-supply sector. These partnerships shall strengthen knowledge exchange, promote global good practices and provide access to emerging technologies, while aligning with India's decentralized and community-led implementation model.

i. Priority Areas for Collaboration

- a. Access to global good practices, structured

Table 8: Disbursement Linked Indicators and Performance Deliverables for JJM Professor Chair

DLI No.	Activity / Deliverable	Annual Target per Chair	Verification Protocol	Disbursement Trigger
DLI-1	Applied research studies addressing operational challenges in rural water supply	Minimum one applied research studies annually in the identified thematic area	Study report submitted and peer reviewed by National Technical committee	Acceptance of research report
DLI-2	Technology pilots / demonstrative models / field validation studies	At least one field pilot or prototype validation per year	Field validation report, supporting documentation, State certification	Completion of pilot activities
DLI-3	Technical notes, guidance documents or policy inputs	Minimum one technical guidance documents annually	Document endorsed by DDWS/SPM-NIWAS	Publication or adoption in sector advisory
DLI-4	State/UT technical engagement	Engagement with minimum two States/UTs annually through workshops or advisory sessions	Submission of engagement summaries	Completion of engagements
DLI-5	Policy or programme integration	At least one research output integrated into JJM guidelines, advisory notes, or pilot programmes	Certification by DDWS/SPM-NIWAS	Adoption into programme implementation



knowledge exchange and emerging technologies.

- b. Technical and advisory support from multilateral institutions, partner countries, research bodies and development agencies.
- c. Focus on water quality surveillance, digital monitoring systems, climate-resilient infrastructure and utility-oriented service delivery models.
- d. Use of platforms such as Indian Technical and Economic Cooperation (ITEC) and other cooperation frameworks for capacity building, knowledge management and international exchange.
- e. Engagement with countries experienced in community-based management, greywater reuse, private sector participation and performance-based O&M systems.
- f. Sharing JJM's scalable practices: community-led planning, tech-enabled digital monitoring (Sujalam Bharat), water quality monitoring and surveillance through community testing and source sustainability convergence.

D. Knowledge Engagement Framework

SPM-NIWAS shall develop an annual calendar of conferences, workshops and knowledge-exchange platforms focused on water technologies and R&D, aligned with priority issues identified by States/UTs. The activities shall include:

i. National and State Knowledge Platforms

Structured knowledge exchange and sector learning shall be facilitated by SPM NIWAS through periodic knowledge-engagement platforms. An annual framework shall include the following activities.

ii. State level thematic workshop:

These workshops shall be convened periodically, focusing on:

- a. Low-cost treatment technologies including water quality monitoring and surveillance.
- b. Source sustainability.
- c. Digital monitoring systems.
- d. Climate-resilient infrastructure.
- e. Energy efficiency in water supply.

iii. Stakeholder Consultation with Industry and Startups:

Structured engagement with water-tech startups, relevant departments, industry bodies such as DPIIT, FICCI and CII and technology providers shall support innovation promotion, technology transfer, standardization and scaling opportunities. The objective of the consultation shall be:

- a. Promote technology transfer.
- b. Encourage indigenous manufacturing.
- c. Identify scaling pathways for innovations

iv. Knowledge Exchange Forums:

Collaboration with IITs, NITs, CSIR laboratories, JJM Professor Chairs and other national and international research institutions shall enable the exchange of R&D outcomes, emerging technologies and applied research relevant for field adoption. Knowledge exchange forums shall further strengthen cross-learning and dissemination of sector-wide experiences. The objectives include:

- a. Disseminating research outcomes
- b. Sharing implementation experiences
- c. Facilitating cross state learning

v. Innovation Grand Challenge:

Launch of problem statements in coordination with NJJM, academia and States/UTs to support development, prototyping and field validation of innovative and cost-effective solutions. The objective of the Grand challenge shall be:

- a. Identify solutions to specific operational challenges.
- b. Support prototype development.
- c. Enable field testing of innovative technologies

vi. India International WASH Conference:

India International WASH Conference shall serve as a national platform for knowledge exchange and sectoral innovation. The conference shall facilitate:

- a. Presentation of research papers.
- b. Technology demonstrations.



- c. Global knowledge exchange.
- d. Dissemination of innovation compendiums.

E. Knowledge Management and Dissemination

A national knowledge repository shall be maintained to systematically document and disseminate sector knowledge, innovations and implementation experiences. The repository shall include:

- a. Technologies and innovations relevant to rural water supply
- b. Outcomes of pilot projects and demonstration initiatives
- c. Case studies and best practices from States/UTs
- d. Technical standards, guidelines and reference materials

2.11.8. Cross Learning and Field Assessment

The Mission shall promote continuous cross-learning and robust field assessments by leveraging the diverse expertise of academic programs, national WASH partner networks and specialized line departments.

A. Jal Seva Adhyayan Programme

To strengthen evidence based improvements in service delivery, the 'Jal Seva Adhyayan' initiative shall be conducted in coordination with Ministry of Education (D/o Higher Education) through AICTE impaneled engineering institutions to support field assessment and experiential learning.

- i. **Key Features of Jal Seva Adhyayan Programme**
 - a. This shall be conducted with AICTE-empaneled engineering institutions to utilize academic and technical expertise for field-based learning and assessments.
 - b. At least two field assessment cycles shall be conducted each year to enable regular learning and feedback from implementation sites.
 - c. Faculty-led student teams shall be undertaken structured technical field studies as part of experiential learning and sector knowledge generation.

ii. Areas of Assessment

- a. Scheme functionality: Assessments shall examine functionality, performance and technologies used in drinking water supply schemes.
- b. Water quality management: Reviews shall cover monitoring, treatment processes and compliance with water quality standards.
- c. Source sustainability: The studies shall examine measures taken to ensure long-term sustainability of water sources.
- d. Operation and maintenance practices: The studies shall review existing O&M practices to identify gaps and improvement areas.

iii. Findings of the Jal Seva Adhyayan shall be utilized to:

- a. Improve design protocols: Insights generated through the program shall be leveraged to inform improvement and refinement of technical designs and planning approaches.
- b. Strengthen O&M frameworks: It shall be used to support strengthening of operation and maintenance frameworks and institutional arrangements.
- c. Identify replicable best practices: The program shall facilitate identification and documentation of good and replicable practices for wider dissemination.

B. Rural WASH Partners' Forum (RWPF)

RWPF shall be leveraged as a national knowledge and collaboration platform for:

- i. Promoting Innovation: RWPF shall facilitate exchange of experiences and ideas to support development, piloting and adoption of innovative and context-appropriate solutions in the rural WASH sector.
- ii. Development of knowledge products: RWPF shall support preparation and dissemination of guidance documents, technical notes, case studies and other knowledge products.
- iii. Capacity building of sector institutions: RWPF shall enable knowledge sharing, peer learning and exposure to good practices aimed at strengthening institutional capacities at State and local levels.



- iv. Dissemination of evidence-based practices: RWPF shall support sharing of research findings, field evidence and documented good practices to inform policy, program design and implementation under the Mission.

C. Cross learning in convergence with line Department and other Ministries.

- i. **Ministry of Housing and Urban Affairs (MoHUA) (Atal Mission for Rejuvenation and Urban Transformation (AMRUT) and Smart City Mission)** – Use of GIS mapping, IoT-based monitoring and data-driven for smart water management.
- ii. **Department of Water Resources, River Development & Ganga Rejuvenation**– Sustainable water use, conservation, surface and groundwater planning and management, including river basin organizations and Water Resources Information System (WARIS).
- iii. **Ministry of Science and Technology** – Advanced water treatment technologies like membrane filtration and oxidation processes.
- iv. **Ministry of Electronics & IT (MeitY)** – Digital tools, real-time analytics and IoT-based water monitoring.
- v. **Ministry of Environment, Forest and Climate Change (MoEFCC)** – Sustainable watershed management and climate-resilient infrastructure.
- vi. **State Water Departments & Panchayats** – Localized water conservation and decentralized management models.

2.11.9. Outcome-Based Financing for R&D Proposals

Outcome-Based Financing (OBF) shall be adopted to link financial support for selected R&D projects with the achievement of defined milestones and measurable results. This approach encourages structured progression from concept development to field validation and ensures that technologies demonstrate verifiable performance.

i. Key Principles

Funding for research and innovation projects shall be linked to the achievement of clearly defined

outcomes and milestones. This principle emphasizes measurable progress, encourages efficient use of resources and aligns financial support with demonstrated result.

ii. Indicative Financing Structure

The States/UTs shall earmark support funds for research and piloting initiatives in the rural water supply sector, with approval, oversight and tranche release undertaken at the State/UT level, including clearance by the State Technical Committee. Such R&D proposals shall be integrated into the State/UT Annual Action Plan (AAP) to enable structured planning, financial provisioning and coordinated implementation and shall align with the Comprehensive Implementation and Reforms Plan (CIRP), which serves as a strategic, time-bound roadmap for the Jal Jeevan Mission by outlining utilization of earmarked central and state funds and key governance, service delivery, sustainability and implementation milestones and deliverables, with adherence linked to the release of payments.

- a. **Proposal approval:** Initial seed grant (up to 30%)
- b. **Prototype development:** Milestone-based tranche.
- c. **Field validation:** Disbursement linked to pilot performance.
- d. **Independent evaluation:** Performance verification by premier technical institutions such as IITs, NITs, CSIR labs, any other institutes as recommended by NTC.
- e. **Scaling recommendation:** Final tranche upon demonstrated impact.
- f. **Outcome indicators**

R&D proposals shall be assessed based on clearly defined outcome indicators that demonstrate measurable improvements in cost efficiency, service quality and system performance. Outcome measurement shall be benchmarked against baseline values established at the start of each R&D project to enable consistent and transparent assessment of progress. These indicators shall include:

 - a. **Reduction in capital or O&M costs:** extent



to which the solution lowers infrastructure, operational, or lifecycle expenditure.

- b. **Improvement in water quality compliance:** ability to enhance adherence to drinking water standards through better treatment, monitoring, or contamination control.
- c. **Enhancement in energy efficiency:** reduction in energy consumption in pumping, treatment, or distribution systems.
- d. **Improvement in system reliability:** contribution to consistent service delivery and reduced operational disruptions.
- e. **Reduction in non-functional schemes:** impact on restoring or sustaining functionality of rural water supply systems.
- f. **Replicability across States/UTs:** suitability of the solution for adoption in diverse hydro-geological and operational contexts.

iii. **Monitoring of R&D Projects:**

The R&D portal shall be utilized as a central platform for monitoring the progress of approved R&D proposals. It shall enable systematic tracking of milestones, documentation of field progress, submission of periodic updates and review of technical outputs. The portal shall maintain digital records of evidence, pilot data, verification reports and audit trails to ensure transparency.

iv. **Verification Mechanism:**

Outcomes shall be validated through evidence-based processes to ensure transparency, technical robustness and credibility of results. It shall include:

a. **Independent technical evaluation**

External assessment of the technology's performance, design adequacy and suitability for rural water-supply conditions.

b. **Pilot performance data**

Analysis of quantitative and qualitative data generated during pilot implementa-

tion to assess operational performance under field conditions.

c. **Field validation reports**

Structured on-site assessments documenting functionality, user experience, efficiency and overall system performance.

d. **Certification by the State Technical Committee and NJJM**

Formal technical review and endorsement confirming alignment with sector standards and readiness for wider adoption.

2.11.10. Key action points for the States/UTs

- i. Notification of the STC with clear TORs for appraisal, pilot approvals and scale up recommendations aligned to NTC
- ii. STC notified with approved Terms of Reference aligned to NTC.
- iii. Launch of structured program for State led R&D, product development and pilot implementation using an Outcome Based Financing (OBF) approach
- iv. Call for Research Proposal Outcome Based Financing (OBF) approach, covering concept development, prototyping and field validation.
- v. Approval of Research Proposal.
- vi. Verification, standardization and phased scaling of the developed product across suitable hydro geological regions, based on independent evaluations and approvals from the State Technical Committee.
- vii. Prototype development and piloting
- viii. Independent evaluation report completed.
- ix. STC approval/certification
- x. Documentation including Technology Protocols and SoPs.
- xi. Scale up plan implemented in at least one cluster.

The implementation of research, innovation and knowledge eco system shall be done as per SOPs issued by the Department.